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2022 K-Innovation Partnership Program with Perú

S&T Planning and Technological Forecasting for Prioritized Sectors in Perú

SCIENCE AND TECHNOLOGY POLICY INSTITUTE

CONSEJO NACIONAL DE CIENCIA, TECNOLOGÍA E INNOVACIÓN

2022 K-Innovation Partnership Program with Perú

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In Cooperation with	National Council of Science, Technology and Technological Innovation of Perú (CONCYTEC)
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Executive Summary

Peru is a wealthy country in terms of natural resources, and its mining industry is the most important sector for economic growth. On the basis of these resources, we can say that Peru has great growth potential, while the Peruvian government has tried to bolster national competitiveness by linking advanced technologies with core industrial sectors.

In order to promote the country's scientific foundation and technological capacity, the Peruvian government reformed the Law on the National System of Science, Technology and Innovation (STI) in 2021, the main purpose of which was to establish a national system composed of universities, public research institutes, and companies, as well as all entities and human resources related to STI. The Peruvian government has confirmed the National Council of Science, Technology and Technological Innovation (CONCYTEC) as the governing body, which is responsible for planning economic development by formulating STI policies using modern methodologies such as foresight, technology forecasting, and technology roadmapping.

Against this background, CONCYTEC requested STEPI to share Korea's experience, knowledge and skills in planning S&T (Science & Technology) policies, monitoring R&D activities, and building technological capacities. In this respect, STEPI's main activities consist of policy consultation, capacity-building training, foresight and forecasting practices, and workshops.

The main objective of the K-Innovation Partnership Program with Peru is to strengthen the capacities of Peruvian civil servants in order to gain a competitive advantage in prioritized sectors based on advanced knowledge. The STEPI team will conduct foresight and forecasting practices in order to build a technology roadmap and devise mid-to-long term technology strategies focusing on prioritized sectors. Then, this project aims to develop a national R&D program based on the technology roadmap for the prioritized sectors and establish a permanent center to take charge of technology foresight and forecasting.

To achieve these goals, STEPI and CONCYTEC have engaged in various types of activities together. In February 2022, STEPI and CONCYTEC held a preliminary meeting to introduce the research team from each side, discuss the scope and direction of the project, and arrange the project timetable, as well as preparing the MOU and TOR. During this meeting, CONCYTEC suggested holding the Korea-Peru S&T (Science and Technology) Seminar in October in Lima, Peru. The two parties have agreed the project details including the annual timetable.

In June, STEPI and CONCYTEC held the official kick-off meeting and started the in-depth study via a virtual meeting. The main objective of the kick-off meeting was to improve understanding of the project details, including the outline, scope, local consultant, terms of reference, and schedule. The two parties identified the project stakeholders and their roles, the overall scope of the research, and the specific needs of each side, as well as discussing data collection and information sharing. They also reviewed the details of the ToR signed between STEPI and CONCYTEC for the K-Innovation Partnership Program with Peru.

In September, STEPI and CONCYTEC jointly held the Capacity Building Workshop via a virtual meeting. On September 28, STEPI gave lectures on Korea's STI policies and technology forecasting, and both parties participated in the follow-up discussion. On September 29, the STEPI team delivered in-depth lectures on technological forecasting methodologies, and both sides participated in the follow-up discussions. The main objective of this workshop was to improve the Peruvian participants' understanding of technology forecasting methodologies and the context of South Korea's STI policies during the last fifty or so years.

From October 18 to October 20, STEPI and CONCYTEC held the Dissemination Workhop in Lima, Peru. The two parties invited Peruvian experts to a technology foresight practice session for three days in order to build practical capacity and share South Korea's experience. On the first day, the STEPI team introduced the basic concepts of technology foresight and provided guidance on foresight practices. Then, the participants selected the main drivers based on the results of a STEEP analysis, drafted scenarios, and developed the vision and goals of the technology roadmap.

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Project Overview

S&T Planning and Technological Forecasting
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Chapter 1. Project Overview

1. Introduction

The K-Innovation Partnership Program is one of South Korea's most representative ODA programs for the development of science, technology, and innovation (STI) policies. The program aims to provide STI policy consultations that meet the needs of South Korea's partner countries, particularly with regard to the use of Korea's highly-recognized STI development models and ample experience, as well as offering capacity-building training programs in order to effectively design and implement the STI policy of its partners for sustainable development. These policy consultations are carried out after conducting a selection process for a number of projects presented by the applicant countries, Peru being one of the selected countries which presented a project.

Peru is a wealthy country with abundant natural resources, and the mining industry is essential to the country's economic development. Based on these resources, Peru has strong growth potential, and seeks to strengthen national competitiveness by linking science and technology policies to sectors of the national economy with strong potential.

With the aim of promoting the STI system and its capacity, the Peruvian government last year reformed the Law on the National System of Science, Technology and Innovation (2021), whose main purpose was to establish a national STI system composed of universities, public research institutes, and companies, as well as all the entities and individuals related to STI. The newly reformed law also confirmed CONCYTEC as the governing body, whose role is to contribute to the planning of the country's development by formulating STI policies using modern planning instruments such as foresight, technology forecasting, and technology roadmapping.

Peru aims to promote technological development across the country, as evidenced by its National Competitiveness and Productivity Policy (2018), one of whose prioritized objectives is to generate the development of the capacities required for innovation, adoption, and the transfer of technological improvements.

The following sectors have been prioritized to foster a productive environment and innovation via scientific and technological activities:

1) Agroindustry and food processing (synthetic food)
2) Forestry (timber, non-timber, cellulose)
3) Textiles and apparel
4) Mining and related manufacturing (composites, nanotechnology)
5) Advanced manufacturing (automation, specialization, software, biotechnology)
6) Ecotourism, restaurant and catering, and creative industries

Therefore, it is important both to improve the capacity of public employees to make better plans and to use technological forecasting in order to make a major change in the construction of Peru’s future.

This project aims to strengthen skills in the planning and monitoring of technologies and observe their application, and skills in building technological forecasting scenarios.

With this background, CONCYTEC requires STEPI to share Korea’s experience and knowledge in such areas as policy consultation, capacity-building training to establish STI policy, and technological forecasting planning for those areas of the country with great potential.

2. Objectives

The main objective of the project is to strengthen the capacities of Peruvian civil servants at the national and sub-national levels to carry out planning and technological forecasting in order to build competitive advantages based on knowledge in the sectors prioritized by the country.

For this reason, the project was selected for the K-innovation Partnership Program. The participants will cooperate for the successful implementation of the project, which consists of the following activities:

1) Activity 1(2022): Developing CONCYTEC's capacity and institutional building plan to improve the foundations for S&T planning and technological forecasting for the prioritized sectors.

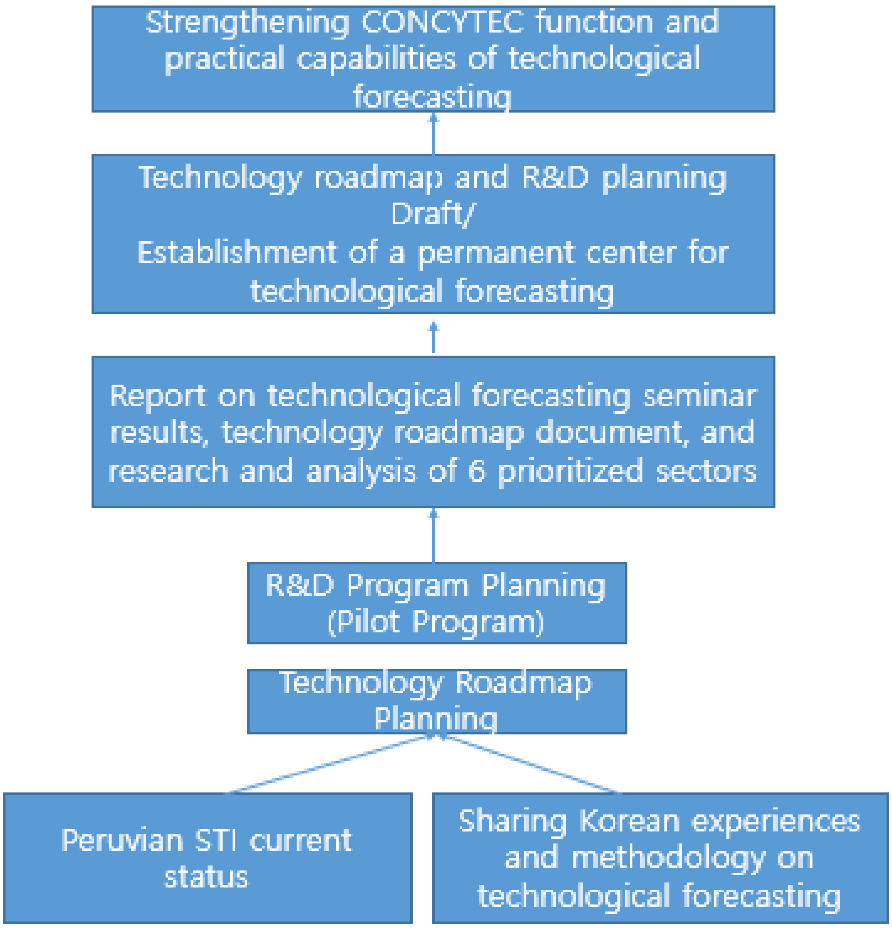
2) Activity 2(2023): Building the technology roadmap and devising the mid-/long-term technology strategy based on technological forecasting for the prioritized sectors.

3) Activity 3(2024): Developing the R&D program based on a technology roadmap for the prioritized sectors and establishing a permanent center for technological forecasting.

3. Project Framework

The project framework is configured for the purpose of strengthening Peru's capacity for S&T planning and technological forecasting in the prioritized sectors. It will be carried out over a period of three years with the goal of surveying the current status of Peruvian STI and sharing the technological forecasting methodology, establishing a technology roadmap and guidelines for the mid-/long-term technology strategy, planning R&D programs based on the technology roadmap and establishing a specialized institution for technological forecasting.

[Figure 1-1] Analytical Framework



[Table 1-1] Research Framework

Year	2022 (1st year)	2023 (2nd year)	2024 (3rd Year)
Annual Objectives	Develop CONCYTEC's capacity and institutional building plan to improve the foundations for S&T planning and technological forecasting for the prioritized sectors.	Build the technology roadmap and mid-/long-term technology strategy through technological forecasting for the prioritized sectors.	Develop the R&D program based on a technology roadmap for the prioritized sectors and establis a permanent center for technological forecasting.
Specific Objectives	1. Share the technology foresight methodology. 2. Current Status of Six Prioritized Industries:* *① Agroindustry and food processing ② Forestry ③Textiles and apparel ④ Mining and the related manufacturing ⑤ Advanced manufacturing ⑥ Ecotourism, restaurant and catering, and creative Industries	Establish the technology roadmap for 6 key sectors.	1. Support the establishment of the technology foresight center. 2. R&D planning (draft).
Main Activities	• Share Korea's experience of technology foresight. • Analyze the current status of 6 prioritized sectors. • Improve the understanding of CONCYTEC researchers, local experts in 6 key sectors, and other key stakeholders on S&T planning and technological forecasting methodology	• Determine the detailed technology priorities in 6 key sectors. • Derive the technology acquisition methodology. • Share the R&D planning methodology.	• Practical R&D Planning • Proposal for planning the Technology Foresight Center

4. Project Team

4.1 Research Team

The teams of Korean and Peruvian researchers will conduct joint research to derive policy recommendations, and Korea intends to build practical capacity building and assist Peru with internalizing its own capacity to establish feasible plans.

[Table 1-2] Korean Research Team

Organization	Position	Name	Main Role
STEPI	Senior Research Fellow	ChiUng Song	PM • Strategies and policy analysis for Peruvian science and technology policy planning . • Strategies and policy analysis for technology foresight planning in Peru.
	Senior Researcher	So Hyun Kwon	PC, PI • Preliminary analysis of the current state of science and technology in Peru. • Strategies and policy analysis support for Peruvian science and technology policy planning. • Research for technology foresight planning in Peru.
SUNY (State University of New York)	Professor	Johng-Ihl Lee	Analysis and development of the technology foresight methodology, and conduct of training on technology foresight.
KISTEP (Korea Institute of Science&Technology Evaluation and Planning)	Senior Research Fellow	Hyun Yim	Practical training on technology foresight.
Kyung Hee University	Professor	Ahreum Hong	Analysis of technology foresight.
Deliverdy	CTO	Sehee Park	Analysis of technology foresight, and training on technology foresight.
KISTEP	Research Fellow	Seok-ho Son	Research on technology foresight oveview and Korea's experience.

[Table 1-3] Peruvian Research Team

Affiliation	Position	Name	Key role
CONCYTEC	Director of Research & Studies	Fernando Ortega	Project Manager(PM)
	Senior Specialist	Alfonso Rodríguez	Project Coordinator(PC)
	Senior Specialist	Camilo Figueroa	Deputy Project Coordinator(PC)
	Senior Specialist	Oscar Diez	Logistics Support
	Head of International Cooperation Office	Rocio Casildo	MOU, International Cooperation
National Agrarian University La Molina	(former) Chief of Budget and Planning Office	Alberto E. Valcarcel	Local Consultant on the Agroindustry and the Food Processing Industry
Environmental Investigation Agency Inc.	Forestry Specialist	Ines D. Orbegozo S.	Local Consultant on the Forest Industry
National University of Engineering(UNI)	Professor	Beatriz Gloria Orcón Basilio	Local Consultant on Textiles and Apparel
Pontifical Catholic University of Peru	Scientific Researcher	Jorge Dulanto Carbajal	Local Consultant on Mining and its Manufacturing Industry
University of Engineering and Technology	Assistant Professor, Bioenginering Department	Paul Cárdenas Lizana	Local Consultant on the Advanced Manufacturing Industry
ESAN Graduate School of Business (AACSB–AMBA)	Professor	Otto Regalado-Pezúa	Local Consultant on the Ecotourism, Restaurant and Catering, and Creative Industries

5. Main Activities

1) Kick-off Meeting and In-depth Study

Prior to the Kick-off Meeting, the Launch Meeting (Preliminary Meeting) was held in February 2022, when the Korean expert group met with the relevant government officials and experts online to improve understanding of the project and clarify its scope.

During the Kick-off Meeting and In-depth Study conducted in June 2022, the Korean expert group met the relevant government officials and experts online to identify the policy priorities, finalize the research topics, and acquire additional necessary information and data. The meeting aimed to cover the following topics:

- Discussion between Korean experts and the related organizations in Peru to finalize the research topics and collect relevant data and information.
- Selection of the most suitable counterpart organizations for further cooperation in conducting The Project with Peru.
- Recruitment of Peruvian experts as Local Consultants for each designated research topic.
- Presentation of the Korean experts' research proposals at the Launch Meeting (Preliminary Meeting) to build common ground from the starting point.
- Online interviews conducted by the Korean delegation with stakeholders in the relevant institutions of Peru in order to gain in-depth knowledge of the situation in Peru and to meet the relevant experts to gain knowledge associated with the research themes.

2) Policy Practitioners’ Capacity Building Training Workshop and Interim Reporting

The Policy Practitioners’ Capacity Building Training Workshop and Interim Reporting was held on September 2022, in the form of an online meeting, to share the interim policy recommendations. Policy practitioners from Perú were invited to this event. Two important subjects were addressed:

- Presentation of the interim research results and feedback by the Peruvian experts, which will be reflected in the final research results.
- Introduction of the relevant Korean institutions to the Peruvian delegation, including government ministries, relevant organizations, and companies.

3) Dissemination Workshop and Final Reporting

The Dissemination Workshop and Final Reporting was held in person in Lima from October 18-21, 2022, during which the final policy recommendations were presented to various associated participants and Peru’s high-level policymakers.

- Presentation of the final policy recommendations based on the final research findings.
- Presentation by Korean experts and Peruvian experts on the final policy recommendations to various participants and opinion leaders from the government, private sector, academia, and media during the Dissemination Workshop.

4) Publication of the Report

Upon completing the project annually, STEPI will publish the final reports on the policy recommendations, which will be shared between the Participants. The policy consultation report will be uploaded in English and Korean on the STEPI website, unless classified as confidential.

5) Monitoring and Evaluation of the Project Results

During and after the completion of The Project, both parties hall directly participate in the monitoring and managing of the results of The Project. This involves conducting interviews with the related stakeholders from Peru, collecting data and evidence of the results, and reporting the findings concerning:

- Implementation of the policy recommendations proposed through The Project with Peru, such as introduction or modification of policies, laws, institutions, systems, etc.;
- Exploration of the follow-up cooperation.

6. Project Schedule

In 2022 the project was carried out according to the following schedule. The project activities were carried out in a hybrid (offline/online) form due to the COVID-19 pandemic during the first half of the year, and the dissemination workshop was held in Peru in the second half of the year.

[Table 1-4] 2022 Project Schedule

Activities	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.
Project Planning and Launching												
Preliminary Meeting (Online)												
Discussion and signing of MOU												
Kick-off Meeting& In-depth Study (Hybrid)												
Local Consultant Contract Process												
Consignment Research Agreement												
Interim Reporting												
Policy Practitioners' Capacity Building Workshop (Hybrid)												
Dissemination Workshop in Peru												
Final Reporting												
Korea-Peru S&T Cooperation Forum and Seminar (Online/Hybrid/Peru)												

7. Deliverables

- **Deliverable 1:** Provide policy consultation through joint research to assist Peru's capacity to conduct S&T planning and technological forecasting in the policy areas agreed upon by the Participants.
- **Deliverable 2:** Form a research team composed of experts, which may employ "Local Consultants" from Peru to support the research.
- **Deliverable 3:** Conduct meetings, workshops, interviews, and field studies on the previously agreed-upon topics and provide logistical support to coordinate the visits of Peruvian public officials and experts to Korea to attend seminars, workshops, and training courses for the purpose of exchanging knowledge and experience.
- **Deliverable 4**: Exchange information and data on matters of mutual interest upon request.
- **Deliverable 5:** Seek opportunities to influence policy, decision-making processes, and follow-up projects and programs to achieve the mutually agreed objectives.
- **Deliverable 6:** Monitor and evaluate the implementation of the policy recommendations, and draft an annual report on the progress of the results of The Project.
- **Deliverable 7:** Provide recommendations to connect the Korean and Peruvian innovation ecosystems for activating collaboration among innovation actors.
- **Deliverable 8:** Submit the Final Report.

CHAPTER 2

Introduction
of Technology Forsight

Chapter 2. Introduction of Technology Foresight

1. Theoretical Background

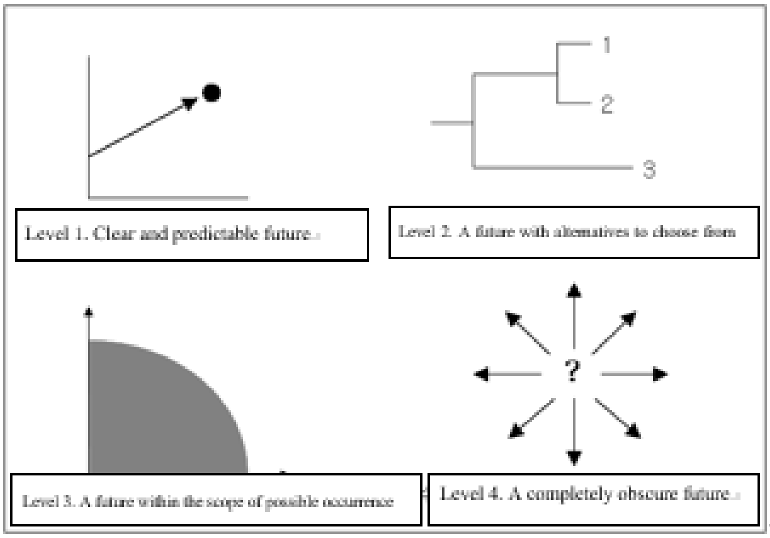
1.1 Necessity of Technology Foresight

Amid the increasingly fierce competition for technological supremacy among developed countries, many countries around the world are making earnest efforts to occupy promising fields preemptively. Who secures a cutting-edge technology and maintains the super-gap is becoming the key to national competitiveness. On the other hand, given the reality that the cycle of technological innovation is accelerating, it is becoming ever more difficult to find an answer to the question of which technologies to secure and with what strategy. Also, even if a country has identified the technologies it needs, it cannot necessarily afford to invest in the resources required to develop all the technologies it wants. Therefore, strategic judgment at the national level is necessary to ensure the efficient use of inevitably limited financial resources. The starting point for overcoming these limitations is the implementation of science and technology policies, including strategic plans based on future prospects. Technology foresight has been driven by the private and public sectors' strategic need for rapidly changing and developing science and technology. Technology foresight began to develop into a systematic means of exploring the future of technologies in 1949 with massive sponsorship from the U.S. government (Jeong Geun-ha et al., 2002). Its development was accelerated by the demands of the times, such as the Space Race, as well as the arms race, between the United States and the Soviet Union during the Cold War. In attempting to predict the development trend of technologies more accurately, countries sought to know specifically when they would be able to achieve their desired goals in connection with their unique social needs. For the next decade, technology foresight was largely focused on predicting the pace of technological change accurately. Against this backcasting of the needs of the times, research on technology foresight was actively carried out by various countries such as Japan, the EU, United States, and Korea.

Technology foresight reflects the changes in future society and the needs associated with it, and presents reasonable results on the direction, speed, and scope of technology development. Based on these results, countries can set a desirable direction for the development of their science and technology programs. More specifically, they can obtain the information they need for long-term planning in the field of advanced technology with high social utility. For this reason, technology foresight is widely used not only by governments as an important policy tool at the national level, but also by companies seeking to establish mid-to-long-term development plans, while technology foresight itself keeps going through various changes. The focus is shifting from the developmental direction of technologies to the socio-economic ramifications. As such, Japan, the EU member states, and other countries are actively introducing and utilizing methods that allow a wide range of stakeholders in society to participate in the technology foresight process. Korea is also moving away from the method that focused on predicting future technologies and the timing of their realization, and is now placing greater emphasis on predicting future social changes and the associated socio-economic needs.

In addition, while the future only leads to one specific result, changes associated with the result do not occur as set at the time of forecasting. In other words, no one can be completely sure that things will come true exactly as predicted. With the conventional, orthodox method of prediction, attempts were made to predict the future as accurately as possible through experts' intuition or the Delphi method. For this reason, orthodox technology foresight focused on a limited scope of specific phenomena or influences rather than seeking broad insights into the future. The process excluded any uncertainties that could arise from emphasizing a specific scope. As a result, the results of prediction did not adequately capture the future changes, or they only explained specific phenomena, revealing limitations in terms of policy use. Consequently, efforts were made to figure out a method of considering various changes that might occur in the future and to verify the factors that could influence and affect them. From trying to accurately predict a phenomenon that will occur at a specific point in the future, a shift is being made to strategic technology foresight that can recognize various possibilities of change and capture them rationally and logically.

[Figure 2-1] Four levels of uncertainty in Hugh Courtney's "Strategy Under Uncertainty"



Source: Yim Hyun et al., 2010.

1.2 Concept and Evolutionary Direction of Technology Foresight

Technology foresight is defined[1] as "the process of comprehensively looking into the future of science and technology and socio-economic changes from a long-term perspective in order to select future-based technologies and strategic research areas that are expected to create the greatest economic and social benefits."

Technology foresight predicts and explores the direction of future technological development with the aim of providing countries and corporations with necessary knowledge and information when they establish R&D plans and strategies. In other words, it enables them to design specific policies for the development of science and technology

[1] Based on "Special Issue on Government Foresight Exercises" (OECD, 1996) and the definition of Martin, B. (1995), who first introduced the concept of foresight in the UK's 2nd technology foresight survey.

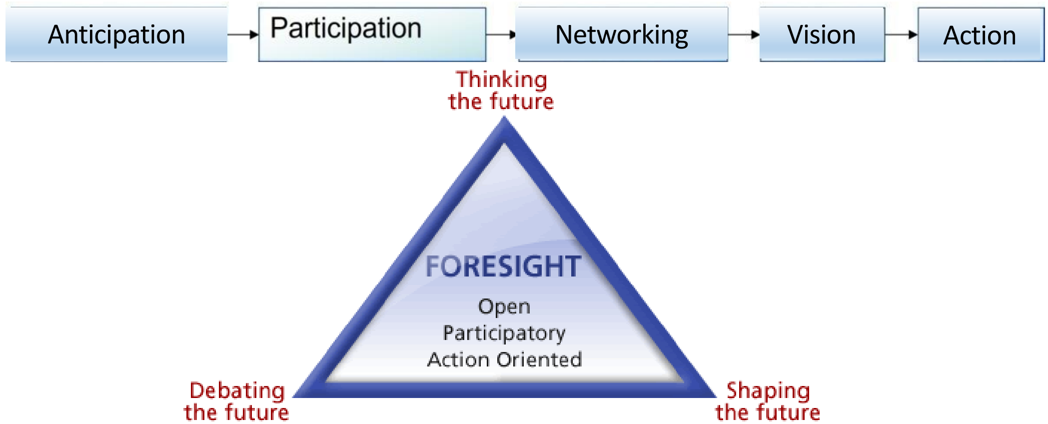
by using technology foresight to foresee future changes that are highly likely to occur and linking the results with economic and social needs.

Technology foresight explores the direction and prospects of future technological development with the aim of providing countries and companies with the information they need to establish their research and development plans. In other words, by forecasting the future situation - including future economic and social needs - more precisely, the development of science and technology can be effectively linked with economic and social development.

[Reference] The EU's Concept of Technology Foresight

Foresight is a systematic, participatory, future-intelligence-gathering and medium-to-long-term vision-building process aimed at enabling present-day decisions and mobilizing joint actions. It can be envisaged as a triangle combining “Thinking the Future”, “Debating the Future” and “Shaping the Future”. Foresight is neither prophecy nor prediction. It does not aim to predict the future – to unveil it as if it were predetermined – but rather to help us build it.

< The Concept of Technology Foresight as Defined by the EU>



Source : JRC IPTS[2], 2022.

[2] Institute for Prospective Technological Studies (IPTS) is one of the seven scientific institutes of the European Commission's Joint Research Centre (JRC).

Based on the various definitions mentioned above, the purpose of technology foresight can be summarized as follows.

"Activities that produce knowledge and information that are useful for mid-to-long-term planning in cutting-edge fields by conducting rational forecasts on the direction and speed of technological development in light of future social environmental changes and needs, and suggesting desirable directions for scientific and technological development."

In order to achieve this goal, technology foresight itself has undergone continuous change and development. Its development can be divided into three generations: The first generation was led by experts in specific fields. The second generation was led by industry and academia to predict market changes and the development of science and technology. The third generation is transitioning to a problem-solving method that includes social factors with stakeholders from a wide range of backgrounds involved in the process. (Georghiou, 2001).

The third-generation technology foresight differs from the previous approaches in terms of how we see the future. Assuming that the future could be chosen by technology, the orthodox technology foresight posited a single future and sought to accurately predict the timing of the technological and social realization of a particular technology. In other words, the technological foresight of the third generation is a much more dynamic process than that of the first and second generation approaches, as it involves various activities to explore the interaction between S&T and society as well as present and future[3].

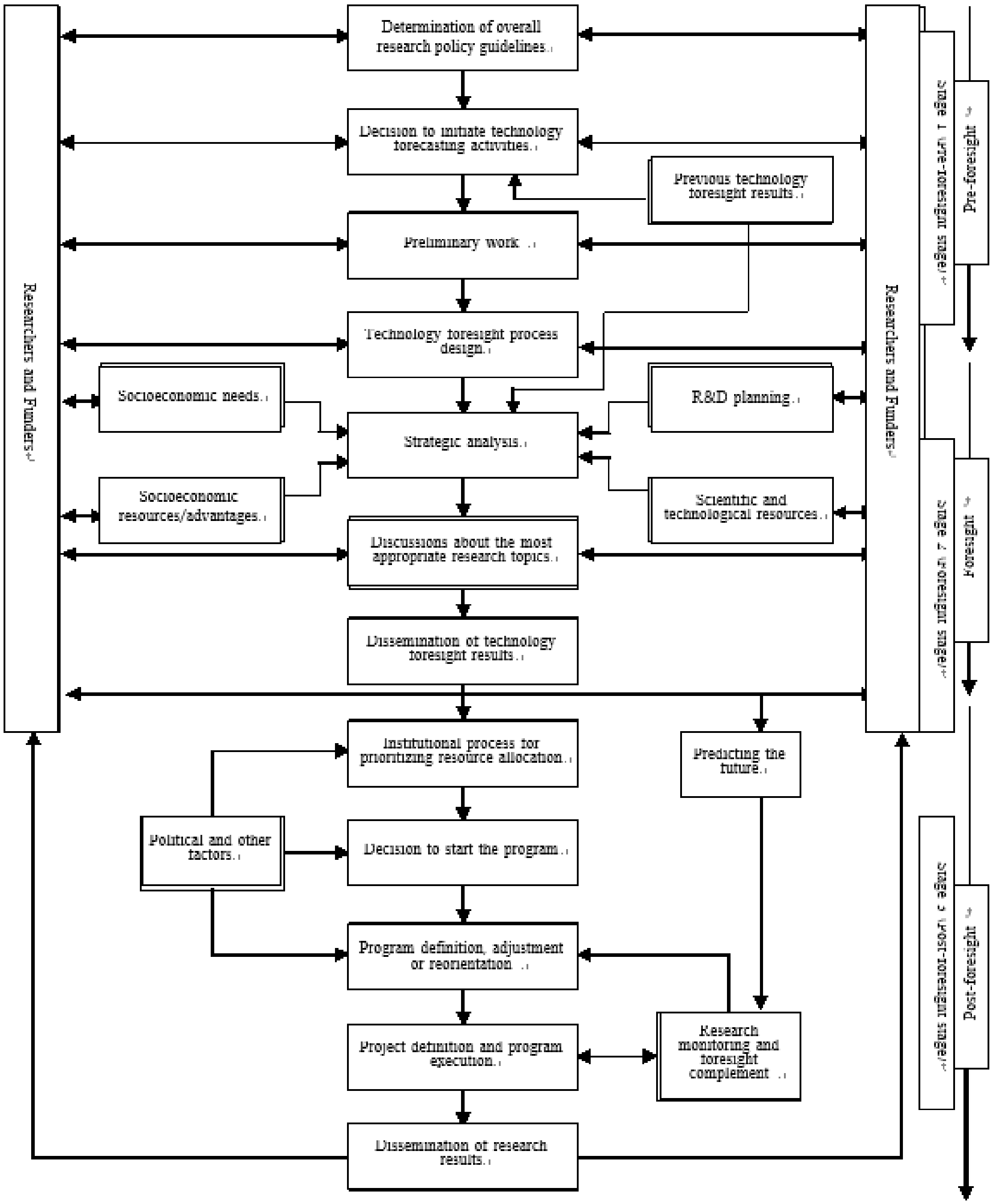
This shift in technology foresight is due to the increased need for policymakers to prioritize support for science and technology with limited resources. The principle of selection and concentration is essential for the efficient allocation of limited resources for economic and social development. To this end, the policy needs to gather the opinions of experts and identify promising technologies that are expected to emerge in the future.

[3] While the forecast was suited to the 1st and 2nd generations, foresight represents the technology foresight of the 3rd generation. Although they differ slightly in the basic philosophy, purpose, and process, their methods are highly similar, so they are collectively referred to as (technology) foresight (Jeong Geun-ha et al., 2002, partially modified after citation).

The third-generation technology foresight is a series of processes that systematically analyze the current opportunities and options and determine future selective outcomes at the present point in time (Grupp, 1999).

The biggest feature of such technological foresight change is the recognition of various possibilities of change in predicting the future in line with technological development. In the past, technology foresight focused on the prediction of the future in a single direction by reflecting the results of technology development based on statistical analysis, but the approach itself has changed. The new concept of technology foresight acknowledges the possibility of much more diverse and dynamic future changes in relation to the interaction between present and future, and technology and society (Jeong Geun-ha et al., 2010). It is an organized and systematic effort to carry out future prospects more effectively based on the acknowledgment that there is not only one type of future, but that there are many possibilities. Through technology foresight, countries and organizations can secure key basic data that can be used in establishing their mid-to-long-term R&D plans. Moreover, it can attract the active interest and participation of more scientists and engineers in the implementation of research and development. By using the results of technical foresight, it can reduce the trials and errors that may be caused by inaccurate information.

[Figure 2-2] Procedures and Components of Technology Foresight



Source: B. Martin(1989)

Not only is technology foresight fundamentally aimed at a highly uncertain future, but it is also subject to confusion as it is compared to the different results produced by other forecasting activities at the same time. Further, technology foresight has various limitations, such as the inherent inaccuracy of the process itself. On top of that, many experts have negative or conservative opinions about rapid changes in science and technology, while most scientists tend to be too optimistic about the technological and social applications of their R&D results. Also, the fact that some experts tend to underestimate the rapid pace of development of science and technology serves as a limitation in actual technology foresight activities. [Figure 2-3] sums up the technology foresight process based on this concept. The process is divided into three stages: the pre-technology foresight stage, the technology foresight stage, and the post-technology foresight stage. Various stakeholders, such as researchers and R&D investors, participate up to the technology foresight stage (Martin, 1989).

The pre-technology foresight stage aims to prepare for successful implementation. First, overall guidelines for foresight implementation are prepared. Based on past technology foresight experiences and results, matters that need to be checked when carrying out full-scale execution are determined, followed by the design of the foresight process. In the technology foresight stage, an analysis is conducted, based on the designed process, of economic and social needs, opportunities for R&D on future technologies, and S&T policy implementation strategies, and the results are disseminated and shared. In the post-technology foresight stage, the most important goal is to establish and implement a strategy for allocating resources to high priority areas analyzed through foresight. At this stage, an institutional foundation for allocating R&D resources is established, and the R&D program is implemented after prior adjustment and review of the scope and method of technology development. Further, even after implementing the R&D program, systematic monitoring is promoted and continuous enhancements are carried out to ensure that the program meets the set goals.

2. Key Technology Foresight Methodologies

The methods used for technology foresight have evolved into very diverse forms according to science and technology R&D systems that reflect each country's political, economic, industrial, social, and cultural characteristics because the future with high uncertainty is the subject of research, and the complexity of reflecting diverse interests has to be addressed. The method mainly used in the early days of technology foresight was the Delphi method, which allowed various experts to produce a consensus on the resulting value. In other words, it was appropriate for realizing the goal of predicting technologies at the time, which was to establish an accurate picture of future change. Since then, techniques such as scenario building, environmental scanning, and simulations have been introduced and utilized. The main factor of these changes was to secure reasonable explanatory power by considering more comprehensively the various social and economic characteristics leading to technological development. In other words, technology foresight efforts were made to look at the long-term future of science, economy-industry, and society. In this context, if we look more closely, technology foresight methodologies are dominated by efforts to capture social changes as the analysis of market outlook has strengthened in recent years. That is, technology has begun to be included as a potentially important element.

[Table 2-1] Major Methodologies Used for Technology Foresight

Category	Methodology	Concepts and Methods
Probabilistic method	Delphi	• Assemble Expert panels are assembled and feedback is gathered through repeated surveys. • Surveys are usually repeated at least twice.
	Cross impact analysis	• A method of analyzing ripple effects by grasping the interaction between various factors that influence future change.
Qualitative method	Scenario building	• A hypothetical situation in the future is depicted in various ways. • Multiple scenarios can be conceived based on other foresight techniques.
	Expert panel	• A panel of a certain size is formed based on a list of experts in various fields. • A method of deriving strategies through discussion and consultation by setting a period of activity and holding periodic meetings to discuss the future of target topics.
	Environmental scanning	• A method of deriving, classifying, organizing and analyzing information related to future changes based on documents and digital data. • Useful for dealing with topics or issues that have arisen in the relatively recent past.
Quantitative method	Trend extrapolation	• A method of predicting a trend by drawing an extension line to a series of data with time-series characteristics. • Recently, with the rapid development of computing technology, various new mathematical and statistical attempts are being made.
	Dynamic modeling	• A method of modeling and analyzing the interrelationships of variables using a mathematical model with simulation techniques.

Source: Son Seok-ho et al. (2008) with partial corrections and supplements made.

As the demand for future economic and social prospects increases, technology foresight processes and methodologies are becoming important. [Table 2-1] above describes the concepts and methods of the main methodologies of technology foresight, while [Table 2-2] below compares the advantages and disadvantages of each methodology.

[Table 2-2] Comparison of Advantages and Disadvantages of Key Methodologies Used in Technology Foresight

Methodology	Advantages	Disadvantages
Delphi	• With anonymity and independence, participants can express their opinions freely and approach issues that require judgment objectively. • Possible to perform various purposes depending on the structure of the survey. Ex: Upon establishing a strategy, it is possible to provide both problem recognition and value criteria about priorities among goals, the appropriateness of policy alternatives, and the feasibility of execution.	• Concerns about misjudgment due to experts' overconfidence. • Possibility of easily modifying one's opinions according to statistics provided in the multiple survey process. • Possibility of simplifying the future. • Difficulty in forming a group of experts that will respond to surveys.
Cross impact analysis	• Easy to analyze and identify the characteristics of each field and their correlations. • Easy to infer and grasp the core causes of future changes.	• Possibility of becoming a meaningless procedure depending on the level of detailed analysis (hindsight determi-nation).
Scenario building	• Possible to propose an alternative future. • Can be used for various purposes such as grounds for discussion, problem sharing, early warning, evaluation indexes, and establishment of strategies. • Enables organizations to quickly establish strategies when changes occur in the external environment.	• Difficult to rule out the influence of participants' subjective views. • Difficulty in decision-making and execution due to lack of specificity. • Possibility that future society scenarios with high feasibility may be overlooked when focus is placed on the future society that policy makers want.
Expert panel	• Possible to make quick decisions while eliminating prejudice through discussions and exchanges of opinions.	• Possibility that meeting outcomes can be impacted by influential experts.

Methodology	Advantages	Disadvantages
Environmental scanning	• Useful for surveys on emerging areas. • Useful for investigating and analyzing drivers of environmental changes. • Possible to collect useful information suitable for the preliminary stage of comprehensive foresight research. • Easier performance due to advances in web, information and communication, etc.	• Need to apply additional procedures to derive a detailed list of future technologies. • The scope of data reference is important, as useful information can be produced only when surveys are conducted continuously and extensively.
Trend extrapolation	• Easy to grasp development trends based on a series of data.	• Possibility of excessively seeking patterns where there is no pattern. • Possibility of excluding meaningful data lying outside the pattern. • Based on the assumption that current trends will continue into the future.
Dynamic modelling	• Reflects the complex operating mechanisms of systems. • A method that reflects the interaction and reflux structure between various elements. • Possible to derive key strategic variables within the reflux structure.	• An esoteric methodology. • Difficulty in understanding the process for calculating the final result.

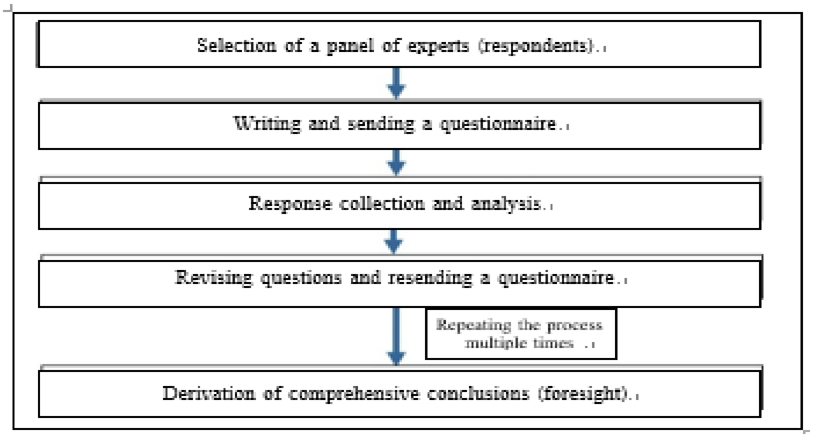
Source: Yim Hyun et al. (2010) and Jeong Eui-jin et al. (2019) with partial corrections and supplements made.

This report has examined in more detail the Delphi method, the scenario building method, the cross impact analysis method, the environmental scanning method, and the trend extrapolation method, since they are more frequently used in technology foresight than the rest.

2.1 The Delphi Method

The Delphi method is a survey-based method of technical prediction. It is designed to draw a group consensus without having the participants gather in one place for debates. It is an alternative to methods that seek a collective consensus in one place. This method's biggest advantage is that it can effectively tap into intuition based on experts' knowledge and experience. In other words, it is a way to reach a consensus through experts' intuition when the use of generalized or standardized data is limited. The Delphi method is conducted in the form of repeatedly exchanging survey questionnaires and the analyzed results among individual experts to give them opportunities to revise their response values and find the convergence value of opinions. In this process, the respondents to a survey can freely present their opinions as individual subjects on an independent, equal footing. The responses are processed anonymously so as not be influenced by outsiders. The method also repeats the controlled feedback process, leading to sustained interest in the topic. The response results are aggregated and processed. The opinion ranges and differences within each group can be confirmed. It is also possible to analyze minority opinions. In general, it does not involve the collection of opinions from a large number of the public, so it is not a process that leads to statistically significant results.

[Figure 2-3] Delphi Method Procedures (example)



The Delphi method starts with the selection of a panel that can provide expert opinions on a foresight topic after the topic has been established. Once the panel is formed, the participating experts present their opinions through an open-ended questionnaire; consequently, closed-ended questions must be constructed. The completed closed-ended questionnaire is sent to the panelists, and the results are collected and analyzed. After supplementing and revising the questions if necessary, the summarized and analyzed results are sent to the same respondents to present their response values and have them re-evaluate their previous answers with reference to the opinions of other experts. The re-evaluation process is repeated several times until the opinions among the experts are within an appropriate range. Based on the final convergence results, the foresight values are derived and the causes of disagreement among the experts are discussed.

The Delphi method can be applied to issues that require objective judgment in situations of high uncertainty, such as foresight. As it guarantees the anonymity and independence of the responses, this method has the advantage that experts are not swayed by others' opinions.

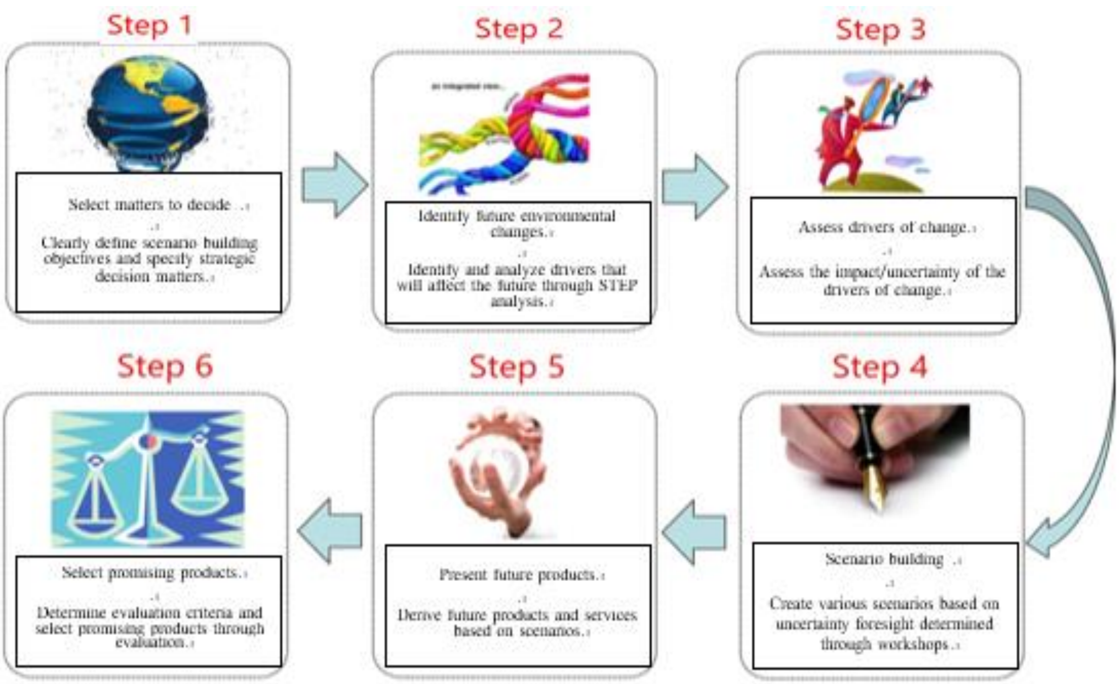
In addition, a large number of experts can participate, which means that the method can be operated cost-effectively. On the other hand, the disadvantages of the method include the fact that it takes too long to reach a conclusion and that the future can be oversimplified due to experts' excessive confidence. Some experts tend to unconsciously follow statistical results. In addition, it lacks intellectual stimulation compared to face-to-face meetings, resulting in a lack of discussion among experts with dissenting opinions.

2.2 The Scenario Building Method

The scenario building method is a foresight technique that helps us to clearly understand various aspects of the future by conveying various situations that can happen in the future in the form of a "story", like a play script (Yim Hyun et al, 2009). Most notably, scenario building methods that can systematically consider future uncertainties are gaining interest not only for technology foresight, but also in other areas as tools to support strategic decision-making.

Scenarios analyze and present the trends in various variables and the interactions between events in order to present a comprehensive picture of the future. In addition, by examining the consistency and systematicity of a series of forecasts, this method aims to describe possible future situations. It describes the process in which various situations develop on the premise that various unexpected situations may occur in the future, and presents important variables and their interrelationships, the order of occurrence of events, and the decision-making process. Therefore, it has the advantage of overcoming the error of categorical foresight because it takes into account all structurally different future uncertainties.

[Figure 2-4] Six Steps[4] of the Scenario Building Process (example)



In general, the scenario building process follows the six steps shown in Figure 2-5 above. Step 1 is designed to clarify the main purpose of the scenario, i.e. the final decision-making issue and its duration. A group that wants to build a scenario sets up a scenario agenda by organizing various issues of interest. Step 2 largely consists in figuring out the drivers that

4 It is a reconstruction by researchers of the Korea Institute of Science & Technology Evaluation and Planning based on the scenario building procedures proposed by SRIC-BI (Stanford Research Institute Consulting Business Intelligence).

will affect future changes. "Change drivers" refers to external environmental factors that determine the future direction and value of decision-making matters. Based on the STEEPV[5] guidelines, the change drivers that are the key factors in creating the future are derived. Step 3 evaluates the drivers of change to identify those which are highly influential for future changes while nonetheless being uncertain, so that the number of cases needs to be weighed. Through this process, the change drivers that need to be looked at with particular attention are reviewed when predicting future changes. Step 4 constructs the framework by choosing the logical structure that forms the skeleton of the scenario. The scenario is created using the most appropriate method of expression centered on the constructed framework. Step 5 analyzes the opportunities and threats that will arise in the future, based on the assumption that each scenario has become a reality. Based on the results, specific countermeasures (promising products/items) that can take advantage of opportunities and minimize threats are presented. Lastly, Step 6 prioritizes the abovementioned countermeasures (promising products/items) that should be implemented.

2.3 The Cross Impact Analysis Method

The cross impact analysis method started from the fact that the major factors that cause future changes influence each other, and that the prediction results are also correlated. The cross impact analysis method quantitatively identifies how a given factor causes a specific event, as well as how that factor affects the changes in other factors if a certain factor causes a particular event. The quantitatively calculated value is included in the analysis of the prediction results.

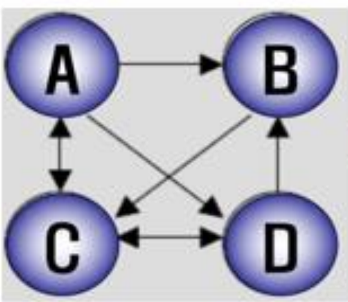
Factors that cause future changes influence each other and cause events. The Delphi method ignores the interaction between future change factors and makes predictions based on the intuitive judgment of experts. The cross-impact analysis is designed to solve this problem. It is a method of analyzing the probability of occurrence of a factor or

[5] STEEPV is an abbreviation of six fields including social, technological, economic, ecology, politics, and human values.

making a judgment on the interaction that exists between other future change factors while understanding the ripple effect of each factor related to the subject to be predicted in light of the judgment.

The cross impact analysis method, like the Delphi method, derives numerical results based on the opinions of experts. However, there is a difference in that it predicts the probability of future changes according to certain factors or makes judgments about the interactions that exist between other factors. One of the main ways to use cross impact analysis is scenario writing. The participating experts predict the likelihood of the occurrence of various future change factors related to the forecast topic and predict the likelihood that each factor will be affected by changes in other factors. Cross-impact analysis focuses on the causes, effects, and directions of influence. For example, if there are future change factors called x, y, and z, the effect of x on y and the effect of y on z are analyzed. This analysis process can be represented as a matrix, which shows the likelihood of a scenario that can be created by a combination of change factors through mathematical analysis.

[Figure 2-5] Analysis of the Impact of Future Change Factors



	A	B	C	D	Total impact
A	-	1	1	1	3
B	0	-	1	0	1
C	1	0	-	1	2
D	0	1	1	-	2
Total dependency	1	2	3	2	8

[Figure 2-5] above shows an example of performing a cross impact analysis on four future change factors, A, B, C, and D. The direction of the arrow in the figure on the left indicates that one factor is affecting anothser. The number 8 in the lower right corner of the matrix is the sum of the relationships (influences or dependencies) between each factor A, B, C, and D. The total sum of the rows represents the total amount of direct impact of certain factors on the system, whereas the total sum of columns is the total amount of direct dependency that certain factors have within the system.

2.4 The Environmental Scanning Method

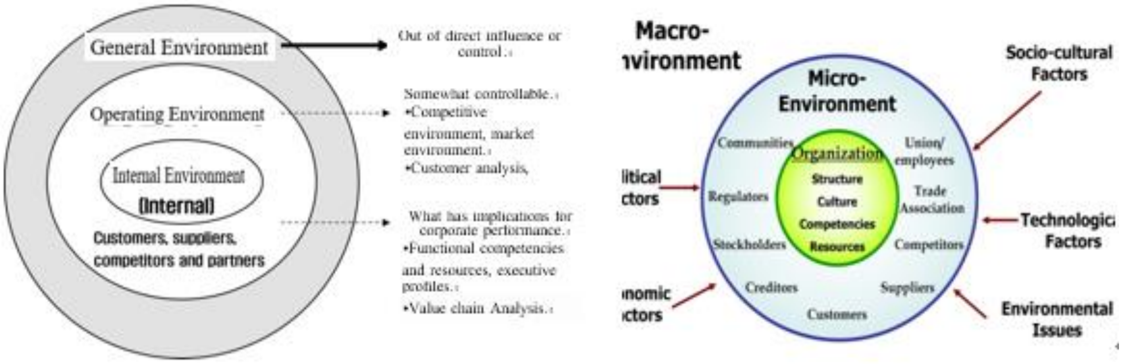
Environmental scanning is a method of finding and analyzing new issues. It is the starting point for almost all technology foresight. In general, this method is adopted as the most basic one among various methods to grasp information and knowledge about the fields or subjects of technology foresight. Environmental scanning is carried out through Internet searches and the systematic analysis of document data such as reports and papers written by experts and conference records. Internet searches based on databases, including press releases, are usually used. Information on various social phenomena related to the subject is extracted, classified, and organized.

Environmental scanning is useful for identifying recent issues or topics. This is because traditional methods such as the Delphi method and expert panels are limited in gathering sufficient data or information on new fields. With the growth of the Internet, it has become possible to find, organize, and categorize information on various topics.

Environmental scanning techniques are classified into several types as follows (Yim Hyun et al., 2010):

- Passive scanning: A method of continuously obtaining information by reading newspapers, magazines, periodicals, etc., or watching TV about an area of interest without any particular opinion.
- Active scanning: this method strives to expand the range of the scanning area as much as possible while regularly and continuously scanning specific information sources.
- Directed scanning: An organic and selective method of scanning for a specific purpose.

[Figure 2-6] Scope and Analysis Area of Environmental Scanning (example)



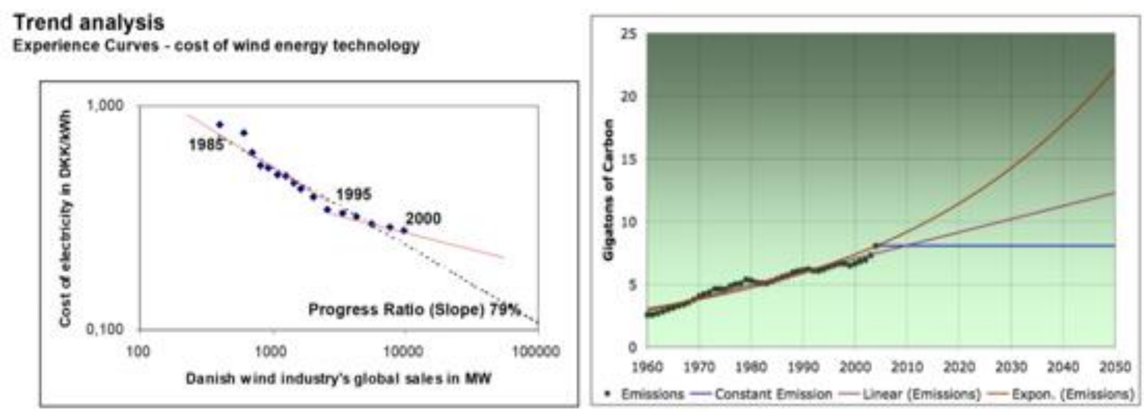
[Figure 2-6] illustrates the scope and analysis area in which environmental scanning is performed. It may be applied to the internal environment, such as the functions, capabilities, and resources of a group. On the other hand, it can be performed by extending the scope to the operating environment, which is largely within a controllable range, or to the general environment, which is beyond one's control. However, since most of the future problems that a group wants to predict are influenced by factors beyond its control, scanning is usually applied to the general environment, including internal and operating environments.

2.5 The Trend Extrapolation Method

Trend extrapolation is one of the most widely chosen methods for technology foresight activities. Foresight results based on the kinds of expert judgments that we encounter in reality are most likely to be the results of trend extrapolation. Trend extrapolation predicts the future based on the regularity with which a phenomenon is consistently observed within a certain period of time. It is also widely used because it is relatively easy to use and explain. A trend is a phenomenon that can be expressed quantitatively and has a visible pattern of development. Demographic changes and advances in technological capabilities can be good examples of this. "Trend" refers to patterns of change in the past, with extrapolation meaning that data related to these changes will be reflected in the future.

Trend extrapolation derives the direction of change from a long-standing trend. Given enough time, a short-term rapid change becomes a small phenomenon in the whole section. As such, it is possible to exclude, to some extent, future appearances that can be predicted incorrectly due to the trend of unusual phenomena over a short period of time. However, this requires solid evidence to give credence to the assumption that current trends will continue into the future. Thus, the trend extrapolation method has several limitations. First of all, there is a possibility of occurrence of many problems due to inaccurate data. For example, if historical data are essential for analysis while no value has actually been measured, expected estimates may be used to compensate for missing values. In such a case, inaccurate data from the past are used to predict future trends several decades later, resulting in inappropriate foresight results. It may also fail to take into account changes in the most important factors, and may sometimes lead to incorrect foresight for those factors. For example, a disruption of long-term energy consumption trends was anticipated based on changes in oil prices in the 1970s.

[Figure 2-7] A Case of Trend Extrapolation (example)



3. Case Analysis of the Science and Technology Foresight Program in Korea

3.1 Overview of Korea's Technology Foresight Programs

Korea's science and technology foresight programs are conducted every five years according to the Framework Act on Science and Technology. The first science and technology foresight program was launched in 1994 by the Science and Technology Policy Institute. This decision was influenced by foresight programs that Japan had been conducting since the 1970s. Since then, regular science and technology foresight programs have evolved in response to demands for expansion of the social role of science and technology. In 2001, the Framework Act on Science and Technology made science and technology foresight mandatory. The third program was the first to be carried out at the national level. In 2007, in order to secure linkage with the Basic Plan for Science and Technology – the top-level national plan for science and technology, the timing was adjusted and the contents were supplemented, with the results of the third foresight program re-announced.

Thanks to these efforts, the science and technology foresight programs conducted since then have been recognized as the core basic data for top-level science and technology planning at the national level.

Since the first science and technology foresight programs, continuous improvements and developments have been made in methods, systems, and processes. During the first program, when Korea benchmarked its performance against Japan's performance in the technology foresight, focus was placed on deriving future technology tasks and identifying the timing of their realization through the participation of experts. Since the third science and technology foresight program, many efforts have been made to predict not only the development of science and technology, but also the future societal changes driving its development. Through such efforts, future demand has been presented and scientific and technological tasks to meet the demand have been identified. In the fourth science and technology foresight program, on top of basic information such as the timing of the realization of future technology tasks, future changes were presented in scenarios alng with illustrations designed to make it easier for the general public to understand and access them. In the fifth program, participation by the general public was greatly expanded in an effort to evolve it into a policy formation model in which the opinions of various members of society could be reflected so as to overcome the limitations of the previous foresight programs, which had depended on the collection and analysis of experts' knowledge. In the sixth program, focus was placed on efforts to improve policy utilization. In addition to presenting various future technology challenges, new attempts were made to provide concrete answers to the question of which technologies should be chosen strategically to enhance national competitiveness.

By the end of 2022, six national-level science and technology foresight programs had been conducted. This report presents a detailed analysis of the national science and technology foresight programs conducted over a period of almost 30 years in Korea.

[Table 2-3] Comparison of Science and Technology Foresight Cases in Korea (6 programs)

Category	1st program	2nd program	3rd program	4th program	5th program	6th program
Launch date	1994	1999	2004	2012	2017	2022
Methods	• Expert brainstorming • Delphi (mail)	• Expert brainstorming • Delphi (mail)	• Environmental scanning • Delphi (online) • Scenarios for the future world	• Environmental scanning • Google search-based network analysis • Delphi (online) • Scenarios for the future world	• Environmental scanning • Google search-based network analysis • Knowledge map analysis • Delphi (online) • Technology diffusion point analysis	• Environmental scanning • Word Cloud • Expert brainstorming • Delphi (online) • 15 future innovative technologies (technology diffusion point analysis)
No. of future technologies	1,174	1,155	761	652	267	241
Survey areas	15 (technology areas)	15 (technology areas)	8	8 (issue areas)	6 (issue areas)	6 (issue areas)
Forecast period	20 years (1995-2015)	25 years (2000-2025)	25 years (2005-2030)	25 years (2012-2035)	25 years (2016-2040)	25 years (2021-2045)

Category		1st program	2nd program	3rd program	4th program	5th program	6th program
Promotional bodies		• Technology Foresight Committee • 12 Subcom-mittees (15 areas)	• Technology Foresight Committee • 15 Subcom-mittees (15 areas)	• Technology Foresight Committee • Technology Analysis Committee • 8 Technology Subcommittees • 2 Scenario Building Subcommittees	• General Committee • 3 Prospects Committees • 8 Subcom-mittees • 8 Evaluation Committees	• General Committee • Future Prediction Committee (3 subcommittees) • Future Technology Committee (6 subcommittees)	• Future Prospects Committee • Future Prospects Subcommittees (5) • Future Technology General Committee • Future Technology Subcommittees (6)
Key Features		• Derive future technologies based on the results of expert brainstorming. Centered around science and technology experts.	• Derive future technologies in consideration of R&D areas, purposes, and Korea's situation at the Technology Foresight Committee. • Centered around science and technology experts.	• Derive future technologies in consideration of future society's prospects and needs. • ·Trial introduction of the scenario building method. • ·Including humanities and social science experts.	• Trend analysis and introduction of Korean-style future demand analysis. • Investigation of potential negative effects. • Writing scenarios, illustrations, and technical briefs. • Including humanities and social science experts. • Evaluation of the realization of existing foresight surveys.	• Standardization of future technology names and technology descriptions. • Derive future technologies to effectively respond to future social issues. • Conduct surveys to collect public opinions in the process of selecting future social issues.	• Separate operation of the Future Prospects Committee and the Technology Committee. • Investigation of technology trends through analysis of academic papers. • Selection of 15 future innovative technologies and analysis of technology diffusion points.
Delphi Recovery Rate	1st	1,590 responses out of 4,905 (32.4%)	1,844 responses out of 4,500 (40.7%)	5,414 out of 32,411 responses (16.7%)	6,248 out of 68,991 responses (9.1%)	4,133 responses	2,147 responses
	2nd	1,198 out of 1,590 responses (75.3%)	1,444 out of 1,833 responses (78.8%)	3,322 responses out of 5,414 (61.4%)	5,450 out of 6,248 responses (87.2%)	3,621 out of 4,133 responses (87.6%)	1,617 out of 2,147 responses (75.3%)

* Adjusted according to the cycle of the Basic Plan for Science and Technology.

Source: Jeong Eui-jin et al. (2019), revised to reflect the latest information.

3.2 Case Analysis of Science and Technology Foresight Programs in Korea

The first Science and Technology Foresight Program, published in 1995, was Korea's first national-level study. In South Korea, where R&D was the main strategy for benchmarking developed countries, there was an increasing social need for future technologies that it had to meet by itself. The science and technology foresight program was launched to explore the long-term development direction of science and technology and meet the needs of the times. Based on the national-level science and technology foresight program model promoted by major competitors such as Japan, efforts were made to reflect Korea's social, economic, and industrial characteristics. However, due to the country’s lack of experience in scientific and technological foresight programs, the Japanese method was adopted to a large extent. In the first program, a system was prepared based on building the Technology Foresight Committee, which included authoritative experts from each field. The expert-centered committee used the Delphi method to select technical tasks and analyze technologies. In order to derive future technology tasks, ideas were collected from about 25,000 domestic and foreign experts by mail, etc., and the future technology task candidates were narrowed down to about 9,000. Subcommittees were formed in fifteen fields, and after holding an average of 3-4 meetings, they derived 1,174 future technology tasks. An itemized evaluation was conducted for each future technology task over the next 20 years, including such items as the timing of realization, importance, R&D methods, and the domestic level. The first science and technology foresight program laid the foundations for developing and introducing Korea’s unique promotion processes and methodologies.

The second science and technology foresight program, published in 1999, did not differ significantly from the first program in terms of its process and methodology. The Technology Foresight Committee was formed to select 1,200 future technology tasks in fifteen fields[6] as in the first one. The selection criteria applied at that time were as follows (KISTEP·STEPI, 2000).

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[6] ① Information-Electronics-Communication, ② Machinery and Industrial Processing, ③ Materials, ④ Chemicals, ⑤ Life, ⑥ Agriculture-Forestry-Fisheries, ⑦ Medical, ⑧ Energy, ⑨ Environmental·Safety, ⑩ Mineral·Water

- Significant technologies currently in the foundation, development or nascent stages worldwide and those tasks expected to lead the technology field down the road.
- Tasks suitable for solving special domestic issues .
- Except for those that have already been prototyped or will soon be developed at home or abroad.
- Balancing of short-, mid-, and long-term technical tasks.
- Balancing by technology field, etc.

Starting from the second science and technology foresight program, the time range of the forecasts was expanded to twenty-five years to explore the direction of technological development from a more long-term perspective and to compare the R&D results with those of major developed countries.

The biggest feature of the third program was the fact that, compared to the previous programs, social and economic needs were considered in light of the prospects for future social changes. That was to reflect the expanding social impact of science and technology and the fact that technological development was also driven by changes in society. To that end, in the selection of future technology tasks, the focus was on conducting social and economic needs surveys extensively and deriving technologies that could meet them. The basic approach was expanded from the technology-centered one. First of all, the composition of the technical tasks to be surveyed was expanded from the existing technology classification system. The experts participating in the science and technology foresight programs were induced to use their imagination and creativity for the future as much as possible, while keyword-centered discussions were encouraged to consider fully the acceleration of technological development, the enormity and expansion of technological development, and technological convergence. In addition, the scope of the

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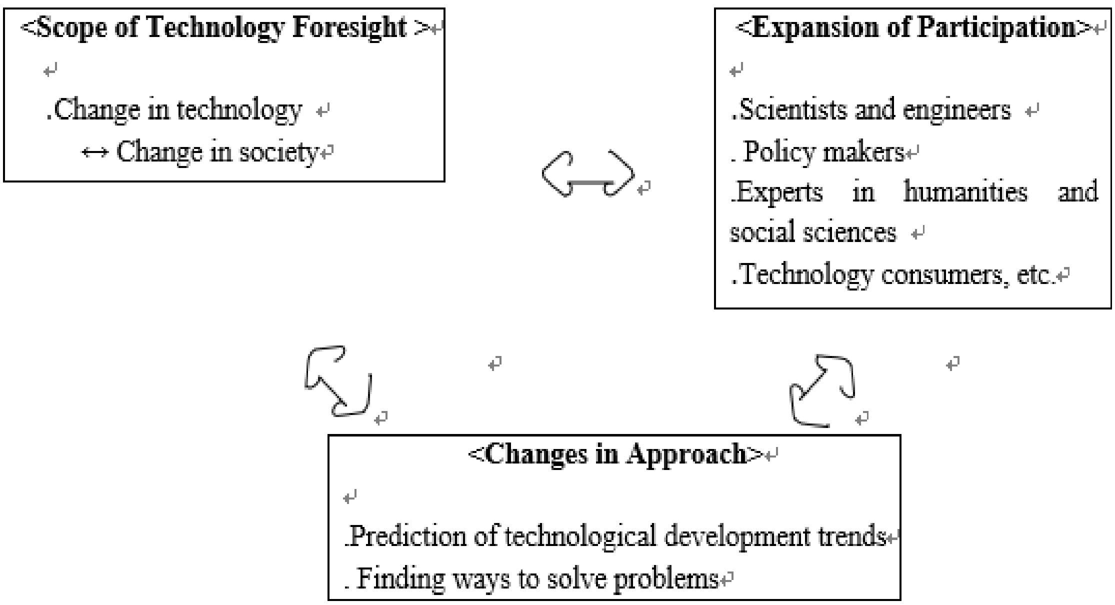
Resources, ⑪ City·Construction·Civil Engineering, ⑫ Transportation, ⑬ Ocean·Earth, ⑭ Astronomy·Space, ⑮ Extreme Technology.

Delphi survey was greatly expanded compared to the second foresight program, whereas the postal survey was replaced with an Internet-based online survey.

The biggest change in the third science and technology prediction program was that it presented the prospects for changes in the future society in terms of politics, economy, society, and culture. Contents expected to be an issue in the future society were included, while scientific and technological countermeasures by Korean society were presented. This expansion of the scope of foresight from technology to society in the third science and technology foresight program became the standard for all subsequent programs. Although the methods and objectives for the future social outlook changed with each program, the third program was important in that it established the idea that the development of science and technology should be predicted along with social change.

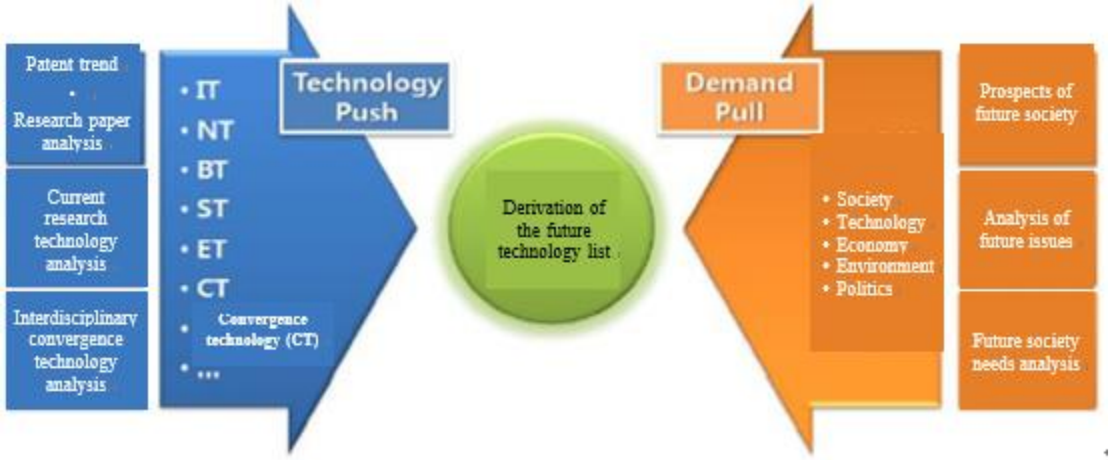
[Figure 2-8] Key Features of the Third Foresight Program compared to the First and Second Programs



Source: The 3rd Science and Technology Foresight Program, 2005.

The fourth science and technology foresight program was an effort to reflect the derived future social needs more actively. It derived 601 technologies to solve the needs of the future society (Demand Pull), as well as 51 technologies that were expected to emerge due to the natural development of science and technology apart from social needs (Technology Push).

[Figure 2-9] Methods of the Fourth Foresight Program to Derive Future Technologies



Source: The 4th Science and Technology Foresight Program, 2012.

Among the trends analyzed to derive future social needs were intensifying globalization, intensifying conflicts, changes in demographic structure, increasing cultural diversity, depletion of energy and resources, intensifying climate change and environmental problems, and the rise of China (KISTEP, 2012). Future technologies were derived for eight sectors including machinery, production, aviation, space, astronomy, agriculture-forestry-fisheries, city·construction, and transportation. The details are shown in [Figure 2-10] below. For the fourth program, the Technology Foresight General Committee, Technology Foresight Subcommittees (8 technical fields), Future Forecast Committees (3 fields) and Future Technology Evaluation Committee were all formed and operated according to the survey contents and procedures. In addition, the Future Technology Evaluation Committee was separately formed to analyze and evaluate whether the future technology tasks presented in the first and second foresight programs had actually been realized.

The fifth science and technology foresight program made great efforts to standardize technical names and descriptions compared to the previous programs. Thus, the names of future technologies were simplified—focusing on key contents such as target technologies, technology use, improvement plans, and future keywords—but the description of future technologies was written in more detail in consideration of complaints that the previous future technology tasks proposed up to the fourth program were somewhat vague or highly uncertain in terms of their content. Efforts were also made to solve such problems as duplication between technologies and a lack of differentiation between the descriptions of the technologies by presenting the names of future technology tasks as specifically as possible.

The biggest feature of the sixth program is that it predicts and presents the tipping point[7] along with the timing of the domestic and international realization of future technologies. The diffusion point is to identify more systematically at what point a realized technology will contribute to changes in our society. Once the diffusion point has been identified, it can be effectively used in the realistic design of policy responses to the future that the development of science and technology will bring. In the sixth program, out of 241 future technology tasks, 15 were selected as future innovative technologies through the consensus of experts (committee members) based on the results of the Delphi survey, in response to the opinion that the future technologies proposed in the last science and technology prediction surveys had been presented without clear priorities and hence were limited in terms of their use in policy. In other words, important technologies expected to have a greater impact on the development of our future society were selected with additional analysis and presentation of the technology diffusion point and the subsequent development prospects and policy proposals for realizing technology diffusion as well as future social changes.

[7] The point at which a technology begains to spread rapidly throughout society.

[Figure 2-10] Relevance between Future Trends and Technology Sectors in the Fourth Foresight Program

Future trends		Number of technologies related to future trends by sector								Total
		Machinery/Production/Aviation/Space/Astronomy	Agriculture-Forestry-Fisheries	City Construction/Transportation	Life/Medical	Materials/Chemicals	Energy/Resources/Extreme Technology	Information/Electronics/Communication	Environment/Earth/Ocean	
Deepening globalization	Integration of the world market	4	3	11	0	0	1	14	1	34
	Multipolarization of the international order	24	0	1	0	2	1	1	0	29
	Globalization of workforce mobility	4	0	6	1	0	0	3	0	14
	Expansion and diversification of the concept of governance	0	0	0	0	0	0	0	3	3
	Rapid spread of infectious diseases	0	20	0	22	8	0	0	2	52
Escalation of conflicts	Intensification of conflicts between peoples and nations	19	0	0	2	2	1	1	0	25
	Increasing cyber terrorism	6	0	0	0	0	1	13	1	20
	Increasing risks of terrorism	25	0	1	6	1	3	3	0	40
	Deepening polarization	0	0	0	4	0	0	0	0	4
Changes in population structure	Continued low birth rate/rapid aging	9	15	7	69	14	0	3	0	117
	Growth in the world's urban population	9	2	23	1	3	1	0	4	43
	Changes in the family concept	2	1	3	3	0	0	1	0	10

Future trends		Number of technologies related to future trends by sector								Total
		Machinery/Production/Aviation/Space/Astronomy	Agriculture-Forestry-Fisheries	City Construction/Transportation	Life/Medical	Materials/Chemicals	Energy/Resources/Extreme Technology	Information/Electronics/Communication	Environment/Earth/Ocean	
Increased cultural diversity	Increasing cultural exchanges and multicultural society	2	0	6	2	0	0	19	0	29
	Improving status of women	2	2	1	2	1	0	0	0	8
Depletion of energy·resources	Increasing demand for energy·resources	23	8	25	0	33	49	7	7	152
	Intensification of water and food shortages	1	29	9	0	2	0	0	14	55
	Weaponization of energy·resources	22	7	23	0	29	48	7	7	143
Intensifying climate change and environmental issues	Intensification of global warming and increasing abnormal climate phenomena	7	15	5	0	2	6	0	24	59
	Increasing environmental pollution	5	19	10	0	11	6	1	48	100
	Changes in the ecosystem	0	37	6	0	2	1	1	36	83
Rise of China	China's growing economic influence	15	12	20	5	37	17	22	6	134
	China's growing diplomatic and cultural influence	1	0	1	3	0	0	5	0	10
Total		180	170	158	120	147	135	101	153	1,164
Average number of related trends per future technology										

Source: "The 4- Science and Technology Foresight Program," KISTEP, 2011.

[Table 2-4] Results of the Sixth Science and Technology Foresight Program for 15 Future Innovative Technologies (Technology Diffusion Period)

Technology	Expected timing of technology diffusion point		Technology	Expected timing of technology diffusion point	
	World	Korea		World	Korea
Fully autonomous aircraft	2031(USA)	2036	AI semiconductors	2028(USA)	2030
Fully autonomous vehicles	2030(USA)	2033	Autonomous robots	2028(USA)	2030
Personalized vaccines	2025(USA)	2029	Small nuclear batteries	2030(USA)	2035
Hydrogen energy	2030(USA)	2032	Natural disaster prediction	2029(USA)	2033
Hyper-personalized AI	2029(USA)	2031	Carbon neutral fuel	2030(USA)	2034
Biochips	2030(USA)	2033	Carbon cycle observation technologies	2029(USA)	2033
Complex disaster response system	2029(USA)	2032	Cell reprogramming technologies	2030(USA)	2034
Quantum cryptography communication technologies	2030(USA)	2034			

Source: The 6th Science and Technology Foresight Program, 2022.

The Delphi method was chosen as the main methodology of all six science and technology foresight programs conducted at the national level in Korea due to the need to collect diverse and extensive information on all fields of science and technology and large-scale future technology tasks. In other words, a large number of experts have determined that the Delphi method is the most appropriate way to efficiently express their opinions and produce a consensus result in the process. However, the component items are differentiated according to the purpose of using the foresight survey results for each program, the form of the required result values, and the method of using them in conjunction with other methodologies such as scenarios. [Table 4] summarizes the analysis items of future technologies examined through the Delphi method in each survey. Items such as the 'importance and timing of technology realization', 'technology level' and 'major R&D entities' have been continuously surveyed since the first survey, whereas items such as 'policy measures necessary for the realization of technology', 'respondents' expertise', 'negative impacts of technology development', 'innovation' and 'uncertainty' are either added or excluded to reflect the policy situation at the time of the surveys.

[Table 2-5] Comparison of Future Technology Analysis Items (Delphi survey) of the Science and Technology Foresight Programs (Surveys 1-6)

Survey	Analysis Items
1st	• Professional degree (high/medium/low) • Importance (high/medium/low/unnecessary) • Timing of domestic realization • Confidence in the subjective probability of the prediction time (high/medium/low) • International comparison of current level of domestic R&D Method of R&D promotion (private capital, government-led, industry-academia cooperation, international cooperation) • Factors inhibiting realization (technical, institutional, social/cultural, capital, research manpower, etc.)
2nd	• Professional degree (high/medium/low) • Importance (high/medium/low/unnecessary) • Timing of domestic/global realization • Confidence in the subjective probability of the prediction time (high/medium/low) • International comparison of the current level of domestic R&D • Method of R&D promotion (private capital, government-led, industry-academia cooperation, international cooperation) • Policy measures (HR training, cooperative exchanges, infrastructure construction, research funding, institutional improvement, etc.) • Factors inhibiting realization (technical, environmental, safety, social/cultural, etc.)
4th	• Importance of selecting government-focused investment Technologies (science and technology/public interest/economy/industry) • Country with the most advanced technology level • Technology level of Korea • Timing of technological realization and social dissemination • Desirable domestic research entity for technological realization • Necessity of domestic industry-academia-research and international joint research for

Survey	Analysis Items
	technological realization • Necessity of government investment for technological realization • Measures that the government should prioritize for technological realization • Importance to future society (scientific and technological/public interest/economic and industrial) • Possibility of social dissemination that negatively affects society, culture, environment, etc. • Institutions that actively conduct related research in Korea and abroad • Relevance to future trends
5th	• Necessary government policies (nurturing human resources/activating cooperation/establishing infrastructure/expanding research funds/improving systems) • Leading research entities (academia/research institutes/industry) • Timing of technical realization • Innovativeness • Uncertainty • Importance in future society (scientific and technological/public/economic and industrial) • Technological competitiveness of Korea • Country with the most advanced technological competitiveness • Possibility of negative impacts
6th	• Future technology characteristics (innovativeness/uncertainty/possibility of negative impacts) • Technological competitiveness of Korea • Countries with advanced technological competitiveness • Importance in future society (scientific/public interest/economic and industrial) • Timing of technical realization • Leading research entities (academia/industry/research institutes) • Necessary government policies (nurturing human resources/activating cooperation/establishing infrastructure/expanding research funds/improving systems) • Domestic and foreign research institutes

Source: Jeong Eui-jin et al. (2019), revised to the results of the 6th S&T Foresight Program.

4. Implications and Suggestions

Since the launch of the first national science and technology foresight program in 1995, Korea has been conducting forecast programs every five years. The program direction, method, process, and form of results have been set according to changes in Korea's policy environment. Such changes have also been observed in Japan and other countries that regularly conduct scientific and technological foresight programs because such programs do not apply standardized and generalized methods and processes, but rather design and apply promotional methods in consideration of the respective countries' policy environment, social and cultural characteristics, and economic background. Countries wishing to introduce science and technology foresight programs need to pay attention to this fact. In particular, if a country's economic and industrial development is at the stage of a developing nation, it is likely to lack experience and foundations in the formation, implementation, and management of its science and technology policies. In this respect, it is necessary to check the matters that need to be reviewed before the full-scale introduction and implementation of a science and technology foresight program. Based on case analysis and experiences in Korea, some of the points that need to be checked in advance are summarized as follows.

First, it is very important to clarify the purpose of science and technology foresight programs. Basically, foresight programs are only one part of the effort to create the future we want. The key is to set the desired goal based on the foresight results, and to devise and implement policies and strategies to achieve it. Therefore, where to place the purpose of foresight programs is highly correlated with future science and technology policy decisions. Foresight programs are an activity that encompasses a wide variety of purposes, in addition to identifying the science and technology needed for the future. How the purposes of science and technology foresight programs are set determines the process and method of their implementation, as well as influencing their future uses in policy and strategy setting.

Second, it is necessary to design an appropriate process and select a method that takes into consideration social, economic, and cultural backgrounds. Science and technology foresight is the process of implementing decisions related to the future according to a designed method involving various experts. The process and methodology of generating prediction results are also a matter of who participates, what choices to make, and how to operate the process. All foresight activities are related to networking, communication, cooperation and decision-making among participants. Therefore, it is important to analyze in advance what kind of economic, social, and cultural characteristics the country or organization concerned has in order to design and select an appropriate foresight program process and method. For example, in Japan, there is no great resistance to participants actively taking part in Delphi surveys and linking the results to policies, while some European countries do not trust the reflection of Delphi survey results in policies without following any social consensus process. Therefore, the policy implications can be very different for countries and organizations that have applied the same method.

[Figure 2-11] Considerations in Designing the Foresight Program Process and Methodology (example)

[Foresight survey stage]	[Key considerations]
Pre-foresight	- Specification of survey purposes (time span/policy uses of results/survey target areas/survey scope, etc.) - Size and scope of foresight experts - Major decision-making measures such as the promotion
Foresight	-Foresight survey process design and selection of application methodologies - Ways to support participating experts' activities (networking, cooperation, etc.) - Ways to derive key trends for future social changes - Ways to discover future technology tasks, etc.
Action Plan	- Designing a method for setting priorities for future technology tasks - Preparing ways to identify policy tasks needed to secure future technologies - Establishing detailed strategies for technology development and policy realization, such as a technology roadmap - Decision-making related to resource allocation, such as investment strategies

Third, efforts should be made to secure resources and infrastructure for the implementation of science and technology foresight programs. Most of the activities performed in foresight program research consist in making consensus decisions using the experience and knowledge of the participating experts. Furthermore, if the Delphi method is applied, two or more large-scale surveys and analyses may be included. There are also cases in which it is necessary to analyze various kinds of global information related to survey fields. This process usually results in situations that require more resources, manpower, and time than expected. Therefore, if due consideration of the resources required for the process is neglected before the full-scale promotion of a foresight program, various limitations will be encountered during its implementation. Therefore, to successfully promote, complete, and link science and technology foresight programs to policies, a systematic and detailed strategic approach to appropriate resource input plans and security measures is required from the planning stage. [Figure 2-11] summarizes other matters that need to be considered at each stage of a foresight program.

CHAPTER 3

Overview of the Status of S&T in Peru

Chapter 3. Overview of the Status of S&T in Peru

1. Economic Overview

Resembling the tendency of insufficient R&D investment typically seen among developing countries, Peru's R&D investment, which accounts for less than 0.5% of GDP, has not been sufficient to bring about economic change. Despite continuous political instability, the Peruvian government has been making efforts to promote science and technology activities and to revitalize the innovation infrastructure. Peru is also well aware that science and technology are the foundation of economic success, as proved in Korea's case. Nonetheless, Peru shows limited innovation capacity stemming from its low execution capacity as a developing country, like its Latin American neighbors, intermittent policy implementation, and insufficiently innovative workforce and organizations.

1.1 GDP per capita

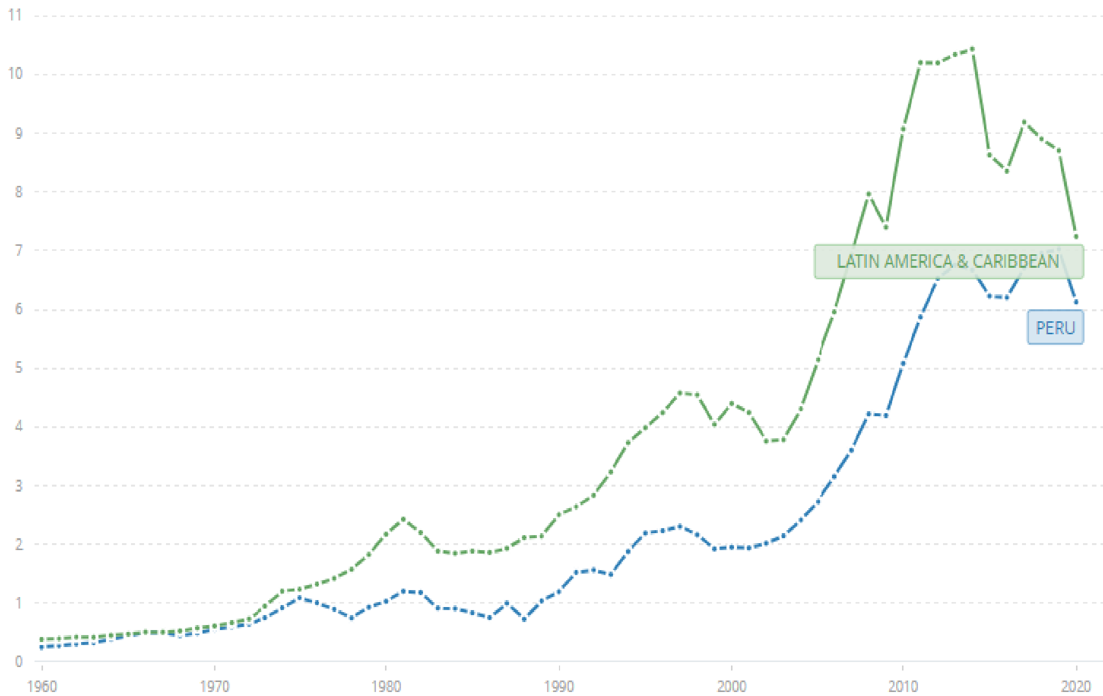
Peru's per capita GDP grew by about 3.5 times from 2000 to 2013, but thereafter its economic growth rate declined gradually between 2014 and 2020 in the wake of the global financial crisis of 2013. Its per capita income was $6,100 in 2020, which is slightly less than the average of $7,200 among neighboring countries in Latin America. After overcoming the financial crisis in 2016, Peru's GDP per capita grew again, but the economic growth rate over the past three years has remained negative due to the impact of the 2019-2020 COVID-19 pandemic.

It is worthwhile noting that as Peru's per capita income differs significantly from neighboring countries that are actively promoting technological cooperation with Europe, such as Chile ($13,000) and Argentina ($8,600), it is necessary to increase investment in science and technology to increase the country's growth potential. Chile and Argentina joined the EUREKA Network, an international cooperation program for technology commercialization in which European and non-European countries participate, as associate members in 2017 and 2019, respectively. Even South Korea has joined the EUREKA, which is actively promoting international technology cooperation.

However, Peru's economic recession was not as bad as that of its neighboring countries between 2014 and 2020, which proves that the fundamentals of Peru's economy are stable.

[Figure 3-1] Per capita income of Peru and Latin American countries, 1960-2020

(unit: USD 1,000)



Source: The World Bank[8]

[8] https://data.worldbank.org/indicator/NY.GDP.PCAP.CD?locations=PE-ZJ(accessed on November 1, 2022)

[Table 3-1] Overview of Peruvian Economics

Category	1990	2000	2010	2020
GDP (US$ billion)	26.41	51.74	147.53	202.01
GDP growth rate (annual %)	-5.0	2.7	8.3	-11.1
Inflation, GDP deflator (annual %)	6,261.2	3.5	5.7	4.2
Agriculture, forestry, and fishing, value added (% of GDP)	8	8	7	8
Industry (including construction), value added (% of GDP)	26	29	36	30
Exports of goods and services (% of GDP)	16	17	28	22
Imports of goods and services (% of GDP)	14	19	24	21
Gross capital formation (% of GDP)	16	20	24	18
Revenue, excluding grants (% of GDP)	12.5	17.9	20.3	19.5
Net lending (+) / net borrowing (-) (% of GDP)	-8.1	-2.1	0.5	-2.2

Source: World Bank Country Profile – Peru [9]

1.2 Major Industries

In Peru's economy, the service industry, which comprises the tourism, finance, and telecommunications sectors, accounts for 59% of GDP, the manufacturing industry, which is centered on mining and textiles, accounts for 33%; and agriculture accounts for about 8% (OECD, 2019a). Peru has become a middle-income country thanks to its

[9] https://databank.worldbank.org/views/reports/reportwidget.aspx?Report_Name=CountryProfile&Id=b450fd57&tbar=y&
dd=y&inf=n&zm=n&country=PER(accessed on November 1, 2022)

rapid economic growth since 2000, but the contribution of each industry to GDP growth has not changed significantly.

Peru's exports and imports account for 49% of GDP, with mineral resources such as copper, silver, and gold accounting for 63% of its exports.[10]

Peru's high dependence on its primary industry, namely mining, implies an economic vulnerability that is sensitive to any changes in international mineral prices due to global economic fluctuations. The textile and clothing industries, which rely on high-quality woollen fabrics - such as Alpaca, Guanaco and Vicuña - produced in Peru, are competitive, while the marine product processing and aquaculture industries, which use abundant water resources, are also classified as manufacturing industries, so there is a high possibility of future growth (Kwon, 2015).

1.3 International Trade

Peru ranks 49th in exports and 51st in imports out of 127 countries worldwide. As of 2019, Peru's total exports amounted to $46.1 billion, while total imports amounted to $42.4 billion. As such, more than half of the economy is directly dependent on trading. Major trade products need to be gradually transformed into high-value-added industries to ensure sustainable growth due to the fact that exports are centered on low-value-added raw materials.

Peru's major trading partners are China, the United States, Canada, and South Korea. It has a trade surplus with most of its trading partners, except for the deficit arising from trade with the United States.

[10] As of 2017, Peru's exports and imports accounted for 49% of Peru's GDP (OECD, 2019a).

[Table 3-2] Exports and Imports by Major Trade Partners in Peru (2019)

Trade Partners	Exports ($)	Export Rate (%)	Imports ($)	Imports Rate (%)	Trade Balance ($)
China	13,545,988,734	29.40	10,265,206,990	24.20%	3,280,781,744
USA	5,748,249,803	12.50	8,808,607,435	20.80%	-3,060,357,632
Canada	2,407,872,138	5.22	677,659,881	1.60%	1,730,212,257
South Korea	2,277,693,573	4.94	961,949,106	2.27%	1,315,744,467
Switzerland	2,266,327,679	4.91	172,547,002	0.41%	2,093,780,677
Japan	1,975,320,361	4.28	1,068,916,058	2.52%	906,404,303
India	1,786,876,817	3.87	880,074,867	2.08%	906,801,950
Brazil	1,441,619,600	3.13	2,430,003,578	5.73%	-988,383,978
Netherlands	1,434,128,135	3.11	245,325,434	0.58%	1,188,802,701
Chile	1,302,851,434	2.82	1,338,760,555	3.16%	-35,909,121
Spain	1,206,247,265	2.61	890,448,473	2.10%	315,798,792
Germany	1,033,530,056	2.24	1,132,502,880	2.67%	-98,972,824
United Arab Emirates	975,104,282	2.11	32,580,370	0.08%	942,523,912

Source: World Integrated Trade Solution[11]

Peru's main trade surplus comes from raw materials, minerals, stone and glass, and agricultural products, while its main trade deficit is from trading capital goods, machinery and electronic parts, and consumer goods.

- Peru is a supplier of valuable raw materials for the global value chain.
- To transform its major trade items into manufacturing/service industries with high added value, it is necessary to establish and implement a strategic industry promotion policy.

[11] https://wits.worldbank.org/CountryProfile/en/PER(accessed on November 1, 2022)

[Table 3-3] Exports and Imports by Product in Peru (2019)

Products	Exports ($)	Export Rate (%)	Imports ($)	Import Rate (%)	Trade Balance ($)
Capital goods	240,925,220	0.5%	5,915,218,725	14.0%	-5,674,293,505
Consumer goods	3,546,515,125	7.7%	7,427,018,290	17.5%	-3,880,503,165
Intermediate goods	7,581,937,640	16.4%	5,233,834,535	12.4%	2,348,103,105
Raw materials	11,696,347,415	25.4%	2,611,936,760	6.2%	9,084,410,655
Animal	624,985,595	1.4%	302,925,255	0.7%	322,060,340
Chemicals	425,903,515	0.9%	2,332,781,620	5.5%	-1,906,878,105
Food Products	1,810,265,920	3.9%	941,941,455	2.2%	868,324,465
Footwear	12,873,020	0.0%	265,272,620	0.6%	-252,399,600
Fuels	1,584,537,730	3.4%	2,982,539,625	7.0%	-1,398,001,895
Hides and Skins	11,629,790	0.0%	81,044,555	0.2%	-69,414,765
Machinery and Electronics	191,914,185	0.4%	4,832,084,985	11.4%	-4,640,170,800
Metals	1,830,408,655	4.0%	1,667,967,765	3.9%	162,440,890
Minerals	8,692,008,280	18.8%	180,080,595	0.4%	8,511,927,685
Miscellaneous	51,580,585	0.1%	910,142,110	2.1%	-858,561,525
Plastic and Rubber	312,907,300	0.7%	1,433,285,120	3.4%	-1,120,377,820
Stone and Glass	3,710,459,900	8.0%	268,506,980	0.6%	3,441,952,920
Textiles and Clothing	696,812,685	1.5%	1,025,230,655	2.4%	-328,417,970
Transportation	41,253,425	0.1%	2,033,287,540	4.8%	-1,992,034,115
Vegetable	2,909,231,230	6.3%	1,292,937,880	3.1%	1,616,293,350
Wood	159,010,550	0.3%	638,188,315	1.5%	-479,177,765
Total	46,131,507,765	100.0%	42,376,225,385	100.0%	3,755,282,380

Source: World Integrated Trade Solution[12]

[12] https://wits.worldbank.org/CountryProfile/en/PER(accessed on November 1, 2022)

2. S&T Fundamentals

2.1 National Innovation System in Peru

Peru's national science and technology innovation system, which began to be built in the late 1960s, was fortunately reorganized in earnest in the mid-2000s (World Bank, 2016).

Despite the growing need for system maintenance and expansion of the support budget, Peru's science and technology innovation capabilities and consequent investments lag some way behind the pace of economic growth, and improvement is needed (KOTEC, 2017). This has led to a lack of manpower to lead science and technology innovation, with education expenditure accounting for only 2.8% of GDP, and a lack of statistical data on those who have completed higher education in science and engineering. Only 1.5 to 2% of Peruvian companies are conducting R&D activities, which are concentrated in the mining sector due to the high concentration of R&D investment in equipment purchase.

In the period 2008-2018, the number of patent applications did not show a marked increase, remaining at a little more than 1,000 per year (Acevedo-Flores et al., 2021). Furthermore, indicators or statistical data that can measure science and technology activities and innovation capabilities are insufficient or absent. However, efforts are being made to expand participation in open innovation networks by utilizing the EUREKA program, a multilateral European joint R&D platform, along with Chile among other Latin American countries.

2.1.1 Governance

The Peruvian government established a system to oversee the science and technology innovation system by granting the Consejo Nacional de Ciencia y Tecnología (CONCYTEC) the authority to develop, promote and coordinate policies related to science and technology innovation through the enactment of the Science and Technology Innovation Framework Act (Act No. 28303). The National Science and Technology Innovation Committee (CONCYTEC) is a decision-making body that is responsible for developing systems to optimize the development of human resources in the field of scientific and technological innovation, as well as for implementing digital government.

Peru's national competitiveness and productivity policies were finalized in 2018. CONCYTEC strengthened its governance functions for science and technology innovation, and further strengthened its capabilities for science and technology planning and technology forecasting.

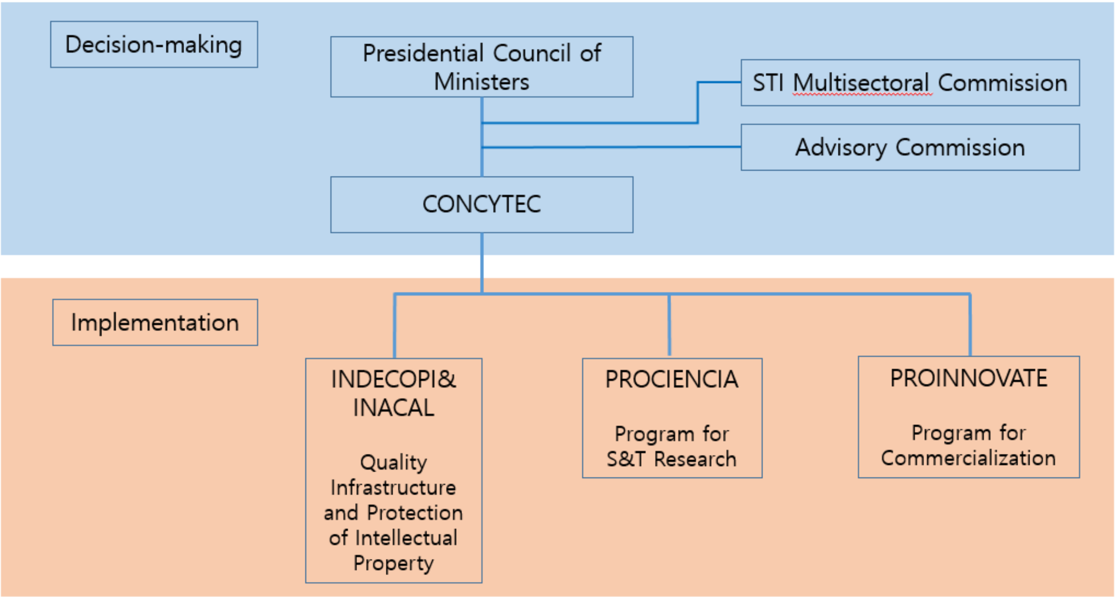
In order to take the initiative in implementing science and technology innovation policies, the FONDECYT was established under CONCYTEC to expand investment in basic and applied R&D support, technology transfer, and higher education and to determine investment priorities in each field according to the National Science and Technology Plan (PNCTI).

To this end, the Science and Technology Innovation Advisory Committee (CONID) has been established to provide advice on science and technology innovation systems as well as investment priorities. The CTI Multisectoral Commission is composed of twelve government departments, and representatives of the National Assembly, local governments and CONCYTEC, whose role is to establish and coordinate national policies and implementation guidelines for science and technology.

Government departments related to the science and technology innovation system include the Ministry of Production (PRODUCE), the Ministry of Economy and Finance (MEF), the Ministry of Education, and the State Council under the President.

Thanks to this reorganization of the national science and technology innovation system, the CONCYTEC’s roles were strengthened and its budget increased significantly from $5 million in 2012 to $39 million in 2015.

[Figure 3-2] Peruvian Science and Technology Innovation System



Source : Author's modification[13]

[Reference] Mission, STI Policy and Main Directorates of CONCYTEC

Mission

To formulate public policies, promote and manage activities for generating and transfering scientific and technological knowledge, as well as technological innovation for SINACYT members, in a timely and efficient manner.

STI POLICY

CONCYTEC, within the framework of its functions and powers, plans to strengthen the institutionality of the science and technology innovation system in the coming years. As national "Science, Technology, and Technological Innovation" (STTI) should be oriented towards the sustainable development of the country, we aim to articulate the research agenda with all SINACYT stakeholders, taking into account territorial differences and country priorities, and we also facilitate technological development and innovation as fundamental tools for competitiveness. In that sense we seek to increase the generation, dissemination and transfer of STI by promoting incentives and mechanisms that facilitate their development.

For this, CONCYTEC's actions adhere to its work to the following principles, which frame the development of its strategies:

Excellence: The activities promoted by CONCYTEC and their results should be oriented towards quality and excellence, for all actors and levels of application.

13 https://cdn.www.gob.pe/uploads/document/file/1869793/Nueva%20gobernanza%20del%20Sistema%20Nacional%20de%20Ciencia%2C%20Tecnolog%C3%ADa%20e%20Innovaci%C3%B3n%20%28Sinacti%29%20-.pdf

Integrity and complementarity: The solutions promoted by SINACYT should adopt a holistic approach involving the various actors from the public and private sector for the development of STI.

Sustainability: The permanence and growth of STTI should be guaranteed through all its actors and the proposed strategies.

Adaptability: The guidelines and strategies should respond to the specific needs and characteristics of the agents involved and of each region based on their potential, and should be focused on the needs of the current and future market.

Equity and transparency: The access, opportunities and services provided by SINACYT should be carried out in a transparent and equitable manner that induces development regardless of its geographical location or its role within the system.

Articulation and cooperation: Articulation and cooperation should be promoted between the various government agents at their different levels, as well as the private sector, in order to generate synergies for the development and competitiveness of the country.

CONCYTEC operates the following three directorates:

STI Programs and Policy Directorate
The STI Policies and Programs Directorate is the body in charge of proposing STI programs for approval. It is responsible for directing and supervising the process of designing and formulating the national policy and plans for Science, Technology and Innovation and Technological Transfer; in addition to leading the formulation and management of national STI programs and special STI programs. It depends hierarchically on the Presidency.

Directorate of Evaluation and Knowledge Management
The Directorate of Evaluation and Knowledge Management is the body responsible for evaluating and monitoring policies, plans and programs in the field of STI, as well as for verifying the consistency of the information develop ed. It is also responsible for supervising the guidelines and technical criteria defined by its organic units. It depend s hierarchically on the Presidency.

Research and Studies Directorate
The Research and Studies Directorate is the body in charge of supervising and preparing research and studies that guide the formulation of STI policies, plans and programs.

2.1.2 Public Programs

Under CONCYTEC, there are three key public programs for S&T innovation:

- Technology transfer platform: LINK[14]
  - Online platform for activating research-to-industry knowledge transfer, providing manuals and guidelines for technology contracts, technology maturity diagnosis, technology transactions and transfers.

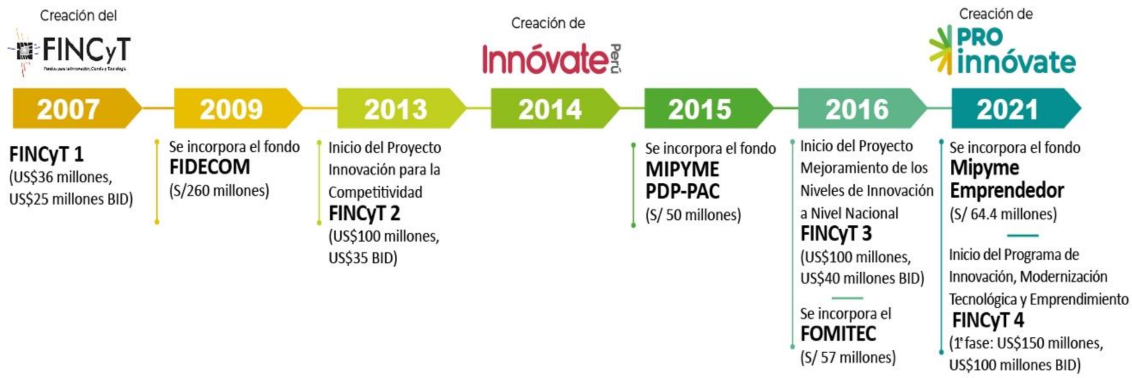
[14] https://vinculate.concytec.gob.pe/

  - Webinar, MOOC (online curriculum) provided free of charge to train experts related to technology transfer.

- Basic R&D support: Pro Ciencia[15]
  - Public funds to collect, manage and execute scientific and technological innovation resources to enhance Peru's national competitiveness.
  - Establishment of higher education and research, establishment of laboratory equipment, establishment of a scholarship system to spread science and technology culture, holding of R&D project contests, equipment construction, networking, etc.

- Commercialization Innovation Support: Pro Innovate[16]
  - National Innovation Program created on March 25, 2021 under the Peruvian Production Department Act 009-2021-PRODUCE.
  - Cooperative funding to support innovative enterprises and entrepreneurship, associations, research institutes, associations and research institutes.
  - As of 2022, there are two types of funds that are newly supported: FINCyT4 and Mipyme.[17]

[Figure 3-3] History of the Peruvian Innovation Program



[15] https://prociencia.gob.pe/

[16] https://www.proinnovate.gob.pe/

[17] It is possible to check the official website for tasks that have been terminated or are in progress from the existing program: https://sistemaenlinea.innovateperu.gob.pe/modules/public/otros/proyectos_4web.php

Source : Ministerio de la Producción[18]

2.1.3 Digital Infrastructure

The digital capability of a developing country is an important indicator of its competitiveness in the future, and plays an important role in the process of performing technology forecasting and obtaining results, as well as affecting the direction of policy development.

Peru was one of the countries that promptly established Internet networks in South America in the early 1990s, and the number of Internet users has increased rapidly since then. The Peruvian government establishes and continuously implements mid-to-long-term strategies to promote digitalization and increase the use of wireless phones, and to promote the expansion of digital markets such as e-commerce, e-government, and digital financial services.

Looking at the main strategies, the 2005 Digital Strategy, the 2011 Digital Strategy, the 2019-2030 Competitive-Productivity Plan, and the 2018 Digital Government Act were developed in an orderly fashion in accordance with the need for a coherent S&T policy. On the other hand, due to spatial limitations such as Amazon, there are restrictions in non-urban areas that require the use of expensive satellite communication due to difficulties in establishing high-speed Internet networks, and the overall investment tends to be delayed or partially delayed by the budget execution by COVID-19.

Thus, the Peruvian government is beginning to make efforts to provide equal digital access to all regions and income classes, whether viewed through a digital-based modernization policy. The Communications Investment Supervision Agency (OSIPTEL), which is in charge of the mobile communication industry, and the Ministry of Transportation and Communication (MTC), which is in charge of building the digital infrastructure and applying information and communication technology, are working together to that end.

[18] https://www.proinnovate.gob.pe/index.php?Itemid=505

2.2 Innovation Capacity

As science and technology become a source of national competitiveness in the knowledge-based economy, it is important to derive policy implications such as the investment direction and the institutional foundation based on an accurate diagnosis of science and technology innovation capabilities. Identifying policy is the first step toward increasing a nation’s innovation capability.

2.2.1 Innovation Capacity Diagnosis

Although the national innovation index released by the International Institute for Management Development (IMD) and the World Economic Forum includes science and technology, there is always a limit to looking at the comprehensive national competitiveness of science and technology properly.

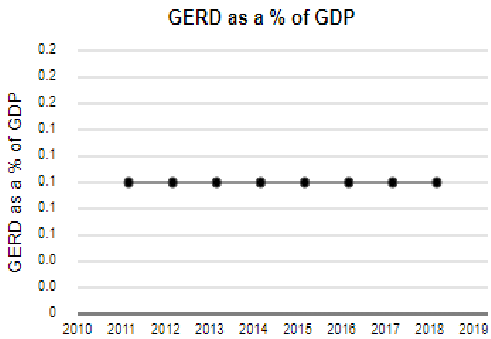
To comprehensively diagnose the capabilities of all sectors of science and technology, South Korea has developed COSTII, which is being used to compare and evaluate the capabilities of each country since 2006 (KISTEP, 2021). Based on Lundvall's (1992) theory of National Innovation Systems, COSTII comprehensively examines all-cycle scientific and technological activities including input, activity, and performance. The COSTII model examines resources, such as R&D investment, start-up activities, human-organization-knowledge, support systems, infrastructure, and culture in the input sector, and analyzes the degree of cooperation activation among R&D activities.

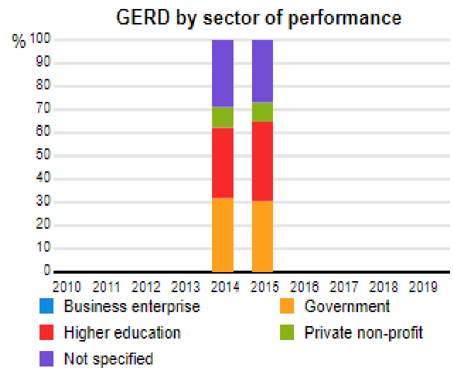
2.2.2 R&D Investment

When comparing the capabilities of science and technology innovation, the distorting effect due to the economic size of each country can be minimized by utilizing the proportion of R&D investment as a percentage of GDP. By comparing the proportions of R&D investment by innovative players within the national innovation system, such as companies, universities, research institutes and governments, it is possible to identify the areas where innovation activities are active and to diagnose those areas that require policy support by priority.

The proportion of R&D expenditure to GDP remains at 0.1%, so R&D activities are not progressing actively when compared with other countries. The most recent data on R&D investment (% of GDP) by innovative entities was 34.23% for universities, 30.45% for the government, 8.37% for companies, and 26.95% for public and university-led R&D investment. In order for science and technology innovation to have a positive effect on economic and social innovation in Peru, a policy for promoting R&D investment by private companies is urgently needed.

[Figure 3-4] Status of R&D Investment in Peru

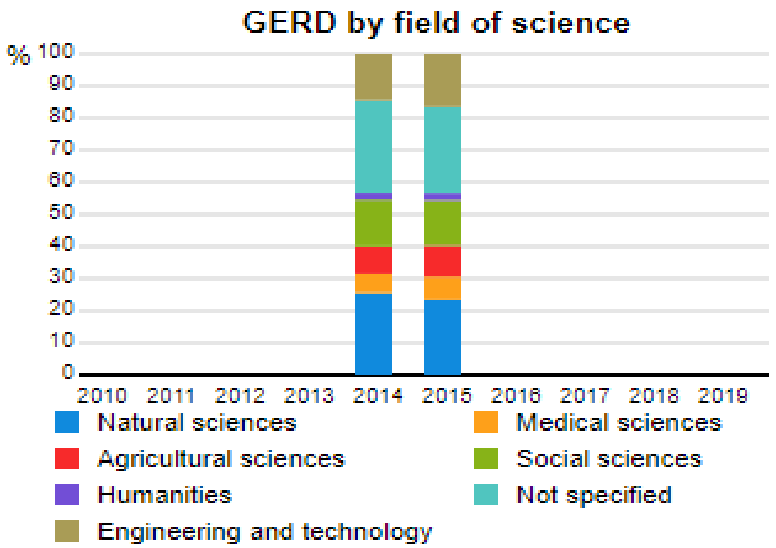




Source : UNESCO Institute of statistics(http://uis.unesco.org/en/country/pe?theme=science-technology-and-innovation)

As of 2015, the proportions of R&D investment by the following sectors were as follows: science and technology, 23.41%; engineering, 16.57%; social science, 14.09%; agriculture, 9.74%; medicine, 7.14%; humanities, 2.21%; others, 26.83%.

[Figure 3-5] Percentage of R&D Investment by Sector in Peru (2015)



Source : UNESCO Institute of statistics[19]

2.2.3 R&D Personnel[20]

■ Overview

Between 2015 and 2020, the number of R&D personnel rose significantly from 3,267 to 7,945, showing an increase of about 2.4 times, and an annual average rate of 19%. Notably, the increase in the years 2019-2020 is striking. The proportion of women among R&D personnel remains at 31%.

[Table 3-4] Peruvian R&D Personnel by Gender (2015-2020)

Gender	2015	2016	2017	2018	2019	2020
Total	3,267	4,201	4,506	4,929	6,661	7,945
Men	2,228 (68.2%)	2,920 (69.5%)	3,125 (69.4%)	3,418 (69.3%)	4,574 (68.7%)	5,470 (68.8%)
Women	1,039 (31.8%)	1,281 (30.5%)	1,381 (30.6%)	1,511 (30.7%)	2,087 (31.3%)	2,475 (31.2%)

[19] http://uis.unesco.org/en/country/pe?theme=science-technology-and-innovation

[20] https://portal.concytec.gob.pe/indicadores/principales/

■ By Tertiary Degree

Since 2015, the number of master's and doctorate degree holders has increased significantly, resulting in qualitative growth of Peru's R&D personnel. In 2015, the holders of master's degrees accounted for the largest percentage of R&D personnel, but since 2016, Ph.D. holders have accounted for more than 40% of all R&D personnel. Compared to 2015, the number of PhD holders has increased by about three times and the number of master's degrees by about 2.2 times.

[Table 3-5] Peruvian R&D Personnel by Degree (2015-2020)

Degree	2015	2016	2017	2018	2019	2020
Total	3,267	4,201	4,506	4,929	6,661	7,945
PhD	1,031 (31.6%)	1,717 (40.9%)	1,925 (42.7%)	2,200 (44.6%)	2,650 (39.8%)	3,423 (43.1%)
Master's	1,116 (34.2%)	1,283 (30.5%)	1,346 (29.9%)	1,460 (29.6%)	2,036 (30.6%)	2,430 (30.6%)
Bachelor's	232 (7.1%)	232 (5.5%)	235 (5.2%)	238 (4.8%)	355 (5.3%)	355 (4.5%)
Diploma	888 (27.2%)	969 (23.1%)	1,000 (22.2%)	1,031 (20.9%)	1.620 (24.3%)	1,737 (21.9%)

■ By Industry

As about 70% of Peru's R&D personnel are working in higher education institutions, it is necessary to encourage them to enter industry. A positive change was observed when the number of healthcare and corporate workers increased significantly from 2019. Various policies are needed to ensure that the number of R&D personnel entering industry rises steadily.

[Table 3-6] Peruvian R&D Personnel by Institute (2015-2020)

Types of Institutes	2015	2016	2017	2018	2019	2020
Total	3,267	4,201	4,506	4,929	6,661	7,945
Higher Education	2,296	3,000	3,246	3,589	4,600	5,718
Public Research	592	636	644	663	752	791
Healthcare and Medical	122	143	154	172	395	389
Non-profit Research	179	199	207	212	329	363
Companies	12	67	81	95	233	275
Research Center	66	126	140	155	236	256
Public Admin	-	30	34	43	116	153

2.2.4 Innovative Startups

Peru's Ministry of Production supports the innovative growth of Peruvian startups through a program called 'Startup Peru' in 2014.[21] According to the Ministry of Production, the sales of some 430 companies supported by the Startup Peru program more than doubled year-on-year to four million dollars as of 2017. The productivity and growth potential of the companies participating in the program demonstrates positive growth potential for the innovative ecosystem in Peru.

2.3 Digital Capacity

2.3.1 Using the Wireless Internet[22]

- As of 2020, about 73% of Peru's inhabitants were using mobile phones, an increase of about 23 percentage points from the 50% recorded in 2013.
- As of 2017, about 83% of the population were using wireless Internet.
  - About 50% of Peru's inhabitants were using the 4G network, and 74% the 3G network.
- 82% of Peru's population is familiar with using wireless Internet networks rather than wired ones, as they generally use mobile phones as a periodic device for Internet access.
  - Although Internet usage is cheaper in Peru than in other South American countries, the inhabitants of non-urban areas, with less than 50% of Peru's average national income, are relatively burdened with the cost of using the Internet, and digital utilization is low.
- Peru's Ministry of Transportation and Communication announced plans to build 5G wireless networks in November 2020, and mobile telecommunication companies were due to start building them in 2021.

[21] Paula Garcia (2019). Perspective of technological innovation and startups in Peru in 2019. https://www.seedstars.com/content-hub/insights/perus-2019-tech-innovation-and-entrepreneur-landscape/

[22] OECD(2019) Digitalisation of processses and regulatory delivery in the Peruvian telecommunications sector: Case study on the Telecommunications Regulator (OSIPTEL), in Shaping the Future of Regulators: The Impact of Emerging Technologies on Economic Regulators. OECD Publishing, Paris

- However, the 5G network is still in the early stages of construction, and in order to build an effective infrastructure, local residents are informed of the advantages of the 5G network and asked for their cooperation.

• The level of wireless Internet utilization is excellent, but the ability to utilize wired Internet-based computers and create contents is somewhat low, so it is necessary to improve education on the use of PCs.

- In particular, it is necessary to provide "one-notebook per person" and distribute high-speed Internet networks to prevent children in non-urban areas being excluded from the reforms of the digital-based curriculum, and the Peruvian government is also trying to address these issues.
- Overall, from the long-term perspective, the lack of a digital talent pool and the difficulty of increasing it in a short period of time are major impediments to the growth of technology-based startups (such as e-commerce and fintech) in Peru.

3. Six Major Industries in Peru

3.1 Agribusiness and Food Processing

Food products and vegetables made up about 10% of Peru's export market as of 2019 (See Table 3-3). The trade balance of agricultural commerce was US $ 2.4 billion in 2020.[23] The main destination markets for Peruvian agricultural exports are the European Union (36%), the United States (34%), Asia, and South America.

The main market is the bloc of European Union countries, along with the United Kingdom. As of January 1, 2021, a new free trade agreement with the United Kingdom and Northern Ireland entered into force, in which Peru enjoys the same benefits as it had under the Trade Agreement signed with the European Union, from which both countries have now withdrawn (Brexit).

The European Union accounts for about 36% of Peru's agricultural exports. Eight countries stand out, namely, Holland, Spain, Germany and the UK (now a former EU member), which are the most important trading partners of Peru after the United States, followed by Belgium, France, Italy and Sweden in descending order of importance. These eight markets represented around 35% of all Peruvian exports in 2020 (US$ 2,682 million), showing a growth rate of 4.1% compared to 2019.

On the other hand, it is noteworthy that the presence of the market of the United States of America that has increased its share since last June to 34.2% in all of 2020, with an increase rate of 5.5% over the previous year. This mega market, individually, has become the most important destination for Peruvian agro-exports, to the extent that there is a Trade Promotion Agreement that allows access to this market without major restrictions, and even allows the entry of a large amount of agro-exportable products free from tariffs.

As such, the two blocs together account for 70% of all Peru's agricultural exports, being very demanding markets in terms of quality, and currently facing extreme pressure due to

[23] Source: Ministry of Agriculture Peru.

the high demand for fresh fruit and vegetables, the same ones that have even increased from Peru, amid the Covid-19 pandemic.

Also noteworthy is the presence of a group of Asian countries, such as China, Hong Kong, South Korea, Japan and Indonesia, which accounted for 8.1% of Peru’s total agricultural exports in 2020; to this should be added the important presence of the member countries of the Pacific Alliance, which accounted for 7.5% of Peru’s exports, and the countries of the Andean Community of Nations, which accounted for 7.2%.

It is also worth mentioning that Peru has signed around twenty trade agreements with more than fifty-six countries in the world, a situation that has allowed Peru to access almost all markets without any major problems. Among the others, only two major markets, Russia and India, have pending agreements.

In Peru, the food industry accounts for 20% of the total manufacturing industry and directly employs 22% of the economically active population (EAP), which explains why it has become a strategic sector in both the economy and the food value chain.

[Table 3-7] Main Export Markets of Peru

(Unit: Thousand US$)

COUNTRIES	2018	2019	2020*	Var. %	Part. %	FTA
				20*/19	2020*	
WORLD	7 033 428	7 462 048	7 678 279	2,9%	100,0%	
UNITED STATES	2 086 968	2 491 528	2 627 874	5,5%	34,2%	Yes
HOLLAND	1 062 081	1 136 475	1 218 246	7,2%	15,9%	Yes
SPAIN	424 131	410 971	452 287	10,1%	5,9%	Yes
UNITED KINGDOM	376 632	355 249	349 648	-1,6%	4,6%	Yes
GERMANY	293 009	266 333	253 082	-5,0%	3,3%	Yes
CHINA	298 097	272 699	252 044	-7,6%	3,3%	Yes
ECUADOR	307 334	304 510	251 951	-17,3%	3,3%	Yes
COLOMBIA	189 327	226 023	234 257	3,6%	3,1%	Yes
CHILE	198 808	199 297	215 234	8,0%	2,8%	Yes
CANADA	150 440	164 920	180 870	9,7%	2,4%	Yes
BELGIUM	144 727	130 940	151 018	15,3%	2,0%	Yes

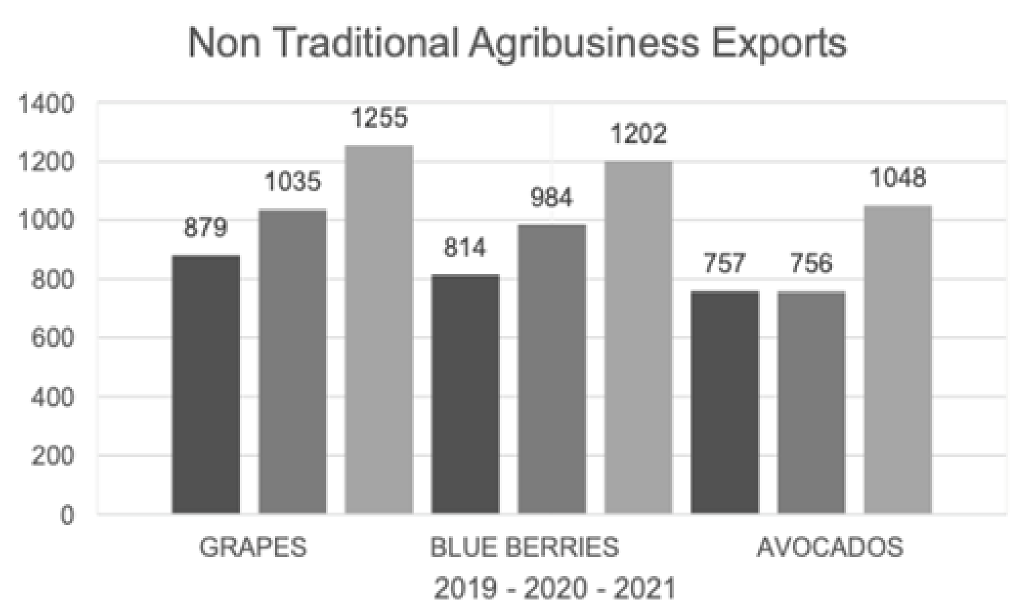
COUNTRIES	2018	2019	2020*	Var. %	Part. %	FTA
				20*/19	2020*	
HONG KONG	114 247	113 727	140 727	23,7%	1,8%	No
MEXICO	110 429	128 415	128 900	0,4%	1,7%	Yes
SOUTH KOREA	155 927	109 639	116 711	6,4%	1,5%	Yes
FRANCE	105 854	114 742	111 015	-3,2%	1,4%	Yes
ITALY	108 743	111 903	99 962	-10,7%	1,3%	Yes
RUSSIA	51 930	62 470	91 006	45,7%	1,2%	No
JAPAN	85 772	85 826	87 644	2,1%	1,1%	Yes
BRAZIL	55 605	70 429	67 159	-4,6%	0,9%	Yes
BOLIVIA	57 769	60 958	66 750	9,5%	0,9%	Yes
PANAMA	46 550	57 758	55 351	-4,2%	0,7%	Yes
SWITZERLAND	48 654	49 474	46 688	-5,6%	0,6%	Yes
HAITI	48 811	43 736	41 687	-4,7%	0,5%	No
DOMINIQUE REP.	36 003	32 302	28 844	-10,7%	0,4%	No
INDONESIA	21 935	46 959	28 229	-39,9%	0,4%	No
OTHER COUNTRIES	453 642	414 763	381 095	-8,1%	5,0%	

Source: Ministry of Agriculture of Peru.

3.1.1 Main Products

Within the wide range of products exported from this sector, the produce that stood out the most were fresh grapes, blueberries, and avocados. Fresh grapes were the main non-traditional agricultural produce exported in 2021, with shipments worth US$ 1.255 million, which translated into an increase of 21.4% over the US$ 1.035 million registered in 2020. As for blueberries, they were almost unknown in Peru until 2014, but since 2019 Peru has become one of the world's leading exporters of blueberries, shipping US$ 991 million worth of produce in 2020.

[Figure 3-6] Non-Traditional Agribusiness Exports



Source: National Taxes Offices

3.1.2 Innovation Trend

Agrotechnology seeks to promote the sustainable management of natural resources and preservation of the environment, in addition to promoting the science of the productive model.

Five types of technology are currently applied to the agricultural sector:

- Agrochemicals: the creation, development, and use of fertilizers, nutrients, pesticides, and phytosanitary processes.
- Mechanics: machinery such as seeders, cutters, fumigators, tractors, collectors, and all other types of equipment that facilitate the work of farmers.
- Biological: the creation of seeds, fertilizers, nutrients, and pesticides from cellular modifications.
- Computing: a tool that uses digital platforms and applications to manage and monitor cultivation processes.

- Robotics: specialized machinery that relies on agricultural software to work with precision agriculture. Satellites and drones generate information (big data) that is analyzed to carry out fertilization, harvesting, and planting processes remotely by using 'intelligent' robots.

Inventory automation and order fulfillment continue to evolve with the aim of optimizing the production process. Technology is the main ally for this tool in helping and facilitating trade. In Peru, Techbrand has developed these systems with marked success.[24]

3.1.3 Growth Strategy

Peru’s agricultural business sector needs to connect and incorporate small farmers into the value chain.

- “The pending task of the agro-export sector is to incorporate small farmers from the mountains and jungle into its productive circuit.” Rosario Bazan, CEO of DANPER (www.danper.com)
- “The great challenge after the bicentennial of the Peruvian agro-export sector is to incorporate thousands of small farmers distributed in the three natural areas of the country into its value chain.” Yoselyn Malamud, CEO of VIRU (www.viru.com.pe)
- “The Beta agro-industrial complex points out that the second step in the agro-export boom of recent years is to massively incorporate farmers from other areas of Peru.” Lionel Arce, CEO of Beta Co (www.beta.com.pe)

Technological innovation is encouraged to increase productivity in the agribusiness sector.

- E-commerce: Online groceries have gained the most funding since AgFunder began tracking investments in the agri-food field.
- Automation, robotics, and non-contact technology: The desire for hygiene has become paramount in the agri-food sector. Non-contact food delivery cabinets, autonomous

[24] Source: https://tbp.com.pe/plataforma-tecnologica-stla

supermarket carts, and robotic waiters or cooks are in high demand.

- Biological inputs: A growing number of bio-inspired crop inputs have emerged since 2020. Farmers, consumers, and regulatory bodies have become more aware of the negative side effects of chemical treatments.
- Carbon market: Farmers practice regenerative farming methods in the form of carbon credits, which will be used to offset the carbon footprint.

Alternative proteins: Consumers in Asia and the US are keen to purchase non-meat-based diets. There is a strongly growing market for the production of vegetable proteins.

3.2 Forestry

According to Supreme Decree No. 018-2015-MINAGRI, Forest Management Regulation (2015), the forestry industry is the set of activities that comprise, among other things, the primary and secondary processes for the transformation of forest resources. Forest products can be classified into timber and non-timber products. According to the type of transformation process to which they are subjected, they can also be classified as first processing products and second processing products. Despite the Peruvian government's efforts to promote the timber/forestry industry, it has been described as incipient by experts such as Marc Dourojeanni (National Forestry Chamber, 2022).

3.2.1 Main Products

The Forest Management Regulation (MINAGRI, 2015) defines primary processing as the first processing process to which forest and wildlife products and by-products are subjected in their natural state. Secondary processing takes as its inputs both forest and wildlife by-products from the primary processing industry to add added value (MINAGRI, 2015).

As shown in Table 3-8, in the period 2015-2021, firewood, roundwood and sawn wood accounted for 73%, 19.6% and 6.3% of national production respectively (SERFOR, 2021), followed by charcoal, parquet and floorboards, veneer and plywood, wooden sleepers, and wooden posts, which collectively accounted for about 1%.

[Table 3-8] National Production of Main Timber Products by Volume (m^3), 2015-2021

Timber product	2015	2016	2017	2018	2019	2020	2021
Firewood	7,028,267.28	7,028,267.28	7,028,267.28	7,028,267.28	7,028,267.28	6,451,893.10	6,451,893.10
Roundwood	1,694,431.21	1,448,366.71	1,656,206.41	1,600,601.84	1,294,501.64	936,466.75	1,282,094.01
Sawn wood	579,079.15	333,265.70	482,320.28	463,823.14	596,547.14	327,849.51	624,834.45
Charcoal	40,695.97	9,094.06	59,564.93	62,065.19	90,281.72	72,499.50	43,061.81
Parquet and floorboards	10,189.54	5,834.36	12,868.56	13,720.11	25,797.26	13,969.57	20,344.47
Veneer and plywood	4,989.12	1,933.02	4,941.70	5,140.13	2,942.57	1,529.86	1,680.91
Wooden sleepers	665	623.27	880.09	824.24	11.61	104.15	178.29
Wooden posts	6.82				7.17	81.15	
Total	9,358,324.09	8,827,384.40	9,245,049.25	9,174,441.93	9,038,356.39	7,804,393.59	8,424,087.04

Source: Servicio Nacional Forestal y de Fauna Silvestre a (2021) Forest Statistical Compendium 2010 - 2020. Lima, Peru, p. 128.

The great majority of the companies in the wood and furniture industry are microenterprises, while there is a minimal number of small and medium-sized companies, and only 0.6% are large companies. The first processing mostly involves small and medium-sized companies, while the second processing (furniture manufacturers and wood joinery) mainly involves micro-enterprises.

3.2.2 Innovation Trend

Most of the Peruvian timber industry produces goods with little added value due to the fact that it uses poorly developed technology (PRODUCE, 2015). Pucallpa, one of the main centers of timber processing in Peru, is characterized by the absence of added value. The absence of quality management systems and technological innovation has the result that timber companies in the Ucayali region are neither sustainable nor competitive (PRODUCE, 2015).

There are three important issues with respect to the technological variable, both at the primary and secondary processing levels:

- The existence of a technological gap in the forestry chain;
- The lack of training programs; and
- The low level of technification of SMEs, which results in low productivity in workshops and carpentry.

The low levels of education and training have a negative impact on productivity in the timber industry. The low productivity FAO (2018) indicates that raw materials and labor represent a very important part of the final cost of wood products in Peru, which is a problem for the competitiveness of companies.

In the primary processing industry, small companies do not have adequate machinery for their operating processes; whereas, in the second processing industry, entrepreneurs have machinery that they make themselves. According to the FAO (2018), only the small group of medium and large companies have sufficient economic solvency and vision to invest in R&D. There is a limited number of companies that have appropriate technology for their activities and trained personnel.

3.2.3 Growth Strategy

In 2020, the Peruvian government developed the National Plan for Forestry and Wildlife Research 2020-2030 (PNIFFS), whose general objective is to increase the adoption of scientific knowledge and technologies generated based on the needs of the forestry and wildlife sector, in order to improve production processes through scientific knowledge and the generation of technologies adopted by the sector's users.

Three strategic objectives have been derived from the general objective, which will be achieved by implementing strategic actions and their respective activities:

- Strengthen coordination among the actors involved in the STI of the forestry and wildlife sector;
- Improve the factors that promote research, technology development, and innovation in the forestry and wildlife sector; and

- Increase technology transfer in the forestry and wildlife sector.

Many microenterprises in the timber industry, generally informal, face the following problems: 1) small-scale production, 2) a lack of efficient equipment and machinery, 3) low production capacity in the face of large volume orders, and 4) a lack product standards (PRODUCE, 2015). As such, the National Forestry Chamber (2022) has highlighted the need to generate and strengthen their associativity

In general, it is necessary to identify the species that meet the technical requirements for the production of products; standardize the processes and furniture pieces; conduct market studies on the products in greatest demand in this subindustry; and invest in constant innovation in designs and finishes, according to fashion trends. Entrepreneurs and workers should be trained in the following areas necessary to boost the sector: (1) specialization of the production process, (2) marketing tools, and (3) machinery management. The renovation of machinery and equipment, the use of modern technologies, financing capacity, and a business run with an entrepreneurial vision are also needed.

The industry needs to install waste processing lines, as in the case of Forestal Otorongo's sawmill, whose main production line is flooring, and the waste is transformed into charcoal; other waste recovery products include chipboard, finger-jointed wooden boards, and the use of short wood, which are already being produced. In the long term, wood sawdust should be transformed into pellets, which are in great demand in Europe. The corner pieces of the trunks could be used in the production of corner pieces dyed with natural dyes that are in demand for garden decoration, while the branches of hardwood trees could be converted into pieces of wood for export as a raw material for the workshops of retirees in the USA and the EU.

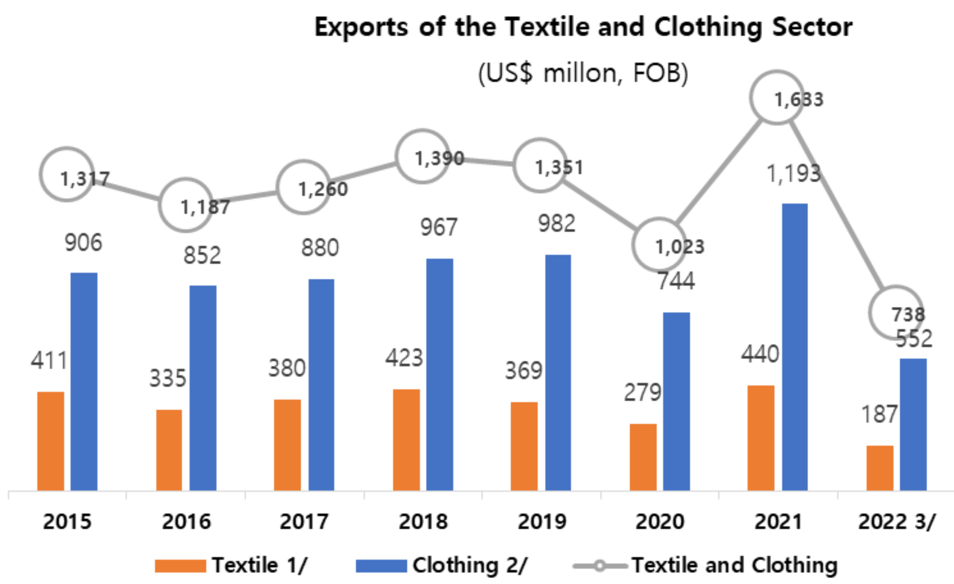
3.3 Textiles and Apparel

The textile industry is defined as the companies that carry out the manufacture of final products, including yarns and fabrics, by such processes as spinning, weaving, and dyeing.

On the other hand, the garment manufacturing industry carries out operations related to the manufacture of finished articles. Peru exports cotton fiber products, mostly clothing and fine alpaca hair products, fiber tops, yarns, and fabrics. The distribution of textile companies in Peru has a very high component of micro and small companies, which affects the technological fragility of the sector.

According to information from the Peruvian Commerce Society, COMEX, Weekly Report No. 1113, exports have been showing a recovery since 2021. As for trade, as per the cited report, the sector runs the risk of being affected by protective measures, which many consider will reduce competitiveness and reduce the benefits for many Peruvian families.

[Figure 3-7] Peruvian Exports of Textiles and Clothing



Source: Ministerio de la Producción (Production Ministry) PRODUCE

3.3.1 Main Products

According to COMEX, Weekly Report No. 1113, the main export destinations of textile products in January 2022 were the United States of America, with a value of US$72.7 million, i.e. 54.7% of the total amount exported; followed by Chile (US$10.2 million, 7.6%), Ecuador (US$5 million, 3.7%), Brazil (US$4.8 million, 3.6%) and Canada (US$3.5 million, 3.4%).

The main export products, according to COMEX, Weekly Report No. 1121, in the first quarter of 2022 were solid color cotton t-shirts with a value of US$ 68.8 million (+47.3% compared to 2021), which represents 14.7 % of the total exported, followed by other cotton t-shirts at US$ 51.2 million (+41.4%), which corresponds to 11% of the total exported. Equally important were knitted undershirts made with other textile materials, which had an export value of US$ 21.6 million (+16.7%), followed by fine hair of alpaca or llama (carded or combed) (US$ 22.9 million, +47.2%), other garments, cotton knit fabric (US$ 14.6 million, +26.1%); and knitted shirts of other synthetic or artificial fibers for men or boys (US$ 16.8 million, +58.7%).

3.3.2 Innovation Trend

The important trends of the global textile industry in 2022 included the increased demand for natural fibers, the introduction of innovative technical fabrics, a shift towards non-woven fabrics, a shift towards sustainable resources and more valued yarns and fabrics, greater preference for purchases online, increased demand for fashion accessories, and growth in home textiles, laser treatment, personalization, and products on demand.

Technology-wise, the main trends include the use of Artificial Intelligence, the Internet of Things (IoT), rapid data analysis for rapid adaptation, virtual and augmented reality (VR), online vector editors, 3D printing, blockchain, and sustainability. For several years, the Peruvian textile industry has been going through a slow process of digital transformation due to the low connection of textile manufacturing devices based on IoT applications and the moderate automation of production and logistics processes.

The few technological innovations acquired from other countries are used to create new business models and opportunities for textile companies. For example, digital textile printing has given businesses and consumers the ability to customize and produce consumers' own designs and ideas quickly and relatively cheaply. Indeed, the Peruvian textile industry is under pressure to be innovative. Many Peruvian textile companies capture R&D from developed countries, which stalls the Peruvian innovation process. The use of advanced materials in textiles has been growing slightly. New materials with antibacterial and antiviral functions are expected to be in higher demand as a result of the Covid-19 pandemic.

Nanotechnology is used in the textile industry to improve such textile attributes as the softness, durability and breathability of fabrics, water repellency, fire retardancy, antimicrobial properties, etc., in fibers, threads, and fabrics; however, in Peru, these technologies are not very developed.

3.3.3 Growth Strategy

The Institute of Economic and Social Studies of the National Society of Industries suggested the following perspectives in a study carried out in March 2021. The study forecasts exports of US$ 2.1 billion at US$ 45/Kilo, which shows an expected increase in the added value since, now, considering broadly, 500 tariff items exported in the period of time from January-July 2022, an export value of US$ FOB 924,401,909 is reached with 49,733,901 Kilos (Information from Promperú) with which a ratio of 18.58 $/Kilo. It is expected to reach 2025 with 340,000 jobs for the textile and clothing sector, starting from 312,000 jobs, which is the number estimated for 2021. Furthermore, these developments are expected to contribution US$ 540 million in taxes to the Peruvian state, while "Made in Peru" is expected to become a world reference (in alliance with the state).

To fulfill these goals, the SNI suggests the following measures.

- Short term: work in the full package mode for the US, which is the main buyer of cotton knitwear.
- Medium term: work on private labels, make use of synthetic fibers, and develop own brands for sale in Peru.
- Long term: work with R&D centers and joint production and purchasing centers.
- According to the same study, as competitive advantages to achieve the goals, the following points are mentioned.

- Understand the customer according to a strategy based on the 3 C's (Customers, Competitors, Company).
- Pursue customer-based innovation.
- Make the PERU brand a sign of Quality with Sustainability.
- Work in collaboration with the government.
- View people and culture as key agents of change.

3.4 Mining

As Peru is a country rich in minerals and metals originating from the Andes Mountain Range, its economy revolves around the extraction of these resources. However, Peru does not have sufficient technological development to use these resources and transform them into final products. The technological process of transforming raw materials into final products or intermediate materials is not feasible in a country that is just starting out on the path to technological development. In Peru, there are two steel producers with an extensive production of steel (SiderPeru and Aceros Arequipa) and a refinery belonging to the Southern Peru mine that manages to obtain copper with purity levels of 99. 7% and 99.99%.

3.4.1 Main Products

Peru is the world's second-largest producer of silver, copper, and zinc. It is also the first producer of gold, zinc, tin, lead, and molybdenum in Latin America; all of which originate from the Andes, the backbone of Peru and a primary source of minerals worldwide. Table 3-8 shows the volume of metal exports by Peru.

Peru has recently discovered lithium reserves, including the deposit for the Corani and Macusani projects. The challenge is that lithium also contains uranium, so an investigation and research aimed at the efficient extraction of the two elements is required.

According to the Ministry of Energy and Mines, mining accoutsn for almost 10% of the national GDP. The mineral-rich areas of Peru are located within the Andes, where mining has been practiced for thousands of years. Mineral resources accounted for over 50% of Peruvian exports in 2000. Gold is the country's most important mining resource, the two most important gold mines being Yanacocha and Pierina, which also yield the most revenue when compared to Peru's other gold deposits. The country's gold and copper deposits total millions of ounces. Mining for silver in Peru dates back to pre-Columbian times. Market price variations, which result in a decline in sector revenues, are one difficulty that the mining business encounters.

[Table 3-9] Metal exports in Peru

VOLUME OF METAL EXPORTS*

PERIOD	COPPER	GOLD	ZINC	SILVER	LEAD	TIN	IRON	MOLYBDENUM
	(Thousands tons)	(Thousands oz tr)	(Thousands tons)	(Million oz tr)	(Thousands tons)	(Thousands tons)	(Million tons)	(Thousands tons)
2011	1,141.0	6,492.2	989.8	6.5	987.7	34.2	9.3	18.9
2012	1,276.7	6,427.1	994.7	6.9	1,169.7	25.5	9.8	17.3
2013	1,324.9	6,047.4	1,059.4	21.2	855.2	23.4	10.4	18.1
2014	1,319.8	5,323.4	1,124.4	17.1	771.5	23.9	11.4	16.5
2015	1,643.8	5,743.8	1,190.3	8.9	938.4	20.8	11.6	17.8
2016	2,317.3	5,936.6	1,102.9	7.2	942.3	18.9	11.1	24.5
2017	2,438.0	6,563.9	1,236.5	6.9	865.5	18.1	11.7	25.4
2018	2,487.9	6,513.3	1,208.0	7.8	793.7	17.1	14.7	27.2
2019	2,535.7	6,096.8	1,187.8	4.7	816.1	19.3	15.8	29.3
2020	2,188.7	4,440.3	1,190.5	4.7	730.2	19.9	14.4	29.1
2021 (Jan-Nov)	**2,118.4**	**5,117.2**	**1,098.0**	**4.2**	**760.7**	**22.4**	**16.3**	**29.5**
Jan.	164.1	399.9	90.9	0.4	63.1	2.5	2.0	2.3
Feb.	182.6	370.5	107.7	0.2	57.2	1.9	1.4	2.1
Mar.	179.6	437.7	110.7	0.4	76.0	1.8	1.4	3.2
Apr.	188.3	458.4	98.0	0.3	77.0	1.8	1.2	2.1
May.	190.9	435.7	104.7	0.5	105.8	1.0	1.5	1.9
Jun.	198.4	492.6	105.1	0.4	53.9	1.5	1.3	2.9
Jul.	190.0	492.0	92.7	0.4	88.4	1.9	0.7	2.9
Aug.	204.7	493.0	125.3	0.4	63.2	2.9	1.6	3.3
Set.	218.5	538.8	87.6	0.4	66.7	2.5	1.2	3.7
Oct.	181.3	506.4	64.7	0.4	46.7	3.1	1.7	2.4
Nov.	219.9	492.1	110.6	0.4	62.7	1.5	2.3	2.7
ACCUMULATED YEAR-ON-YEAR CHANGE - VOLUME* / JANUARY-NOVEMBER								
Jan-Nov 2020	1,971.1	3,981.4	1,037.6	4.3	681.6	18.2	12.5	25.9
Jan-Nov 2021	2,118.4	5,117.2	1,098.0	4.2	760.7	22.4	16.3	29.5
Var%	**7.5%**	**28.5%**	**5.8%**	**-3.1%**	**11.6%**	**23.0%**	**30.7%**	**13.8%**

Source: BCRP, Monthly Statistical Tables. Elaborated by the Ministry of Energy and Mines.
Date consulted: January 19, 2022.
* The table contains data published by the Central Reserve Bank of Peru. The volumes for each mineral are specified below:
. COPPER: The tariff items corresponding to "Copper minerals and their concentrates" and "Cathodes and sections of refined copper cathodes" are considered.
. GOLD: It is considered "Gold minerals and their concentrates", as well as "Gold refined". Includes estimate of gold exports not registered by Customs.
. ZINC: It is considered "Zinc minerals and their concentrates" and "Refined zinc".
. IRON: It is considered "Iron pellets" and "Iron sludge and cakes".
. SILVER: It is considered refined silver and the heading "silver and its concentrates".
The volume is calculated based on the information sent by Customs in kg and transformed to the reference units (that is to say from kilograms to troy ounces, or from kilograms to tons). In the case of copper and other metals, where concentrates are considered, volumes are adjusted by an average grade. Information available at the date of preparation of this bulletin.

3.4.2 Innovation Trend

■ Renewable Energy

One crucial global trend is the use of renewable energy sources such as solar energy. In the case of mining, it is vital because of its benefits, such as reducing pollution and conflicts and offering clean electricity to communities affected by mining. Discoveries of abundant lithium resources and rare-earth elements are essential for developing batteries and electronics. These elements also bring the economic benefit of independence from other producer countries. (i.e. China, rare-earth elements).

Peru can produce more electric energy due to the higher amount of photons (light) per unit area. Peru possesses excellent potential for the extraction of solar energy, a clean source of energy that can be used as a source of electricity for mining settlements and surrounding communities. However, the solar energy produced in Peru is very low.

■ Rare Earth

Rare earth was found by Alturas Minerals at Huancavelica and, in abundant quantities, at Chicama. Cuzco could hold the largest deposit of rare earth in Peru (Ochoa et al., 2022). These elements are strategic because of their uses (catalysts, high-performance magnets, alloys, glasses, and electronics) and are commercially dependent on China. The production of rare elements is known worldwide as a crucial factor for technological growth.

3.4.3 Growth Strategy

The strategy to success in R&D involve the following elements:

- Needs assessment: In accordance with the communities, it is necessary to identify the need or needs that will be satisfied by a R&D project, such as the need for energy, contamination control, clean and potable water, higher purity of metals and minerals, etc.
- Planning: R&D projects can be planned in collaboration with local universities or institutions. It is crucial to determine whether a project is feasible. To that end, it is necessary to have the knowledge and technology. For example, some research

requires certain kinds of special equipment which Peru's local universities do not possess, such as an X-ray diffraction spectrometer (XDR) or energy-dispersive X-ray spectroscopy (EDX). Still, it is possible to use such equipment in much scientific research regarding elements and compounds found in soils.

- Experiment: Develop an R&D project with measurements and essays within the set period of time according to the nature of the project.

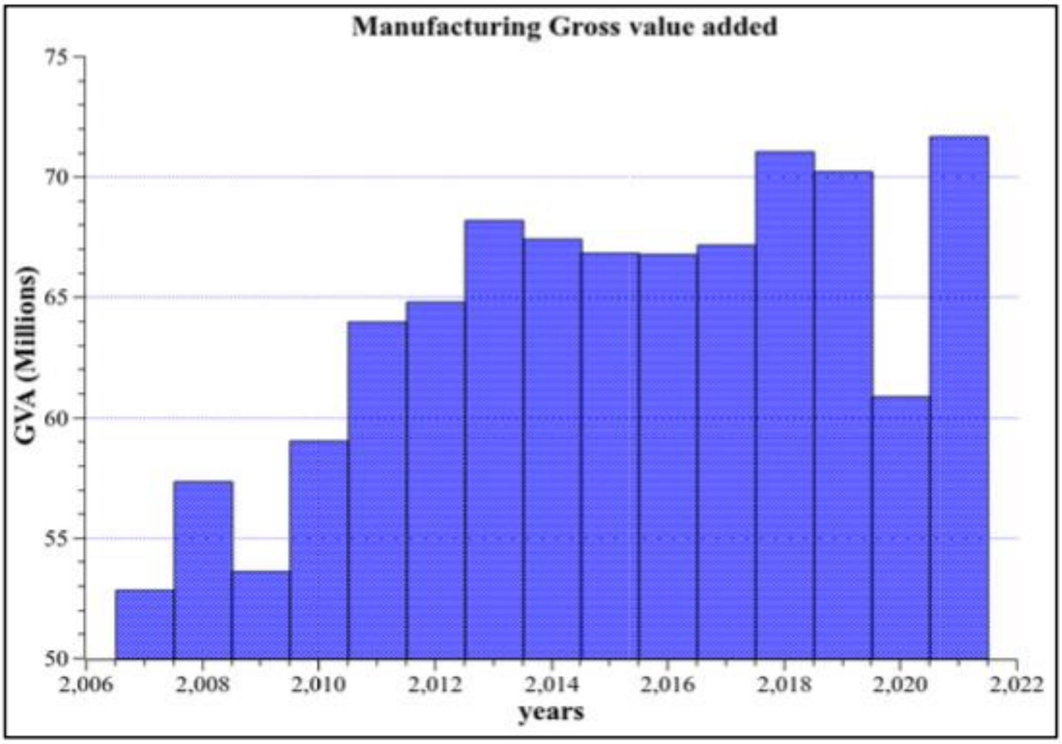
Peru has long been a source of precious metals like silver and gold. In the past, people were exploited without improving their quality of life. Even at this point in time, such communities are still waiting for social improvements, and in many cases they do not accept the construction of new refineries due to the pollution of their environment. For this reason, it is important to note that good trades with communities are the first step. For example, solar energy, clean extraction of metals and minerals, and environmental damage control are well-accepted ideas for these communities.

3.5 Advanced Manufacturing

Advanced manufacturing applies cutting-edge technology to enhance goods or procedures, such technology being either advanced or innovative. It is defined by the rate of technological adoption and the capacity to employ a given technology to remain competitive and add value. Advanced manufacturing not only rapidly incorporates the transfer of science and technology (S&T) into advanced manufacturing goods and processes, but also manufactures products that are characterized as reliable and affordable, and is capable of solving a variety of problems.

Peru is among the world's fastest-growing economies, and is one of the strongest economies in Latin America, enjoying accelerated growth. Currently, its total GDP is $224.73 billion, and its economy is largely driven by industries such as mining, manufacturing, fishing, and tourism. Although the number of industrial activities has increased, Peru's manufacturing Gross Value Added (GVA) has not, typically oscillating between 65 to 70 million dollars since 2011.

[Figure 3-8] Peruvian Manufacturing Gross Value Added



Source: World Bank

3.5.1 Main Products

In Peru, the manufacturing industry is classified as “primary” and “non-primary” (secondary and advanced).

The primary sector is dedicated to the production of precious metals and other non-ferrous metals, the refining of petroleum, fishmeal and fish oil, and, to a lesser extent, the production of milled rice, sugar, and meat products. On the other hand, non-primary manufacturing is the most representative and most important sector, comprising a diversity of branches such as food and beverages, textiles, leather and footwear, chemicals, rubber and plastics; and non-metallic minerals.

Non-primary manufacturing, oriented to the local market, includes the production of printing, milling products, pharmaceutical products, bakery products, malt beverages, soft drinks, and water. Non-primary manufacturing, oriented to the foreign market, includes the production of clothing, the sawing and planting of wood, the production of canned fruits and vegetables, wood products, jewelry, parts and pieces of carpentry, wooden sheets for veneers and panels, and the cutting, carving and finishing of stones.

Manufacturing accounts for 20% of the country's GDP. Peru used to export the majority of its raw commodities. However, raw materials are now treated to add value to them before they are shipped. The textile sector, which exports clothing to Europe and the United States, is an example of a value-added industry. In 2000, the textiles sector shipped clothing worth US$ 700 million to the United States and Europe, making it the industry's most promising sector. Peru signed a treaty with the European Union that resulted in duty-free exports to its member countries. The pact's principal goal was to aid the effort to suppress the drug trade.

3.5.2 Innovation Trend

A growing number of vertical industries in the world are adopting the technologies, concepts and principles of the Fourth Industrial Revolution. Adopting Industry 4.0 is a major leap that primarily implies the use of data to improve efficiency and competitiveness, but the vast majority of entrepreneurs and factory owners in Peru consider this latest industrial revolution as a fad rather than an alternative for process improvements and greater efficiency. To change this perception, it is necessary to show how this will allow industries to achieve goals such as: (a) reducing production losses, (b) increasing production volume, (c) increasing the value added to the product and its quality, (d) reducing production costs and time, and (e) increasing profits. These are the main axes of any industry:

- The design includes the application of collaborative techniques, widespread personalization of products, design that ensures product sustainability, and digital development of the product range.
- Manufacturing includes flexible and efficient methods of production, task automation, decreased environmental impact during production, self-regulating manufacturing processes, and cost savings.

- Logistics: sophisticated models, stock reduction, process automation, less environmental impact, and more demand for shorter machine runs.
- Distribution and sales use the digitalization of channels, multichannel management, real-time connectivity between businesses, products, and customers, efficient and personalized distribution, and post-sale interactions with customers.

Switching from an industry based on traditional technology to Industry 4.0 is a positive step that applies mainly to information management for purposes of efficiency and competitiveness; however, the vast majority of entrepreneurs and factory owners perceive the Industry 4.0 as something similar to another fad rather than an alternative for improving processes, services, and products. In order to change this perception, it is necessary to show the main advantages and goals of implementing Industry 4.0. For instance, it can improve the reduction of raw material losses, increase the speed and volume of production, increase the added value of products and their quality, reduce production costs, and increase profits.

3.5.3 Growth Strategy

The current state of the Peruvian economy in terms of technology, production standards, and access to information or data is still quite precarious in some sectors, making it difficult for businesses to increase their productivity and compete in both the domestic and global markets. As a result, it is necessary to optimize production processes, and many businesses are currently engrossed in a quest to find a solution. Manufacturing industries in Peru must seize the huge development potential offered by the Fourth Industrial Revolution. The COVID-19 pandemic, despite causing significant problems and remaining a persistent difficulty in several industries, should be viewed as an opportunity for businesses to continue growing.

One chronic problem in Peru is that the majority of manufacturers are based in the Lima region, which inevitably slows down development due to the concentration of resources. The government should focus on spreading industrial output by offering incentives and strongly encouraging companies to effectively use the nation's natural resources and achieve self-sustaining growth across the country. Peru also has to diversify its economy, upgrade its infrastructure and manufacturing, and allocate more funding to non-coastal industries if it wants to maintain a sustained growth rate.

Peru is promoting and moving toward digital transformation by allowing all its citizens to access and use digital media. For instance, the number of active mobile and fixed broadband subscribers have risen over the previous decade, while the number of students per computer decreased over a period of three years from 2.2 to 1.4 in 2018, which is similar to the average for Latin America and the Caribbean (LAC). Peru's E-Government Development Index has increased to 0.65, ranking 8th in Latin America and the Caribbean (LAC) in 2018, yet it still lags some way behind the top country, Uruguay, which has an index of 0.79.

To build and maintain leadership in the race to build sustainable economies, industries need well-trained personnel with broad and deep working knowledge of digital technologies and related use cases in order to develop and implement bespoke digital manufacturing strategies. For the future development of the country, it is necessary to promote Industry 4.0 in order to produce products and services with added value.

3.6 Ecotourism, Restaurant and Catering, and Creative Industries

Peru’s cultural and natural resources constitute an important economic and socio-cultural asset. Tourism and restaurants are essential sources of foreign exchange income for Peru. Along with tourism, restaurants are another driving force for economic growth. PromPerú has been the main actor in the promotion and dissemination of gastronomy, carrying out a series of national and international campaigns and events. Tourism and restaurants contribute 11.3% of the GDP in Peru (INEI, 2020).

Creative industries are based on intellectual properties such as the performing arts, books, movies, video games, software, and TV shows. As of 2018, Peru’s creative industries contributed 3.8% (US$ 5.7 billion) of the value added to the national economy.

3.6.1 Main Products

■ Tourism

Peru has considerbale competitiveness in the field of ecotourism as it possesses one of the most biodiverse natural enviroments in the world. Peru has 158 offically designated Protected National Areas (ENP), which cover 16.9% of its total land area. The ENPs are divided into the following categories: National Parks, National Reserves, Communal Reserves, National Sanctuaries, Historic Sanctuaries, Landscape Reserves, Protected Forests, Wildlife Refuges, and Hunting Reserves. The most visited ENP in 2018 was the Machu Picchu Historic Sanctuary, which is regarded as one of the seven wonders of the modern world.

The travel experience can be far more pleasant if services are tailored to the needs of each user. The use of Big Data makes it possible to analyze the large amounts of information collected from travelers via various media and platforms in order to tailor products to suit individual customers. In adapting to supply and demand, travel websites will change these amounts using automation and thereby be able to predict their variations over time more accurately.

■ Restaurants and Catering

In 2021, Peru's restaurant and catering industry recorded higher sales in poultry, fast foods, chifas, tourist restaurants, cafes, Peruvian *cevicher ías*, and sandwich shops (INEI 2022). Slaes of meat and grill businesses, international food, Italian food, Japanese food, Creole food, ice cream, and candy stores all increased. Beverage sales also increased due to the good performance of bars and restaurants, cafeterias, etc. Catering showed strong growth due to the increased demand for public and private events, with new contracts for buffets, executive breakfasts, conferences, conventions, and corporate refreshments (INEI 2022).

■ Creative Industries

Movies, music, theaters, books, and video games are widely consumed in Peru. In 2019, 2.8 million people (33.8%) attended a movie every month. Digital consumption of movies, music, and video games has increased, while the consumption of printed books and digitally stored media (CDs, Blu-ray DVDs) for movies and games has decreased since 2017.

3.6.2 Innovation Trend

■ Tourism

Augmented reality and virtual reality have become firmly established in the travel and tourism industry, thanks to the added value they bring to travel experiences, allowing tourists to interact more closely with the destination or service. Tourists are increasingly aware of social, economic, and environmental sustainability. A sense of responsibility and commitment to local economies drives these travelers to seek alternative activities, such as ecotourism, which focus on valuing the environment and the culture of the destination visited. In terms of technology, efficient transportation, such as trains or low-cost airlines, will increase the productivity of tourism in Peru.

■ Restaurants

The use of digital media has become a valid way of attracting more customers. Social media, such as Facebook, Instagram or Twitter, provide reviews and comments that serve as informed decision-making aids for new visitors. Refined websites also promote restaurants and attract new orders. Optimizing a website for mobile devices is also vital. Digital marketing that uses emails, videos, or influencers is also an effective tool for boosting the revenues of restaurants.

■ Creative Industries

Digitalization drives a new trend in created industries. Through the pandemic, people had to use more online solutions for cultural experiences. E-commerce and E-entertainment inevitably grew fast. The production process of music or film is more digitalized. Computer-generated designs or animations increased. new technologies open up access to digital content, reduce production costs and increase exposure. 3D printers facilitate the creation of prototypes to test new designs or products, and in the fastest way. However, technology can be a threat to handicrafts because design and modeling software can imitate the work and feelings of human beings and can create handicrafts that are difficult to differentiate from the original products, even imitating imperfections made by some artisans. Consequently, the originality of handicraft products can be protected from undue competition through the use of appropriate intellectual property tools.

3.6.3 Growth Strategy

■ Tourism

Peru faces great challenges for rural tourism in general as it was greatly affected by the Covid-19 pandemic, but it also has opportunities for consolidated experiences, such as community-based tourism and ecotourism promoted by the private sector with government support, but above all for enjoying the outdoors. In rural tourism, there are strategic lines of action to boost rural tourism; community- based tourism, agrotourism, etc. One of the challenges facing rural tourism's need to continue growing is sustainability.

The strategies to be implemented should take into consideration the following challenges:

- Safety of the local population: The adequacy of safety protocols and communication strategies will be crucial so that the local population does not perceive the arrival of tourists as a risk to their health.
- New social values and needs: Isolation and confinement at the global level bring with them new needs and priorities for the local population. The value of the human, of the little things, and of emotions must play a central role in the rethinking of rural tourism in the face of Covid-19.
- Tourism knowledge and research: The challenges facing tourism will increasingly require more expert and multidisciplinary knowledge. Resources must be allocated to generate knowledge and research in order to improve decision-making and, consequently, the competitiveness and sustainability of the destination.

■ Restaurants

The Peruvian government announced the Reactiva Plan, which was created to generate economic support for companies that were affected by the Covid pandemic. The tourism and gastronomy sectors benefited the most from the Reactiva Plan:

- Branding: Related to the food industry, a strong brand helps restaurants to emotionally connect with customers.
- Customer DB: Building a customer database is important to deliver personalized value and experience to loyal customers.
- Hygiene: Food preparation, care for deliveries, and non-contact delivery provide trusted safety to customers.

Digital technology will improve the customer experience with digital payments, digital menus, online reservations, etc. Since the pandemic, people have spent more time at home and are concerned about healthy food. Restaurants can adapt to this new trend by turning their menus into meal packs. Delivery services are another new source of income for restaurants.

■ Creative Industries

The Peruvian government has generated more than 60 million soles in subsidies to art workers and bearers of intangible heritage in order to mitigate the negative effects of the Covid-19 pandemic. They subsidized 5,923 projects that benefited more than 22,000 cultural workers. Three festivals were organized: Peru Read Ayacucho 2020, and the Peru Reads Festival for Children and Young People in 2020 and 2021.

Fostering creative industries requires the following:

- A well-articulated education system, including schools, universities, colleges, and training centers.
- Tax incentives and funding programs to encourage R&D in the creative industries.
- A financial arm to provide content to sectors that are often perceived as high-risk segments with better access to venture capital.
- A competition regime that protects intellectual property, provides a balanced copyright regime, prevents abuses of dominant positions, and acts quickly to remedy them.

CHAPTER 4

Six Prioritized Industries in Peru

S&T Planning and Technological Forecasting for Prioritized Sectors in Peru

Chapter 4. Six Prioritized Industries in Peru

1. Agroindustry and Food Processing

1.1 Legal Framework

The Peruvian government has approved Law No. 31110, called the "Agrarian Labor Regimen," which privides incentives for the agricultural, agro-export and agro-industrial sector, guarantees the labor rights of workers and international labor and human rights protection treaties, and contributes to the competitiveness and development of activities in these sectors.

The law covers natural persons and legal entities that develop crops and/or livestock and engage in agro-industrial activities, including agricultural producers who have five hectares of land.

Natural and legal persons must use agricultural products outside the province of Lima and Callao. Agroindustrial activities do not include the production or cultivation of wheat, tobacco, oils and oil seeds, and beers.

Labor contracts may be fixed for a definite or indefinite term. The working week is 8 hours per day and 48 hours per week. Remuneration must not be less than the minimum vital remuneration (set at 1,025 soles per month for the time being).

1.2 Agricultural markets

In Peru, both exports and imports have managed to increase at moderate but sustained rates. By making a cut in the year 2014 to date, the balance of trade has increased at an average annual 25%, as exports have increased at a rate of 6.4% per year, whereas imports have increased at a rate of only 2%.

In 2020, however, despite the global crisis caused by the initial restrictions imposed on international markets in order to neutralize the spread of the COVID-19 virus and prevent a high mortality rate, the balance of agricultural trade yielded a positive value of US $ 2.4 billion, which can be attributed to the dynamism of Peru's agricultural exports.

[Table 4-1] Agriculture Commerce Balance (2011-2020*)

YEAR	IMPORTS	EXPORTS	BALANCE
	CIF (US$ 000)	FOB (US$ 000)	(US$ 000)
2011	4 166	4 777	611
2012	4 519	4 443	-76
2013	4 470	4 427	-44
2014	4 675	5 301	626
2015	4 409	5 285	876
2016	4 511	5 790	1 279
2017	5 171	6 255	1 084
2018	5 167	7 033	1 866
2019	5 126	7 462	2 336
2020*	5 268	7 678	2 411

Source: Ministry Of Agriculture of Peru.

In recent years Peruvian exports have enjoyed a sustained growth rate, reaching US $ 7.4 billion in 2019 and, against all odds, a record figure of US $ 7.7 billion as of December 2020, an increase of 2.9% over the previous year.

At the beginning of 2020, exports were expected to exceed US$8 billion due to the favorable prospects observed in the international market and the good climatic and water conditions. However, the global outbreak of Covid-19 caused countries to initially close their markets, generating some uncertainty during the early days of the pandemic. Despite this, the Peruvian Government established the conditions and protocols required to maintain the continuity of commercial exchanges, and private companies never stopped operating, although some operated to a lesser extent, demonstrating that Peruvian exports were reliable and respectful of all sanitary and quality standards. Peru also showed to the most important customers in the world that it was able to permanently supply all the products, especially fresh fruits and vegetables, that consumers had begun to demand in ever greater quantities.

The main destination markets for Peruvian exports are Europe, North America, Asia and South America. The main market is the bloc of European Union countries, plus the United Kingdom. On January 1, 2021, a new free trade agreement with the United Kingdom and Northern Ireland entered into force, maintaining the same benefits as those previously provided by the Trade Agreement signed with the European Union, from which both British states have now withdrawn due to the Brexit vote of 2016.

The European Union bloc accounts for about 36% of Peru's agricultural exports, with eight member countries standing out, namely, Holland, Spain, the Untied Kingdom, and Germany, which individually are the most important trading partners of Peru after the United States, followed by Belgium, France, Italy and Sweden, in descending order of importance. Together these eight markets accounted for around 35% of all Peruvian exports in 2020 (US$2,682 million), with a growth rate of 4.1% compared to 2019.

On the other hand, it is noteworthy that the United States accounted for 34.2% of all Peru's agro-exports after June 2020, an increase of 5.5% over the previous year. This mega market has become the most important destination for Peruvian agro-exports, to the extent that there is a Trade Promotion Agreement that allows access to this market without any major restrictions and even allows the tariff-free entry of a large proportion of Peru's agro-exportable products.

As such, the two blocs together account for 70% of all Peru's agricultural exports, being very demanding markets in terms of quality, and currently facing extreme pressure due to the high demand for fresh fruit and vegetables, the same ones that have even increased from Peru, amid in the midst of the Covid-19 pandemic.

Also noteworthy is the presence of a group of Asian countries, such as China, Hong Kong, South Korea, Japan and Indonesia, which accounted for 8.1% of Peru's total (agricultural?) exports in 2020; to this should be added the important presence of the member countries of the Pacific Alliance, which accounted for 7.5% of Peru's exports, and the countries of the Andean Community of Nations, which accounted for 7.2%.

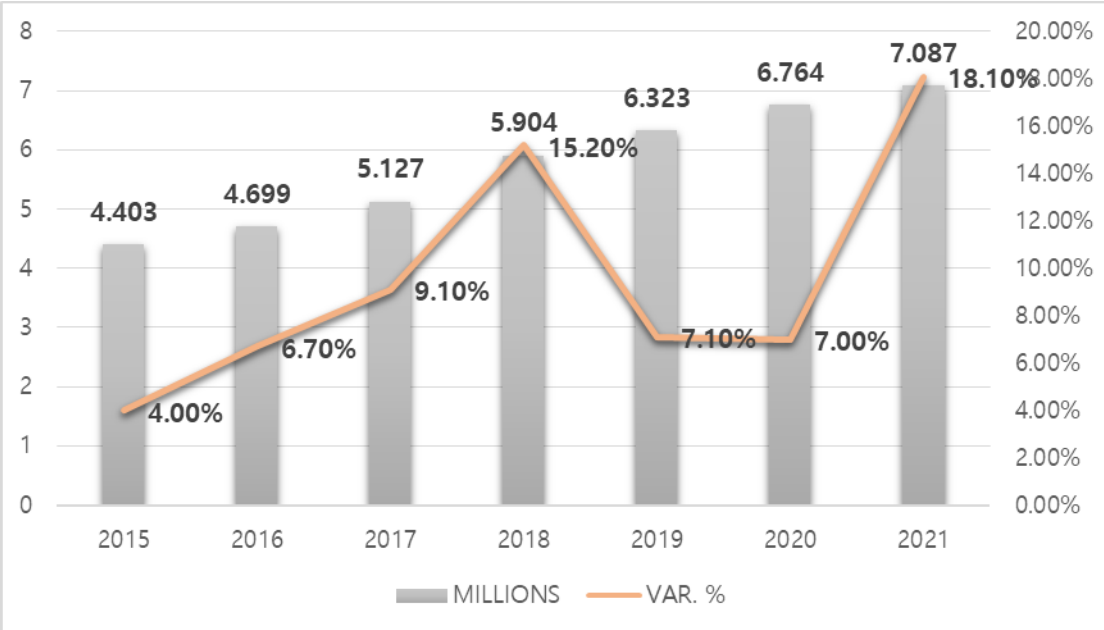
It is also worth mentioning that Peru has signed around twenty trade agreements with more than fifty-six countries in the world, a situation that has allowed Peru to access almost all markets without any major problems. Among the others, only two major markets, Russia and India, have pending agreements.

The economic recovery of 2021 favored the agro-export sector, especially non-traditional exports. According to figures from the Sunat (National Tax Office), non-traditional agro-exports amounted to US$ 7,987 million in 2021, an increase of 18.1% over the previous year.

It should be noted that the US (US$ 2,727 million, +11.7%) and the Netherlands (US$ 1,336 million, +20.4%) were the leading importers of Peru's non-traditional agro-exports, accounting for 50.9% of the total.

[Figure 4-1] Evolution of Non-Traditional Exports

(PERIOD 2015-2021)



Source: SUNAT (National Tax Office).

The sustained dynamism of the sector demonstrates its great potential, as it has grown at an average annual rate of 13.1% since 2009. This has favored the national economy, especially that of the coastal departments. La Libertad (US$ 1,694 million, +12.8%), Lima (US$ 1,664 million, +15.7%), Ica (US$ 1,534 million, +18.5%), Piura (US$ 1,142 million, +12.1%) and Lambayeque (US$ 841 million, +31.5%) accounted for 86.1% of Peruvian non-traditional agro-exports in the year.

Within the wide range of products exported from this sector, the produce that stood out the most were fresh grapes, blueberries, and avocados. Fresh grapes were the main non-traditional agricultural produce exported in 2021, with shipments worth US$ 1.255 million, which translated into an increase of 21.4% over the US$ 1.035 million registered in 2020.

In terms of export destinations, the US was the leader, at US$ 513 million, i.e. 9.4% more than in 2020, accounting for 40.9% of Peru's total shipments of fresh grapes to the world; followed by the Netherlands, with US$ 178 million and a growth rate of 31.2%, accounting for 14.2% of all shipments; and Hong Kong, with US $ 128 million and a growth rate of 34.2%, accounting for 10.2% of the total.

Blueberries were the second largest export product, with shipments amounting to US$ 1,202 million, an increase of 22.2% over 2020.

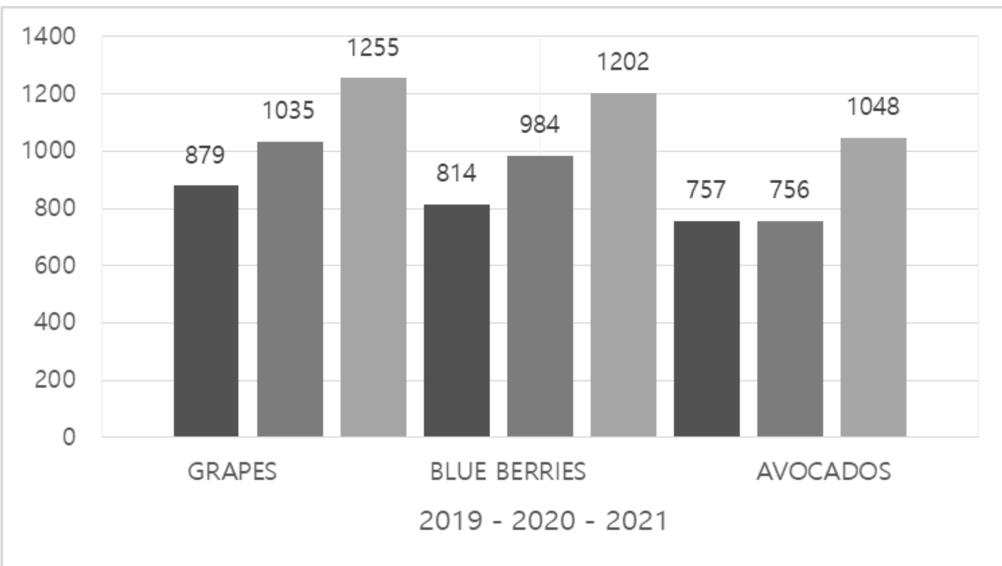
The US ranked first at US$ 639 million, an increase of 23.2% over 2020, accounting for 53.1% of total blueberry exports, followed by the Netherlands, at US$ 281 million, an increase of 12.7%, accounting for 23.4% of total exports; and Hong Kong, at US$ 102 million, an increase of 138.7%, accounting for 8.5% of total exports.

Finally, avocados were the number three agricultural export product, with a total export value of US$ 1,048 million, an increase of 38.7% over 2020.

Specifically, the Netherlands were the main destination of Peru's avocados, at US$ 340 million, an increase of 34.2% over 2020, accounting for 32.4% of the total; followed by the USA, at US$ 185 million, an increase of 17.3% over 2020, accounting for 17.6% of the total; and Spain, at US $ 172 million, an increase of 31.7% compared to 2020, accounting for 16.4% of the total.

[Figure 4-2] Non-Traditional Agribusiness Exports

(unit: million US dollars)



Source : National Taxes Office

1.3 Technological Trends in Agriculture

According to Peru's agricultural network, (www.redagricola.com), “necessity is the mother of invention”. For the food and agricultural industries, this means that technology found its time to shine in 2020. For this reason, the AgFunderNews Portal enumerated the five biggest technological trends to emerge in the agri-food sector in 2020.

1.3.1 E-Commerce

By far, online grocery companies were among the group of companies to attract the most funding since AgFunder began tracking investments in the agri-food space. Two of the biggest deals of the year went to China's Missfresh e-commerce platform, which in July 2020 received $495 million from investors, including Goldman Sachs, Tencent and Tiger Global, followed by a $306 million injection of Chinese state-linked funds in early December of the same year.

Staffdigital.pe is a company that specializes in e commerce in Peru (www.staffdigital.pe).

1.3.2 Automation, Robotics And "Non-Contact" Technology

This trend has been particularly evident in the agri-food sector, where the desire for hygiene was already paramount, and it has only been strengthened by the COVID-19 pandemic. China's XAG raised what may be the largest funding yet for a drone start-up, with $182 million agreed in a funding round led by Baidu and SoftBank.

This trend has penetrated all the way to the end of the value chain, including non-contact food delivery cabinets, autonomous supermarket carts, robotic waiters and grill cooks.

1.3.3 Biological Inputs

A growing number of bio-inspired crop inputs emerged in 2020, as farmers, consumers and regulatory bodies became more aware of the negative side effects of chemical treatments. The Belgian company Biotalys raised $55 million in Series C funding in early

2020 to invest in its C protection technology, i.e. antifungal cultures that mimic the immune response of animals with heavy chain antibodies.

Connectamericas is a company that specializes in biological inputs in Peru.
(https://connectamericas.com/es/company/insumos-biol%C3%B3gicos-per%C3%BA-sac)

1.3.4 Carbon Market in Agriculture

In particular, Cargill has participated in several carbon sequestration initiatives, including the Soil & Water Outcomes Fund, which pays farmers to practice regenerative farming methods in the form of carbon credits, which it then uses to offset its own carbon footprint.

The US startup Nori also revealed its first purchase of carbon credits via its platform, with Shopify among the buyers.

1.3.5 Alternative Proteins

As a result of the Covid-19 pandemic, more people became aware of and interested in alternative proteins. According to market research firm Ai Palette, consumers in Asia and the United States are keen to increase the plant-based component of their diets. The largest round of financing, closed in 2020, was the US $ 500 million dollars raised by the American company Impossible Foods, which is dedicated to the production of vegetable proteins.

The Consumer Defense Institute in Peru has developed a detailed guide to alternative proteins for human nutrition:

https://www.indecopi.gob.pe/documents/51084/2681482/MenuAlternativo_ProductosSustitutos_Indecopi_2021/7f99fce9-935a-4434-34b6-9df7babb0c5b.

1.4 Major Challenges and Suggestions

Representatives of AGAP (Agrarian Producers Association www.agapperu.org) have mentioned that the depletion of traditional markets is beginning to be noticed in Peru's international markets, due to a potential increase in the exportable supply and such factors as the improvement of technological innovation, entry to the optimum level of crop production, and higher crop yields. Added to this is the expansion of the agricultural frontier, with irrigation projects such as Chavimochic III and Majes Siguas II, etc., so it is necessary to think about looking for new markets and where to supply the products, as otherwise a greater exportable supply could bring price problems.

With the outbreak of the Covid-19 pandemic in Peru’s export markets, certain adjustments have been observed in the fresh fruit and vegetable supply contracts that may affect the competitiveness of Peruvian exports, including the following:

- The closure of wholesale markets, for example, in the United States, with marketing only in supermarkets.
- The appearance of consignment purchase contracts, which do not establish minimum prices, but are made according to the evolution of the markets, on the grounds that the market is moving very slowly.

In relation to exports, the WTO (World Trade Organization) estimated that in 2020 there would be a drop of 9.2%, while its estimates for 2021 pointed to a recovery of 7.2%. However, these data were subject to a very high level of uncertainty, to the extent that they depended on the evolution of the pandemic and the measures adopted by governments in response to the situation.

It must be remembered that the negative impact on international trade in 2020 was due to the closure of airports and borders to contain the pandemic, which prevented the exchange of goods and the recovery of trade after the tensions between China and the United States affected the economy.

Peru is, on the one hand, a very competitive country in terms of productivity and its agribusiness exports to markets worldwide. On the other hand, the public budget for science and technology are smaller than those of other countries. Therefore, public policies should be oriented toward increasing the allocation of public resources to science and technology, in line with other Latin American countries such as Colombia and Brasil. The experience of South Korea in this area is very valuable.

2. Forest

2.1 Forest Industry Trends

The main State entities responsible for regulating the timber industry are the National Forestry and Wildlife Service (SERFOR) and the Ministry of Production (PRODUCE). SERFOR is the national forestry and wildlife authority, whose budget is attached to the Ministry of Agrarian Development and Irrigation (MIDAGRI). SERFOR is the entity that regulates the management of the forestry sector throughout the entire value chain of the timber industry, including the granting of licenses for forest harvesting, forest harvesting operations, primary transportation, primary processing, secondary transportation, secondary processing, and exportation. Eamnwhile, PRODUCE registers secondary processing centers.

The Forest Management Regulation (MINAGRI, 2015) defines primary processing as the first processing process to which forest and wildlife products and by-products are subjected in their natural state, and secondary processing as the process by which forest and wildlife by-products are taken from the primary processing industry to add added value (MINAGRI, 2015).

SERFOR (2019) has pointed out that primary processing is mainly carried out in primary processing centers, as well as in utilization areas through the use of mobile equipment. Secondary processing is mainly carried out in secondary processing centers, but it can also be carried out in primary processing centers.

A "first-processing product" is the one that comes from a primary processing plant and does not constitute a final or direct use product, as it is an input for secondary processing centers (MINAGRI, 2015), whereas a "secondary-processing product" can be an end-use product or an input used to produce other products. SERFOR (2019) lists the following primary processing processes: sawmilling, edging, trimming, debarking, drying, sizing, planing, lathing, assembling, finger jointing, sieving, resizing and lathing. Similarly, SERFOR (2019) identifies the following secondary processing processes: molding, tongue-andgroove jointing, cross-cutting, sanding, grooving, tenoning/coupling, reinforcing, gluing/ pressing, planing/painting, nailing, heat treatment, and final sizing.

In 2019, SERFOR presented an updated list of eighty-five timber forest products organized into the following six major groups: 1) fuels and wood residues, 2) roundwood, squared and raw wood, 3) sawn wood, 4) wood-based sheets and boards, 5) semi-finished products, and 6) other manufactured products. Table 1 shows the thirty-one products with the highest demand, organized into the six groups mentioned above.

[Table 4-2] Main timber forest products

Group	Fuels and waste wood	Round, squared and rough lumber	Sawn timber	Wood-based panels and boards	Semi-finished products	Other manufacturing
Products	Firewood	Squared timber	Sleepers	Plywood sheets	Profiled wood for interior flooring	Tableware and kitchenware
	Pellets and briquettes	Round wood	Sawn timber	Decorative foil	Profiled wood for exterior flooring	Wooden handicrafts
	Wood waste	Posts	Parquet	Raw particleboard	Moldings, molded panels	
	Charcoal	Roundwood for construction	Wood obtained by finger jointing	OSB	Cable reels (drums)	
	Lana	Plant stakes		Fiberboard (MDF)		
	Flour			Plywood		

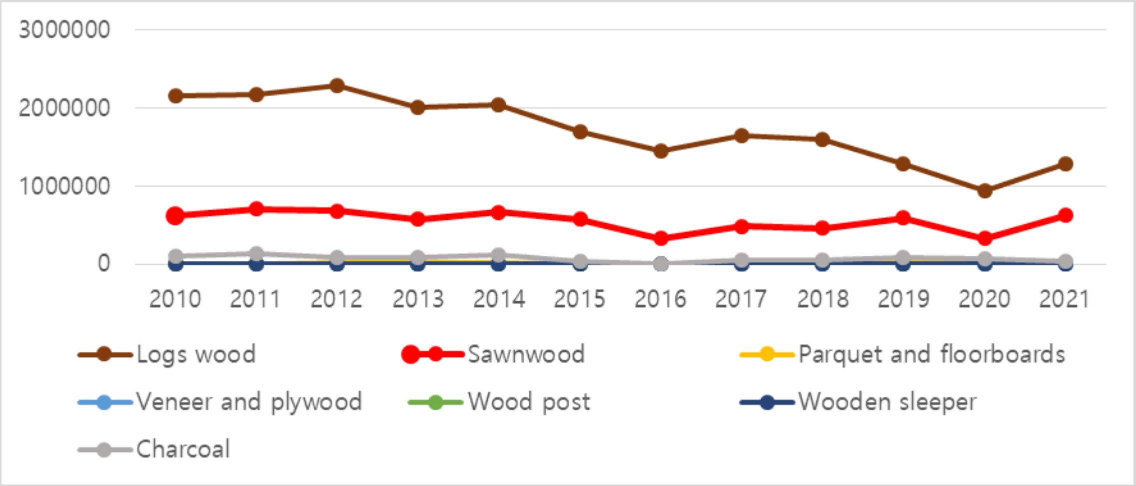
Source: SERFOR (2019).

2.1.1 Trends in domestic production

Sinec 2008, SERFOR has presented data and statistics on forestry information in the form of the annual Forestry and Wildlife Yearbook. In 2021, it presented the Forestry Statistical Compendium 2010-2020, although the information it contains has not yet been ordered according to the six groups identified by SERFOR in 2019, and the lack of consistency in the information along the value chain of the timber industry is a problem for both decision-making in public management of the sector and in strategic business decisions in the private sector.

In the period 2010-2021, firewood, roundwood, and sawn wood accounted for 73%, 19.6%, and 6.3% of national production respectively, while charcoal, parquet and floorboards, veneer and plywood, wooden sleepers and wooden posts collectively accounted for just 1%. It should be noted that the recorded production of wooden posts was not constant during that period, and that the national production of firewood is based on an estimate of annual per capita consumption of firewood, ratehr than on the record of its extraction, so CIFOR (2016) points out that there could be a significant margin of error.

[Figure 4-3] National production of main timber products, 2010-2020



Source: SERFOR (2021).

2.1.2 Trends in domestic consumption

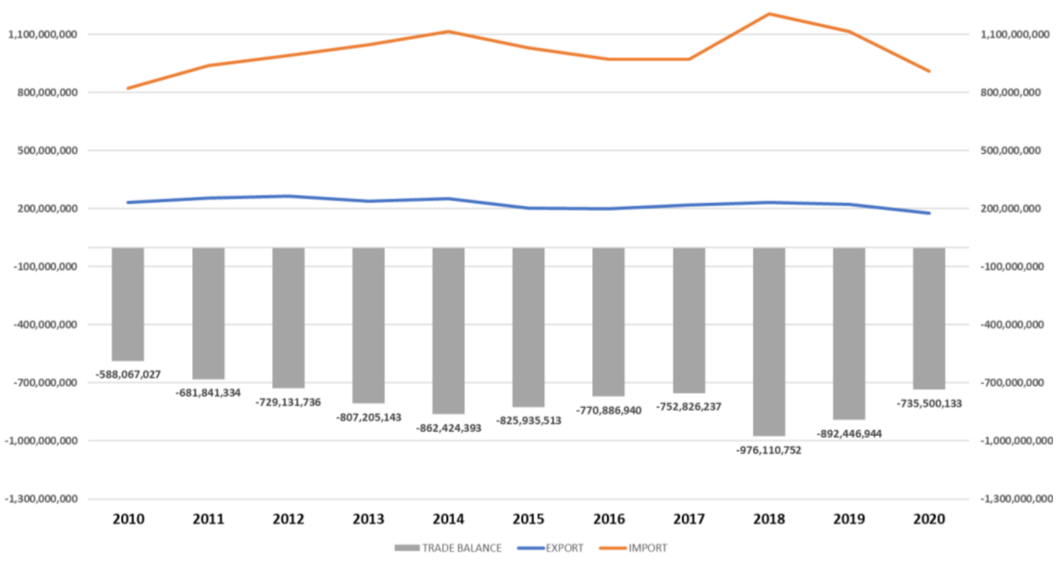
FAO (2018) reported that the domestic market accounted for 90% of industry sales in 2015, of which only 10% was exported. As can be seen in the international trade trends, the national supply does not meet the growing demand of the domestic market.

The demand for roundwood depends on the demand of the multiple productive processes to which it is destined. $SERFOR_b$ (2021) indicates that around 80% of the total roundwood that is processed is destined for sawn-wood production.

2.1.3 International Trade Trends

Peru’s territory covers a a total surface area of 12,852,156,000 hectares, of which 72,083,263 hectares are forest, representing 56% of the national territory. More than 68 million hectares of tropical rainforest are located in the Peruvian Amazon (94.6% of the country's forests), 3.7 million hectares are located along the coast (5.1%), and 0.22 million hectares are in the Sierra (0.31%) (MINAM, 2016). However, the trade balance of timber forest products showed a deficit during the period 2010-2020 (SERFOR, 2021).

[Figure 4-4] Trade balance of timber products 2010–2020



Source: SERFOR, 2021.

2.2 Technology Trends in Forestry

The *Centro de Innovación Tecnológica de la Madera* (CITEmadera) (2008) states that most of the companies in the wood and furniture industry are microenterprises, with a minimal number of small and medium-sized companies, while only 0.6% are large companies. The first processing sector is occupied by small and medium-sized companies, while the second-processing sector (furniture manufacturers and wood joinery) is mainly occupied by micro-enterprises. Sobrado (2022), based on information from SUNAT, identified that in 2021 there were 227 small, medium and large wood industry companies in the regions of Madre de Dios, Ucayali, Loreto, Cusco, Cajamarca and Huánuco. These regions constituted the scope of the study, so it did not include timber companies based in other regions. Also, this number only represents formal companies; it is estimated that 70% of companies in Peru are informal.

Most of the Peruvian timber industry produces goods with little added value, due to the fact that it uses poorly developed technology (PRODUCE, 2015). Estrada (2017, cited in Mori, 2019), agrees with this statement. Regarding Pucallpa, one of the main centers of timber processing in Peru, he points out that it is characterized by the absence of added value, and the absence of quality management systems and technological innovation, with the result that timber companies in the Ucayali region are not sustainable or competitive. (PRODUCE, 2015).

In a study for which SNV (2009) interviewed experts and entrepreneurs from small and medium-sized enterprises (SMEs) in the timber industry, the interviewees all agreed in identifying three important issues with respect to the technological variable, both at the primary and secondary processing levels: 1) the existence of a technological gap in the forestry chain; 2) the lack of training programs; and 3) the low level of technification of SMEs, which results in low productivity in workshops and carpentry.

2.3 R&D Manpower in the Forest Industry

The *Ministerio de la Produccion y Promocion del Empleo* (MTPE) (2019) pointed out that the timber production chain[25] plays a small role in national employment. In 2017, the economically active population (EAP) of the timber production chain was 183,157 workers. Likewise, during the period 2007-2017, the timber production chain had the lowest average annual employment growth rates (-1.0%) in Peru due to a decrease in the production of wood products, and a decrease in investments by the private sector.

Regarding their employment relations, 45.3% of the workforce work in companies with two to ten employees, and 34.1% are self-employed. Most employment is generated by smaller companies (microenterprises). López et al (2017) noted that informal work predominates in the timber industry.

Regarding the educational level of workers in the timber industry at the national level, MTPE (2019) indicates that 61.2% have attained secondary education, and 18.2% have reached a higher level (university: 7.7%; non-university: 10.5%), which results in an activity with low productivity. Another reason for the country's low productivity is the low investment by the State and the private sector in innovation and technology.

2.4 Mid- and Long-term Industries Strategies

It is essential to elaborate a strategic planning process for the timber industry and its sub-industries (lumber, flooring, decking, flooring slats, wooden pallets and boxes, wooden furniture, construction lumber, posts, sleepers, and others), taking into account the fact that approximately 99% of all business units are microenterprises, while only 1% are small, medium, or microenterprises. As the large number of microenterprises in the industry, generally informal, results in various problems including 1) small-scale production, 2) a lack of efficient equipment and machinery, 3) low production capacity in the face of large volume orders, and 4) a lack of product standards (PRODUCE, 2015), the Cámara Nacional

[25] That considers employment from forest management, planting and harvesting, first processing and second processing.

Forestal (2022) has raised the need to generate and strengthen their associativity. However, the FAO (2018) notes that the experiences of the associativity in timber purchasing has not yielded good results, while the Cámara Nacional Forestal (2022) has proposed the formation of forestry clusters. There have been instances of forestry clusters in Ucayali and Madre de Dios; therefore, before promoting the establishment of such clusters, the lessons learned from the two aforementioned clusters should be evaluated. Likewise, it is considered pertinent to implement the policy measures of Objective 6 (Generate conditions conducive to the development of a productive business environment) of the PNCP 2019-2030, which encompasses the imposition of a single tax regime for MSMEs, the National Strategy for the Development of Industrial Parks, the development and implementation of a special public procurement regime (Compras a MYPeru), the formulation of an articulated strategy for business formalization and development, quality standards and technical regulations suitable for a competitive market, administrative simplification instruments, quality standards and environmental sustainability in public procurement. MSMEs in the forestry industry require capacity building for their formalization, technical assistance and training on production organization and business management, and access to soft financing for at least three years (Cámara Nacional Forestal, 2022). One of the components of the financing should be the purchase of appropriate machinery and equipment, where the equipment or machinery is delivered, along with technical assistance and training for its use.

In the commercial field, it is necessary to develop market studies of timber industry products, as identifying and evaluating the demand that will be supplied is of vital importance for the success of the industry. It is necessary to identify products with sufficient demand to ensure the continuous and growing purchase of the products. The market for the eucalyptus roundwood produced by the forest plantations in the Peruvian Sierra, which led the national roundwood production until 2015, was the mining operations that needed boards and stakes. The boom in agribusiness exports in recent years has led to an insufficient supply of domestic stakes, which is why imports of stakes from Chile have increased substantially.

There are also land irrigation projects for agro-exports, which until 2019 were planned to irrigate 440,000 hectares of crop fields (PortalFruticiola,com, 2019). From this perspective, a study of the market for pallets should be carried out with a view to supplying the agroindustry. To supply the raw material, plantations of Tornillo, Huayruro, Cachimbo, Yacusahapana, currently the most widely used species, or other species with good characteristics could be established. For wood processing, a factory could be designed to cut and assemble pieces of pallet wood to supply the demand of the main agroexport centers on the northern and southern Peruvian coast. Similarly, the trend in the mining industry, which also demands wooden pallets, and other industries that need this product, could be analyzed.

The FAO (2018) has pointed out that raw materials and labor represent a very important portion of the final cost of the products, which is a problem for the competitiveness of many companies. For any industry to be profitable, it needs a continuous supply of cheap raw materials and production inputs, which is not the case of the Peruvian timber industry. The main wood used as the raw material of the industry is hardwood from tropical rainforests, which have a great diversity of species with different characteristics for the industry. These issues, combined with the seasonal nature of timber extraction, the increasing dispersion of individual trees of a species caused by the weaknesses in forest management documented by such authorities as the Forest Resources Oversight Agency (OSINFOR), the advance of deforestation and the degradation of tropical rainforests, all serve to make the supply of raw materials for the industry unfeasible in the future. Therefore, in the medium and long term, the supply of timber can only be made viable by harvesting timber from forest plantations. As such, it is necessary to promote forest plantations, primarily by making the legislation more flexible so that they have the same regime as agricultural crops; to ensure land tenure where plantations are established; to implement long-term financial incentives, such as subsidies payable at the time of plantation installation, long-term credits guaranteed by a State fund; and to implement a research and technological development program for forest plantations (Cámara Nacional Forestal, 2022).

Forestry plantations in Chile have been successful thanks to government investment and because the timber produced is destined for three industries, i.e. the sawmill, wood pole and pulp chip industries. The whole chain is intertwined; therefore, if you have a pine plantation and do not supply raw materials to these three industries, it will not be profitable. As Chile, Brazil and Uruguay already lead the market for wood chips to obtain cellulose, it would not be profitable to export chips.

The industry should work with fast-growing species, with rapid growth and cutting shifts, to ensure the supply of raw material. One immediate source of timber is a species known to be a fast-growing species that is found in young secondary forests called "purmas”. The most commonly used species are *Guazuma crinita*, *Ochroma pyramidale* (due to its recent boom as a raw material for wind turbine propellers), and *Calycophyllum spruceanum* among others. The prevailing regulations and punitive actions have not had good results in the sustainable and/or legal management of tropical rainforests for more than twenty years; proof of this are the currently high rates of degradation and deforestation. Therefore, stimulus policies are urgently needed. In addition, it is necessary to analyze whether fast-growing species should be subjected to the same controls as slow-growing hardwood species. A policy aimed at stimulating their use would encourage the same users of these species to reforest them in order to guarantee the economic benefit they receive from them.

With respect to primary processing, such problems as the quality of the wood, the age and lack of precision of the machines used, the lack of maintenance of the machines, the lack of personnel training, and the lack of quality control over the overall process result in uncalibrated sections and cutting defects in the sawn timber, which means that the final product does not have the quality required by the customer, and that there is a high amount of waste and residues. Given these deficiencies, Mori (2019) has pointed out that it is necessary i) to conduct studies to determine the causes of variability in the thickness of sawn timber, within and between pieces, in order to establish measures to solve these causes and thereby reduce the total variation of thickness due to the effect of cutting in sawmills; and ii) to implement a dimension control system for sawn timber boards in sawmills in order to quantify the excess wood that is lost when cutting pieces with excessive dimensions, and to identify the causes thereof in order to design measures for reducing variability in the dimensions of the wood pieces.

As mentioned above, there are also problems with air pre-drying and kiln drying. According to the entrepreneurs themselves, the available ovens are old, the personnel do not apply the drying program properly, and corrective drying is not carried out. The problems extend to the brushing and preservation processes. There is a lack of knowledge of the physical-mechanical properties of wood; and pieces of wood with different shrinkage rates are joined together, resulting in the collapse of their structure. These issues also affect the production of chipboard. Therefore, the technical assistance and training provided by CITEmadera and CITEforestal must be strengthened, as well as the curricula for careers in the forestry science and forestry industry, and for technical careers. New trends in international markets demand standardization and certification of the quality of processes and products. In this context, the need to standardize job training in order to foster human resources that can meet the demands of the productive sector is highlighted (MTPE, 2019).

In general, it is necessary to identify the species that meet the technical requirements for the production of products, standardize processes and furniture pieces, conduct market studies on the products in greatest demand in this subindustry, and invest in constant innovation of designs and finishes, according to the prevailing fashion trends. Entrepreneurs and workers should be trained in the areas necessary to boost the sector: (1) specialization of the production process, (2) marketing tools, and (3) machinery management. Also needed are the renovation of machinery and equipment, the use of modern technologies, financing capacity, and a business run with an entrepreneurial vision.

The industry also needs waste processing lines, as in the case of Forestal Otorongo's sawmill, whose main production line is flooring, and where waste is transformed into charcoal; other waste recovery products include chipboard, finger-jointed wooden boards, and short wood, all of which are already being produced and used. In the long term, wood sawdust should be transformed into pellets, which are in great demand in Europe; the corner pieces of tree trunks could be used in the production of corner pieces dyed with natural dyes, which are in demand for garden decoration; and the branches of hardwood trees could be converted into pieces of wood for export as a raw material for the workshops of retirees in the USA and the EU, where they transform it. (In this respect, we can cite the experience of Mr. King's company in Loreto.) Branches could also be used in the elaboration of fine wooden handicrafts, jewelry, and wooden buttons. To this end, the State, in cooperation with international entities, could invest in the establishment and operation of industrial parks equipped with a multipurpose wood processing plant in the timber production regions, thereby benefiting MSMEs and forestry clusters.

In the furniture industry, the bottom of a product should be made of good quality but low-priced wood, and veneered with fine wood such as mahogany, rosewood, or violet. The furniture production industry should produce standardized carpentry parts for binders, wooden cots, shelves and other products; and these parts should be coded so that when a carpenter receives an order for binders, he can buy the parts, glue them together and assemble the binder. When a customer needs to do maintenance work or change parts, he could request an order of parts identified by their code.

The export market, with the required standards, should be targeted in the long term (15-20 years), without neglecting the domestic market, which can serve as a refuge from market variations and the international economy. One of these markets is the lumber market. Botyriute (2022) has pointed out that there is a global housing shortage, which could help the forestry industry to recover after the fall in lumber sales caused by the COVID-19 pandemic. Another factor driving growth in this market is the approval of changes adopted by the International Building Code (IBC) 2021, a model building code developed by the International Code Council (ICC), which ensures the safety of mass timber buildings and allows the construction of tall buildings with mass timber of up to eighteen stories. This reform is expected to increase the use of wood in U.S. commercial and residential buildings. Meanwhile, mass timber buildings have gained traction in Europe, as the decarbonization of buildings lies at the heart of the EU's Green Deal, which is actively moving toward the goal of ensuring zero emissions by all new buildings by 2030 by promoting the use of more environmentally-friendly materials, such as forest-managed timber.

2.5 Major Challenges and Suggestions

Forestry legislation regulating the technical aspects of forest harvesting and the primary processing industry has not achieved its objective: to promote the sustainable management of forest resources and ensure that forests provide their ecosystem services for future generations. This is because it attempts to protect forests without taking into account the balance between the social and economic aspects of sustainability, and without conducting a technical assessment of the applicability of the law, which has led to a high level of informality and illegality among forest users and the timber industry, as well as low levels of competitiveness in supplying the national and international market for timber products. It is suggested to evaluate the results of the forestry legislation and modify those aspects of the legislation that have not had positive results, in addition to simplifying the administrative processes related to forest harvesting and the timber industry.

A high percentage of the industry is made up of micro, small and medium-sized enterprises, which in turn have high levels of informality with respect to tax, municipal, environmental, forestry and labour legislation. It is suggested that SERFOR, through its Forestry and Wildlife Technical Administrations and the regional forestry and wildlife authorities, provide technical assistance for the implementation of forestry legislation.

There are deficiencies in strategic planning for the sector (for instance, the process of drafting the National Forestry and Wildlife Plan began in 2017, but the Plan has not yet been approved or published), business intelligence, and business management of the timber industry. It is suggested that the State prioritize financial and human resources in finalizing and approving the National Forestry and Wildlife Plan, develop strategic plans for the timber sub-industries, and conduct business intelligence studies and market studies of the main products produced by the timber industry and products with growing current demand.

There is a provision of seasonal timber, with high species diversity and dispersal, and loss of timber source ecosystems due to increased deforestation. It is suggested to analyze the social, economic and environmental impacts of modifying the legislation regulating fast-growing species in order to make it more flexible, such as considering their replacement through reforestation. This would reduce the pressure on hardwood species that inhabit late secondary forests and primary forests that have higher biodiversity indices.

There is little development of forest plantations for commercial purposes and of sources of germplasm for plantations. It is suggested to invest in a national germplasm bank of timber forest species; make legislation more flexible so that forest plantations have the same regime as agricultural crops; ensure land tenure where plantations are established; implement long-term financial incentives, such as subsidies payable at the time of plantation installation, and long-term credits guaranteed by a state fund; and implement a research and technological development program for forest plantations.

The primary and secondary processing industry uses obsolete machinery, and lacks personnel with the ability to operate and maintain it properly. It is suggested to design and implement a program that allows MSME entrepreneurs to develop business plans in order to obtain financing for the purchase of machinery and equipment, and to participate in trade and technology fairs that allow them to identify the latest trends in machinery, equipment and products. With regard to training, it is suggested that incentives and recognition be given to entrepreneurs who train their workers.

The lack of quality control in the primary transformation industry process means leads to uncalibrated sections and cutting defects in sawn timber, with the result that the end product does not meet the quality required by the customer, and there is a high amount of waste and scrap. It is suggested that studies be carried out to determine the causes of variability in the thickness of sawn timber, within and between pieces, with the aim of establishing measures to solve these causes in order to reduce the total variation in thickness due to the effect of cutting in sawmills; and to implement a dimension control system for sawn timber boards in sawmills, which would enable producers to quantify the surplus of timber that is lost when cutting pieces with excessive dimensions and identify the causes thereof, in order to design action plans to reduce the variability in the dimensions of timber pieces.

It is suggested that the technical assistance and training provided by CITEmadera and CITEforestal and access to it should be strengthened. It is also necessary to strengthen the curricula for forestry science and forestry industry careers, as well as technical careers. New trends in international markets demand standardization and certification of the quality of processes, products, and labour training.

In the furniture and office and school furniture industries, studies are needed to identify timber species that meet the technical requirements for the manufacture of products. It is also necessary to standardize both processes and furniture pieces, conduct market studies on the products in greatest demand in this sub-industry, and invest in constant innovation in designs and finishes, according to the prevailing fashion trends.

There are deficiencies in the industrialization of waste from forest harvesting processes and from primary and secondary processing industries. It is suggested to promote, within the industry, lines of waste transformation where charcoal, sawdust and chipboard pellets, chipboards, wooden boards with finger joint joints, corner pieces for garden ornamentation, and branches (as pieces of wood) can be transformed to make handicrafts, jewellery or buttons. To this end, the state and international cooperation could invest in the establishment and operation of industrial parks equipped with a multipurpose wood processing plant in the timber production regions, which would benefit MSMEs and forestry clusters alike.

It is clear that far more private and state funding is needed in Science, Technology and Innovation. In addition to state funding, it is suggested that international cooperation funds be accessed in order to carry out bilateral research projects, training programs and exchanges of researchers with multiple countries. The research thus generated should be used to address the innovation and technology needs of the timber industry.

3. Textiles and Apparel

3.1 Textile and Garment Industry Trends

The textile industry is defined as the companies that carry out the manufacture of final products, including yarns and fabrics, by such processes as spinning, weaving, and dyeing. On the other hand, the garment manufacturing industry carries out operations related to the manufacture of finished articles, in this case garments. There are companies that are only dedicated to manufacturing products for the textile chain, and others that are dedicated to manufacturing garments, as well as those that are vertically integrated to offer the complete package.

However, Peru also exports cotton fiber products, mostly clothing and fine alpaca hair products, and fiber tops, yarns and fabrics.

There are two gaps in the industry: the first gap is in the technology used by exporting companies compared to that used by local producing companies, while the second gap is between companies with a vertically integrated textile chain and those that only manufacture clothing.

The distribution of textile companies in Peru has a very high component of micro and small companies, which affects the technological fragility of the sector.

As a definition, the manufacturing industry is an “industry dedicated to the transformation of different raw materials into products and finished goods for consumption or distribution to final consumers.”

The textile industry is the industrial sector of the Peruvian economy that is dedicated to the production of fibers (natural and synthetic), yarns, and fabrics, and to the manufacture of finished goods. The following is an account of the economic, social, environmental, and national context.

The textile industry is of great importance in the world economy and is a sensitive sector, especially when international trade agreements are defined. The textile industry is essential for the advancement and evolution of any society due to its great contribution to the manufacturing sector, especially for developing countries such as Peru.

The textile industry is characterized by the generation of direct and indirect jobs throughout the entire production chain, which is a considerable support for the improvement and maintenance of the economy, thereby contributing to the improvement of the quality of life of the general population. It is important to consider that the quality and the characteristics of jobs in the sector are diverse since there are differences between the labor needs of a large company and those of micro, medium and small businesses (MMYPES – "*Micro, mediana y pequeña empresas*").

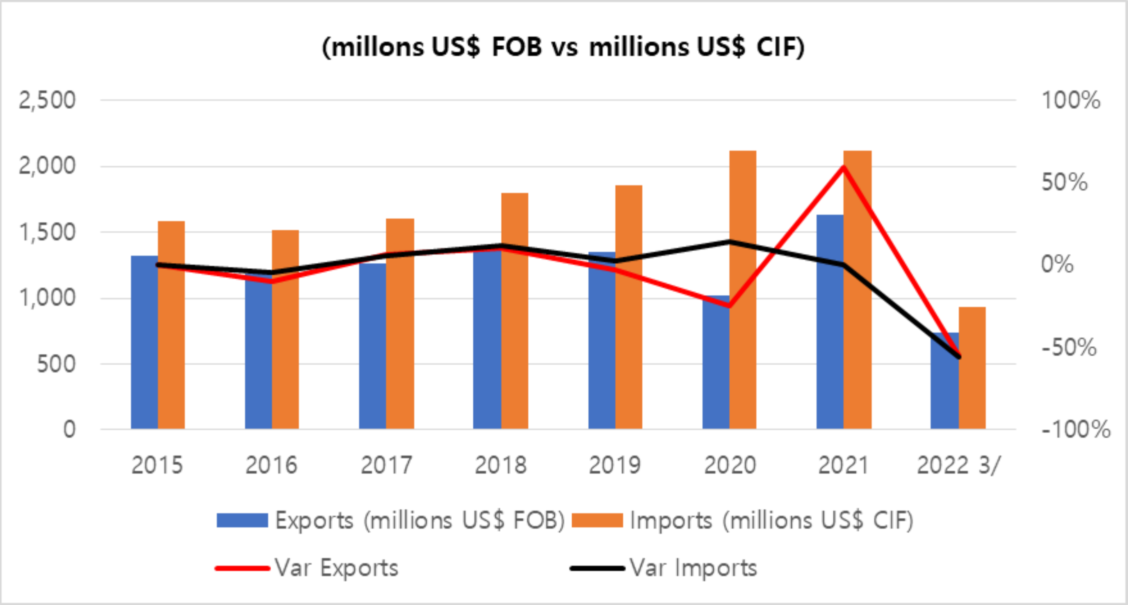
Regarding exports, the large companies are mainly vertically integrated, which offers a competitive advantage. Other favorable points are the quality of Peruvian textile fibers, such as long and extra-long fiber cotton, fine alpaca and vicuña hair. Finally, the geographical proximity to the world's main buyer, the United States, is another competitive advantage. Large exporting companies invest in high-tech machinery and equipment, as well as adequate personnel, in order to obtain the best benefits, and this is one of the characteristics that allow them to offer export products with greater added value.

As will be discussed later, a strong recovery in exports was expected in 2022 as a result of the reactivation of the sector. The main products exported are polo shirts, t-shirts and fine hair fibers. The main market is the United States of America, but there are also exports to other Latin American countries.

3.1.1 Economic Context

The variation of exports and imports shows that exports declined markedly in 2020, whereas imports maintained their growth trend that year, as shown in Figure 4-5. In 2021, both activities recovered, while the figures for 2022 were expected to be equal to or greater than those recorded in the years prior to the pandemic.

[Figure 4-5] Peruvian Exports vs Imports for the Textile and Clothing Sector



1/ Considers Div. 13: Manufacture of textile products. CIIU Rev.4.

2/ Considers Div. 14: Garment manufacturing; except fur. CIIU Rev.4.

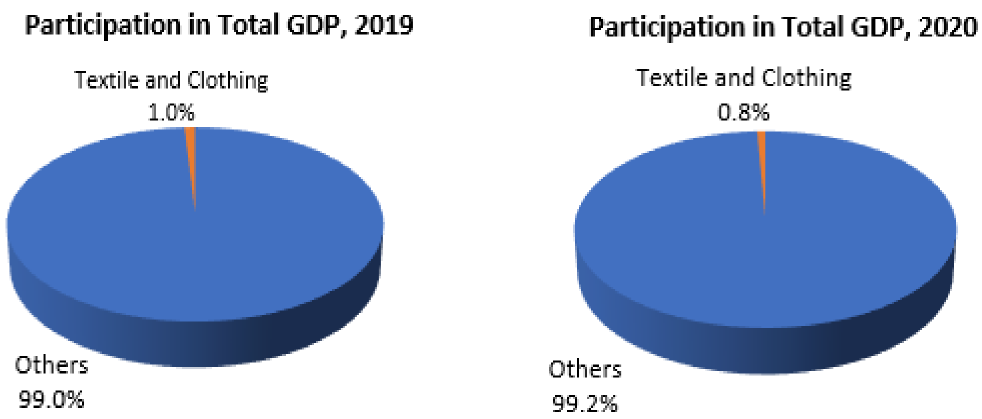
3/ Data collected until May 2022.

Source: *Ministerio de la Producción* (Ministry of Production; also known as *PRODUCE*).

3.1.2 Relevance of the Textile and Clothing Sector

Sectorial relevance can be valued according to a sector's contribution to the Gross Domestic Product (GDP) of the whole country, in this case the manufacturing industry. In 2019 the textile and clothing sector accounted for 1% of GDP, before falling to 0.8% in 2020 (See Figure 4-6).

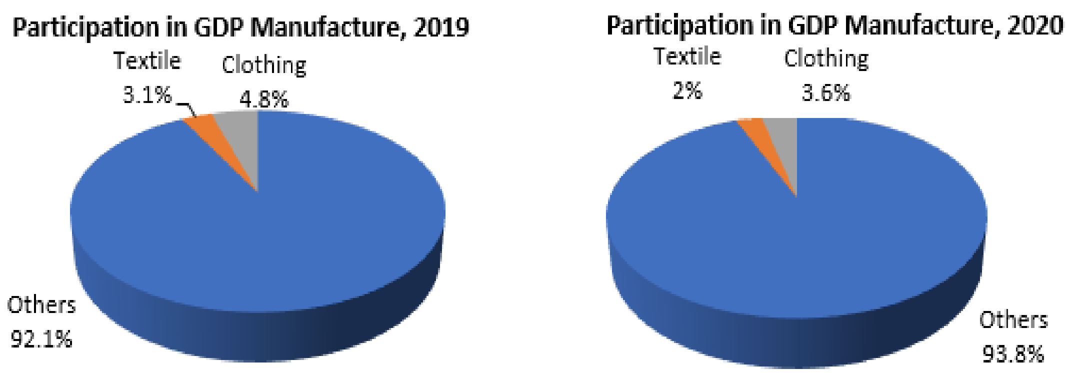
[Figure 4-6] Contribution of the Textile and Clothing Sector to GDP



Source: SUNAT (National Superintendence of Customs and Tax Administration) and PRODUCE.

Correspondingly, in 2020, the share of the manufacturing sector was 3.1% Textile and 4.8% clothing for 2019 and 2% Textile and 3.6% clothing, also showing a decline (See Figure 4- 7).

[Figure 4-7] Textile and Clothing Sector's Share of GDP

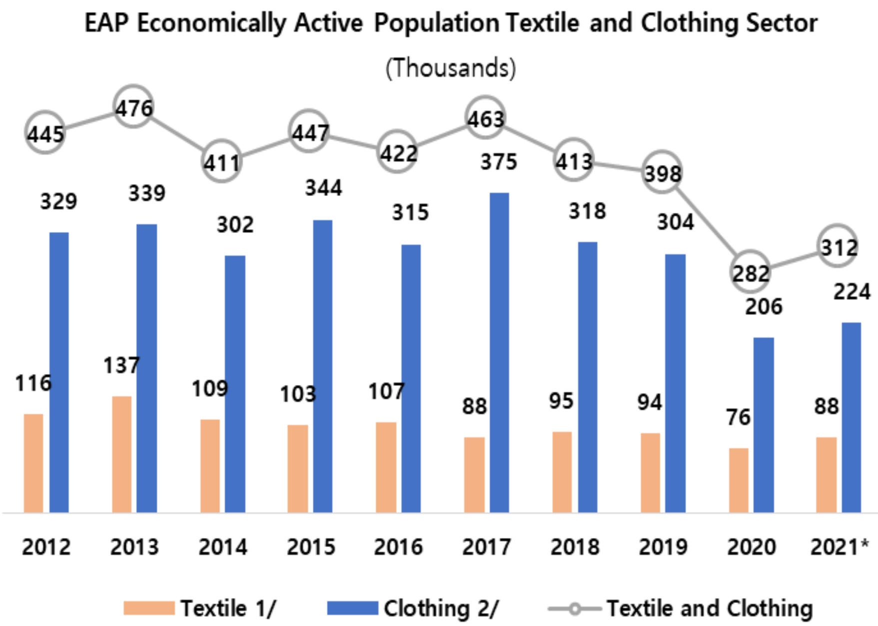


Source: SUNAT (National Superintendence of Customs and Tax Administration) and PRODUCE.

3.1.3 Labor

According to the Manufacturing Production Bulletin (published in May 2022 by the Ministry of Production) concerning the economically active population, Figure 4-8 shows that the clothing sector is labor intensive. However, the number of people employed in the textile sector had fallen before 2020, and showed a slight recovery in 2021.

[Figure 4-8] Economically Active Population in the Textile and Clothing Sector of Peru



(*): Preliminary

1/ Considers Div. 17: Manufacture of textile products. CIIU Rev.3.

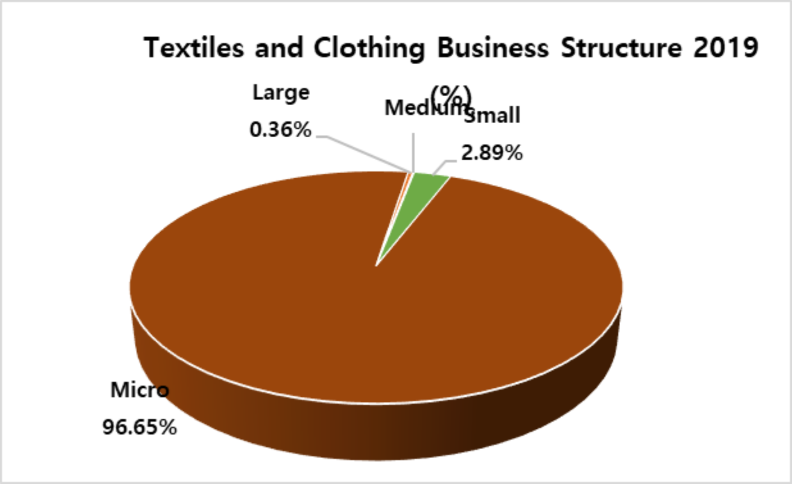
2/ Considers Div. 18: Garment manufacturing; processing and dyeing of fur. CIIU Rev.3.

Source: SUNAT (National Superintendence of Customs and Tax Administration), and PRODUCE.

3.1.4 Structure of the Textile and Apparel Business

As shown in Figure 4-9, the sector is mainly composed of micro-enterprises.

[Figure 4-9] Structure of Peru's Textile and Clothing Industry



* CIIU Rev.3. Div. 17: Textile products manufacture.

** CIIU Rev.3. Div. 18: Clothing; fur process and dyeing.

Source: PRODUCE.

3.1.5 Impact of Textile Protectionism

According to COMEX's Weekly Report No. 1113, a protective tariff is a temporary emergency measure that is applied to imports in the event of an unforeseen increase that could damage domestic production. Its use is exceptional, and there are strict requirements in both national and international regulations. According to COMEX, however, certain interested parties of the Peruvian government seek to apply safeguards to imports of clothing and apparel.

One of the effects mentioned is that the protectionist measures would harm everyone by increasing the cost of clothing imports, which in turn would increase the sales prices for the buyers. Another negative effect of such measures would be the impact on entrepreneurs engaged in the sale of imported clothing, making it difficult for buyers to purchase products.

Finally, the COMEX report stated that protectionism would not solve the problems of productivity, innovation, access to financing, the atomization of companies, or smuggling.

3.2 Technology Trends in the Textile Industry

During the 2022 edition of the XVIII Exporting Textile Forum, organized by the Association of Exporters of Peru (ADEX[26]), various foreign participants gave presentations highlighting the need to adopt innovation and technology to a far greater extent in the textile processes of Peru.

In recent years, Peruvian textile companies have been introduced to e-commerce, which has led to an evolution in their organizational perspective, receptivity to new business opportunities, greater international projection, and pursuit of competitiveness, highlighting the attributes of their products and their manufacture. As such, they now have greater awareness of world trends without having to attend events in person, and are able to implement new marketing strategies and strengthen the company's relationships with its customers through loyalty actions among other measures.

The adoption of digital technologies has simplified some processes and made certain activities previously only conducted face-to-face, such as fairs, meetings and international negotiations, more accessible. This in turn generates a significant cost reduction.

[26] A business institution founded in 1973 to represent and provide services to associated organizations such as exporters, importers, and service providers to trade.

The importance of technology in this sector is key to strengthening the brand identity that characterizes Peru, mainly in terms of cotton and alpaca, which have attracted attention around the world, especially in North America and Europe.

For several years now, the Peruvian textile industry has been going through a slow process of digital transformation due to the low connection of textile manufacturing devices based on IoT applications and only a moderate automation of production and logistics processes. The few technological innovations acquired from other countries create new business models and opportunities for textile companies. For example, digital textile printing has given businesses and consumers the ability to customize and produce consumers' own designs and ideas quickly and relatively cheaply. The Peruvian textile industry is under pressure to be innovative. Many Peruvian textile companies acquire R&D from developed countries, which limits the innovation process in Peru.

The Covid-19 pandemic accelerated the ongoing low digitization trends in the Peruvian textile industry. As a consequence of social distancing, communication within and between companies in the textile supply chain is done through digital means (presenting samples and innovations digitally). The pandemic also brought about urgent changes in communication, leading to more digital exchanges and the use of visualization techniques. The textile industry has played a fundamental role in the era of Covid by providing personal protective equipment with functional textile materials and finishes.

3.3 Major Challenges and Suggestions

1) Weaknesses

- Reduced attitude for the associativity of companies maintaining high fragmentation (more than 99% are MMYPES).
- There is a high level of labor informality in the sector, reaching 82.4% in 2021 according to Weekly Report No. 1120, May 2022, COMEX.
- There is little innovation mainly among MMYPES due to low investment and financing problems (also according to Weekly Report No. 1120).

- There is little knowledge of the competition and little research leading to strategies for accessing markets or marketing strategies.
- Production plants with working facilities have technical deficiencies.
- There is little access to financing for working capital.
- There is limited knowledge of international standards.
- Low productivity, low quality, and reduced responsiveness to customer orders.
- Access to supplies (i.e. fiber blends, special fabrics, trims, accesories, etc.) and intermediate services (i.e. washing, printing, etc.) is restricted.

2) Opportunities

- Preferential access to the world's largest buyer, the USA, characterized by high levels of demand and purchasing power.
- Good political-economic relations with the United States, as well as geographical proximity that facilitates near shoring.
- Access to unexplored market niches, especially products made from fine hair fibers and cotton with extra long fibers and fineness.
- The government has implemented programs designed to support MYPES in order to promote development; while the Ministry of Production (Produce) has created the Business Support Fund for the Textile and Apparel Sector (FAE-TEXCO) with the aim of guaranteeing credits for micro and small companies and productive units (prior formalization) for up to S/100,000. This program is expected to reach 60,000 entrepreneurs.
- Creation of new jobs and increased productivity through training and technical assistance provided to MYPES as a means to reactivate the textile sector, as part of a government program.
- International markets' recognition and positive assessment of cotton raw material with extra-long and long fibers, as well as fine hairs, and positive assessment of the quality of clothing manufacturing.

- Greater motivation to participate in trade fairs and commercial missions that introduce the best of Peruvian textiles to the world, and to obtain contacts and commercial contracts.
- Encouragement of Peruvian designers who include sustainability and aesthetics in their fashion collections, with Peruvian materials such as alpaca fibers, organic cotton, natural dyeing processes, and regenerated and recycled fibers.

3) Threats

■ **Reduced participation in international markets.**

- Exports are concentrated in cotton knitted shirts and T-shirts and, on the fine hair side, mainly carded or combed ribbons.
- High production costs reduce the competitiveness of the sector.
- Increased imports of raw materials and finished products from countries such as China, Bangladesh, etc. at very low prices.
- High levels of informality.
- Marketing of non-original brands at the local market level.

4) Strengths

- Tradition and experience in the textile and clothing sector.
- Labor availability.
- Peru has an abundance of natural resources and raw materials that are valued internationally.
- Close geographical proximity to the main buyer.

4. Mining and Its Manufacturing

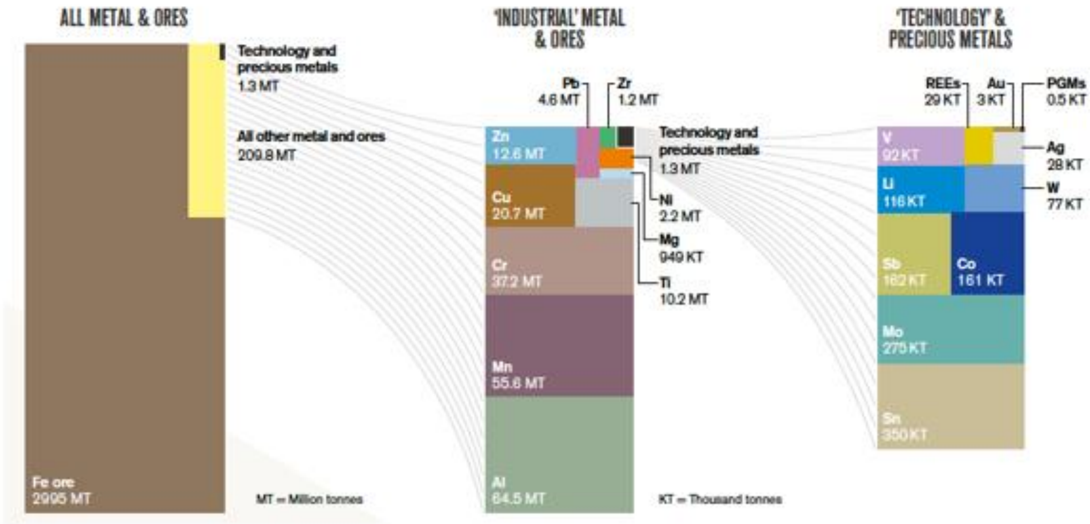
4.1 Minining Industry Trends

As Peru is a country rich in minerals and metals originating from the Andes Mountain Range, its economy revolves around the extraction of these resources. However, Peru does not have sufficient technological development to use these resources and transform them into final products. The technological process of transforming raw materials into final products or intermediate materials is not feasible in a country that is just starting out on the path to technological development. Furthermore, there are two steel producers with an extensive production of steel (SiderPeru and Aceros Arequipa) and a refinery belonging to the Southern Peru mine that manages to obtain copper with purity levels of 99. 7% and 99.99%. Therefore, a South Korea-Peru collaboration would be mutually fruitful if highly developed South Korea shared its technology with Peru, while Peru shared its mineral and metal resources with South Korea.

Peru is the world's second-largest producer of silver, copper, and zinc. It is also the first producer of gold, zinc, tin, lead, and molybdenum in Latin America; all of which originate from the Andes, the backbone of Peru and a primary source of minerals worldwide. Table 1 shows the production of metal in Peru.

Peru has recently discovered lithium reserves, including the deposit for the Corani and Macusani projects. The challenge is that lithium also contains uranium, so an investigation and research aimed at the efficient extraction of the two elements is required.

[Figure 4-10] Global Mining Production



Source: British Geological Survey, 2019; 2022.

Table 3 shows the ten criteria for classifying mining in Peru. It is crucial to consider this classification because it can give us an indication of what type of mining we should aim for in order to make this collaboration more efficient.

[Table 4-3] Peruvian Mining Classification

1. Type of activity	1) Exploration and exploitation 2) Benefit 3) General work 4) Mining transport
2. Nature of substances	1) Metallic 2) Non-metallic 3) Carboniferous 4) Geothermal 5) Oil 6) Precious and semi-precious stones
3. Form of deposits	**Some classify them as:** 1) Layers or mantles 2) Veins or seams 3) Irregular formations **Others classify them as:** 1) Alluvium 2) Rock vein 3) Layered sediment
4. Method of exploitation	1) Surface (open pit) 2) Underground or undercut.
5. Economic value of substances	Variable according to supply and demand
6. Location of Minerals	1) Of ground 2) Underground
7. Denounceability	1) They can be delivered in concession. 2) They cannot be delivered in concession.
8. Size	1) Large 2) Medium 3) Small 4) Artisanal
9. Type of production	Silver, Iron, Copper, Zinc, Lead, Gold, Tin, Others.
10. Legality	1) Formal 2) Informal: 3) Evasive 4) Elusive

Source: Ministry of Energy and Mines of Peru.

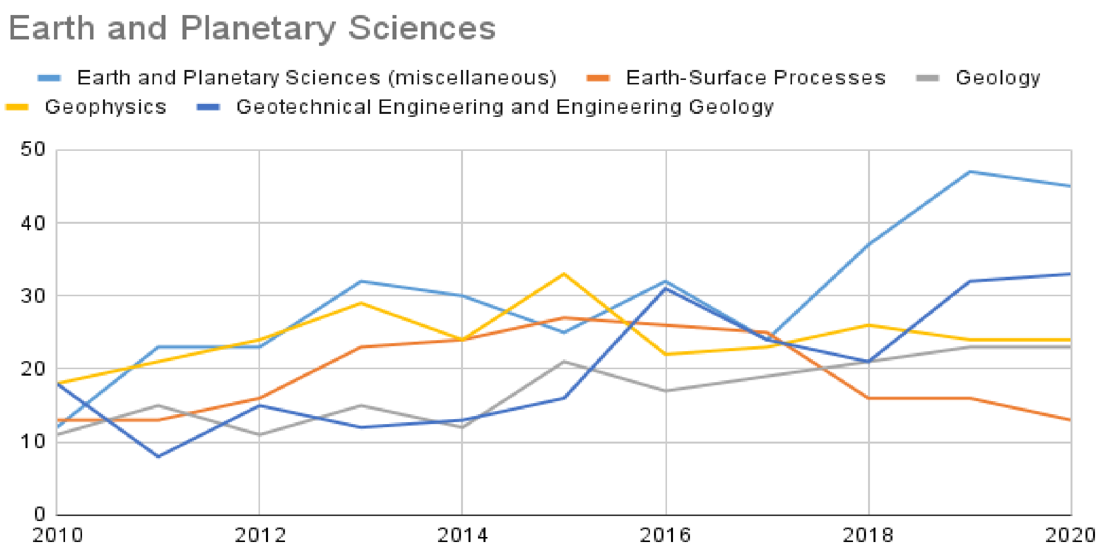
In the mining and earth sciences sector, there is still little development in terms of publications (Table 4-4 and Figure 4-11). The graphs show the number of publications in Peru and its ranking in the world and in Latin America. Regarding publications made in recent years in the field of earth sciences, geology, and related sciences, they contain research on various topics, such as the most efficient methods of extracting metals and minerals and reducing their environmental impact.

[Table 4-4] Scientific Production in Mining-related Subjects

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Atmospheric Science	12	14	15	24	22	19	25	19	28	38	32
Computers in Earth Sciences	0	0	1	2	1	3	1	1	4	3	3
Earth and Planetary Sciences (miscellaneous)	12	23	23	32	30	25	32	24	37	47	45
Earth-Surface Processes	13	13	16	23	24	27	26	25	16	16	13
Economic Geology	2	1	3	3	2	1	5	2	10	11	10
Geochemistry and Petrology	19	16	15	23	18	22	39	22	21	19	14
Geology	11	15	11	15	12	21	17	19	21	23	23
Geophysics	18	21	24	29	24	33	22	23	26	24	24
Geotechnical Engineering and Engineering Geology	18	8	15	12	13	16	31	24	21	32	33
Oceanography	17	16	12	15	23	22	14	32	30	43	38
Paleontology	7	10	11	11	19	23	9	14	14	18	10
Space and Planetary Science	16	17	14	19	18	17	22	25	23	27	25
Stratigraphy	1	1	0	0	2	1	3	8	4	2	2

Source: Scopus (direct access by CONCYTEC) Scientific publications per year in Peru.
https://portal.concytec.gob.pe/indicadores/principales/

[Figure 4-11] Number of Mining-related Scientific Publications per year



Scopus (direct access by CONCYTEC), Scientific publications per year in Peru.
https://portal.concytec.gob.pe/indicadores/principales/

4.2 Technology Trends in the Mining Industry

One crucial global trend is the use of renewable energies such as solar energy. In the case of mining, this is vital because of its benefits, such as reducing pollution and conflicts and offering clean electricity to communities affected by mining. Discoveries of abundant lithium resources and rare-earth elements are essential for developing batteries and electronics. These elements also bring the economic benefit of independence from other producer countries of rare earth elements such as China.

The global trend towards renewable energy sources. Is also significant because a significant part of Peru’s territory has a high level of solar irradiance. A solar panel installed in a high-irradiance area in Peru can produce more electric energy due to the higher amount of photons (light) per unit area. The following graphs show a global irradiance map, a map of Peru's irradiance, and a map of Korea's. Peru possesses excellent potential for the extraction of solar energy, a source of clean energy that can be used as a source of electricity for mining settlements and their surrounding communities. Solar energy could also be used to avoid using other non-renewable sources of energy that could contaminate the exposed territory.

Recent studies of solar systems installed in various areas around the country have revealed how they work under different territories. The study analyzes the production throughout the year. Korean solar cells could be used in such studies. Peru has excellent irradiance in many areas where there are metal or mineral resources (50% more irradiance than South Korea), allowing clean energy consumption for the mines and surrounding communities.

Investment in research and the construction of solar complexes will satisfy the demand for mining-related energy consumption.

Solar energy allows not only the production of electricity but also heat and hydrogen, two other forms of solar-based energy. These three forms have been studied over the years and are still hot subjects of research and development.

Rare earth elements were found at Huancavelica by Alturas Minerals and, in abundant quantities, at Chicama, while Cuzco holds the largest deposits of rare earth elements in Peru. These elements are strategic because of their diverse uses (catalysts, high-performance magnets, alloys, glasses, and electronics) and the current commercial dependence on China.

4.3 R&D Manpower and Facilities

It is difficult to estimate the country’s R&D manpower, but most of the facilities concerned are laboratories inside universities scattered around the country and in the capital. Lima, the capital, has many well-equipped aboratories, and the universities already have scientific publications in mining-related subjects.

The most qualified universities for mining-related research are as follows:

In Lima (capital):

- Universidad Nacional Mayor de San Marcos
- Pontificia Universidad Católica del Peru
- Universidad Nacional de Ingeniería
- Universidad Peruana Cayetano Heredia (health related)
- Universidad Peruana de Ciencias Aplicadas
- Universidad de Ingeniería y Tecnología

Outside the Capital:

- Universidad Nacional de San Agustín (Arequipa, Peru)
- Universidad Nacional San Antonio Abad del Cusco (Cusco, Peru)
- Universidad Nacional de Trujillo (La Libertad)

These universities are the ones with the most scientific production in 2021 and 2022. They have expertise in scientific research and also collaborate with prestigious overseas institutions and universities in the United States, Germany, and Switzerland among others.

4.4 Mid- and Long-term Industrial Strategies

The strategy for success in R&D is as follows:

A. In accordance with the communities, it is necessary to identify the needs that will be satisfied by a R&D project. For example, the need for energy, contamination control, clean and potable water, higher purity of the metals/minerals, etc.

B. In collaboration with local universities or institutions, it is necessary to planify the R&D project. It is crucial to determine whether it is feasible in advance. For this, it is necessary to have knowledge and technology. For example, some research needs special equipment that Peru's local universities do not possess, such as an X-ray diffraction spectrometer

(XDR) or energy-dispersive X-ray spectroscopy (EDX). These types of equipment can be widely used in scientific research on the elements and compounds found in soils.

C. Experimental: It is necessary to develop the project with measurements and essays within a set period of time according to the nature of the R&D project.

D. Conclusion: Analysis of the experimental results, scientific publications, and final reports.

4.5 Major Challenges and Suggestions

- In the special case of mining, the major challenge is not technological; rather, it is diplomacy. Concluding good agreements with the local communities is the starting point for executing R&D projects.
- Geological diversity will be taken into account, as new geological conditions will appear to foreign researchers.
- The transportation of metals and minerals across rugged terrain (mountains, rivers, forests, etc) should be planned carefully.
- In some cases, food, the high land effect (more than 4000 m.s.n.m), and water could be a real challenge.
- In general, logistics are crucial to planning the efficient use of resources.
- The use of solar energy will provide a good incentive for negotiations with the communities

5. Advanced Manufacturing

5.1 Advanced Manufacturing Industry Trends

Throughout the previous century, important industrial developments redefined technology and reshaped our way of life. These events have been characterized as “industrial revolutions” which have shaped our country's technological reality over time. Manufacturing is the process of converting raw materials and resources into completed, saleable commodities using the components of engineering and industrial design. It also includes the process of producing semi-finished goods. Manufacturing is connected with a wide variety of human activities, from handcrafts to high tech, but it is most usually applied to industrial production, which involves the large-scale transformation of raw materials into finished commodities.

Manufacturing is often focused on the mass production of goods and all intermediate procedures necessary for the creation and integration of a product's components. Historically speaking, a specialized approach is used to achieve a high output when working on a specific component. This is a procedure in which each worker does only a tiny portion of the entire task, allowing the worker to specialize, and the economy moves in order to increase ther speed of production. Manufacturing accounts for a sizable share of the global economy and is a wealth-creating component of any national economy, while the service industry is a wealth-consuming sector.

The current state of the Peruvian economy in terms of technology, production standards, and access to information or data is still quite precarious in some sectors, making it difficult for businesses to increase productivity and compete in both the domestic and global markets. As a result, it is necessary to optimize production processes, and many businesses are currently engrossed in a quest to find a solution. Manufacturing industries in Peru must seize the development potentialoffered by the Fourth Industrial Revolution. The COVID-19 pandemic, despite causing significant problems and creating ongoing difficulties for several industries, should be viewed as an opportunity for businesses to continue growing.

The evolution of industries has shifted dramatically over time. We live in a dynamic and unpredictable economy in which we must continually stay ahead of the competition by adapting to new difficulties. Technology is continually evolving and changing, and a new era is fast approaching for businesses and economies, in which the combination of digitalization, connectivity, and data are important pillars. The COVID-19 pandemic has affected all countries and economies and has had a huge impact on all industries, especially in Latin America. According to a CEPAL report, 92% of the industries in Latin American are considered to be facing a significant crisis and could collapse in the near future if adequate actions and policies are not implemented. Under this scenario, technology is a key factor in finding solutions to this crisis, so using the concepts and technology of Industry 4.0 is essential.

Industry 4.0 is a concept that was born in Europe and is synonymous with the Fourth Industrial Revolution. Industry 4.0 refers to the technological and digital transformation of industrial processes to create more efficient and autonomous processes. It is characterized by the connectivity of and communications between all elements that provide information in real time and which can be processed and analyzed for automated decision-making. The Industry 4.0 trend is based on other important, game-chaning technological concepts such as the Internet of Things, machine learning and deep learning, Big Data, cloud computing, wireless networks, cybersecurity, embedded systems, mobile devices, and autonomous robots among others. In Peru's industries, most of these concepts are used, but often in isolation. The World Economic Forum explains how Industry 4.0 is a new concept that is growing exponentially, affecting all productive areas and involvomg not only the transformation of economies but also societies, in which everything is connected, thereby generating data that can be used to improve manufacturing and production processes among many other things.

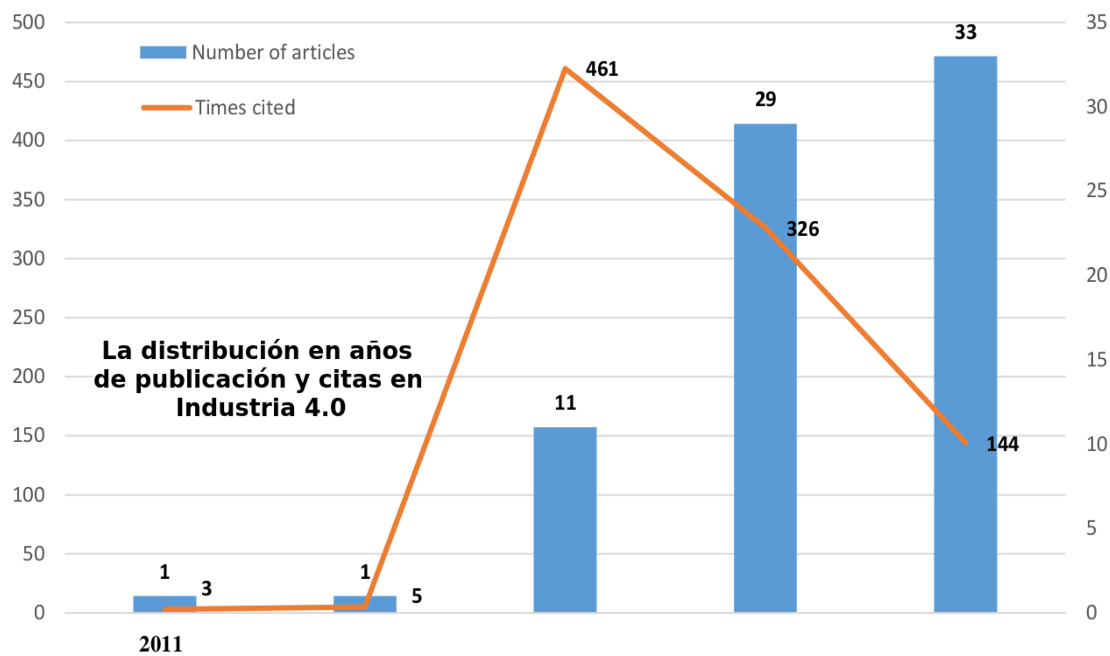
A country achieves industrial competitiveness when there is an increase in productivity based on high-quality human capital, knowledge, and high technology. Furthermore, this can be complemented by other fundamental characteristics, such as the strength of its institutions, accessibility, and the quality of its infrastructure, and it is supported by the commitment to implement a unified comprehensive strategy. The 2021 Peru Bicentennial Plan was established, along with a competitiveness agenda designed in 2014, in order to articulate the actions and factors required to boost the nation's competitiveness. This agenda contains sixty-five prioritized goals in the eight strategic lines, which were defined in conjunction with the public sector, private sector, and academia, as presented below:

I. Productive and business development
II. Science, technology and innovation (STI): increase innovation and knowledge generation capacities, promote alliances between actors, and attract and retain talent.
III. Internationalization of companies: increase access to international markets by articulating global value chains, and facilitating trade and the development of sustainable markets.
IV. Infrastructure: cover the deficit in logistics and transportation based on the generation of articulated logistics service poles that promote regional economic growth.
V. Information and communication technologies (ICT).
VI. Business facilitation.
VII. Human capital: improve the qualifications of the workforce as the basis for increasing labor productivity, wages, and consumption. Efforts will focus on high-quality technical and university training appropriate to the needs of Peruvian industry.
VIII. Natural resources and the environment: optimal usage of energy and resources to ensure sustainable growth, through efficient quality management in the access and use of the resources.

According to the Boston Consulting Group (BCG), there are nine technological trends that form the building blocks of Industry 4.0. The use of big data and analytics in the comprehensive collection and evaluation of data from many different sources is becoming the standard form of support for real-time decision making. Autonomous machines interact with each other and can safely work alongside, and learn from, human beings. The industrial Internet of Things allows more embedded systems to communicate and interact with each other and with more centralized controllers as needed. Meanwhile, the Cloud helps many companies that require more data exchanges between sites and company borders. As a result, machine data and functionality are increasingly be deployed in the Cloud, enabling more data-driven services for production systems. Companies are embracing additive manufacturing, such as 3D printing, which they use primarily to prototype and produce individual components, and augmented reality-based systems, such as picking parts from a warehouse and sending repair instructions via mobile devices.

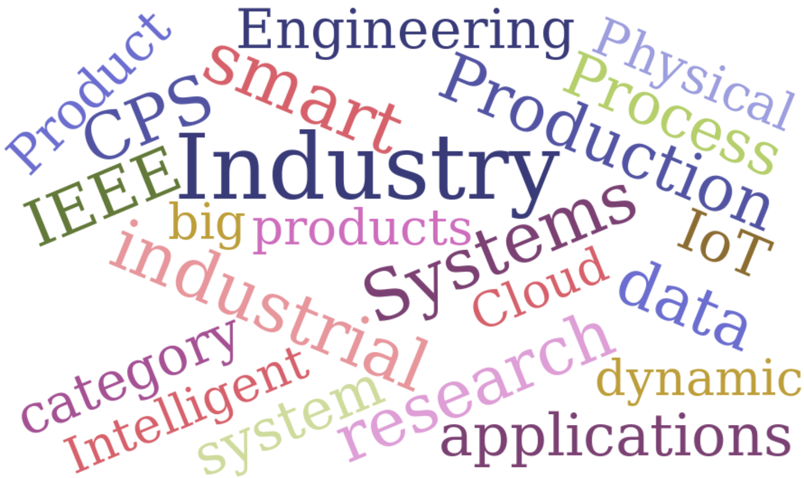
An exhaustive examination of the existing literature on Industry 4.0 and the number of publications has revealed massive interest in publishing research related to Industry 4.0. The Wordcloud (Figure 4-13) below presents the most frequently found terms in most manuscripts, and the relative importance of Industry 4.0 concepts in industrialization processes worldwide is clearly observed.

[Figure 4-12] Distribution of Publications and Citations in Industry 4.0, 2011-2015



Source: Yang Lu (2017).

[Figure 4-13] Wordcloud of Most Frequently Used Terms in Industry 4.0 Publications



Source: Wordcloud (2022)

5.2 Mid- and Long-term Strategies

As the industrial activities in Peru are very diverse, it is important to group them into principal categories that give an overview of industrial enterprises and how they are used. In this classification, the industries that require advanced manufacturing and technologies are identified. Currently, the primitive one is still based on mechanical systems operated with manual labor, while others use assembly lines driven by electric motors, along with intensive manual labor to connect pipes of production, and only a very small number have production processes based on automation and control systems.

Table 4-5, obtained from the available INEI industrial information, shows that most sectors require Industry 4.0 concepts. The table shows that if concepts from advanced technologies are employed, it will improve the productivity and efficiency of processes, services, and products. However, the adoption of this technology is prevented by conventional thinking, i.e. "If it ain't broke, don't fix it", and because new technology could be expensive. If Peru's industries, including logistics and technology, never try to improve or innovate, they will eventually become outdated even though they may be working well at the moment. This is especially true for industries that can be optimized and improved;

however, it is difficult to know when changes need to be undertaken. Changing such paradigms is difficult as they may still hold in very limited cases, so it is necessary to prove that the concepts of Industry 4.0 are the keystones of sustainable prosperity in a continually evolving economy.

[Table 4-5] Industries that require advanced manufacturing in Peru

	Industries that require advanced manufacturing in Peru
1	Food and beverage products
2	Products containing tobacco
3	Textile items, garments, leather processing, and dyeing
4	Processing of wood and derivatives
5	Recordings including editing, printing, and play
6	Products derived from petroleum and gasoline refining
7	Substances and chemicals
8	Products made of rubber and plastic
9	Non-metallic mineral products
10	Base metal processing
11	Electrical and mechanical equipment
12	Manufacturing of medical, optical, precision instruments, and clocks
13	Vehicle parts

Source: National Institute of Statistics and Informatics (INEI).

5.3 Major Challenges and Suggestions

Peru has a medium-sized manufacturing industry which accounts for 23% of GDP and is strongly dependent on mining, fishing, agriculture, building, and textiles. Peru's manufacturing sector is primarily concerned with processing in order to acquire a competitive edge. Textiles, metal mechanics, food and beverages, agriculture, manufacturing, chemicals, pharmaceuticals, machinery, and services are the most promising industrial sectors. A major problem in Peru is that the majority of manufacturers are situated in the region around Lima, the capital, which slows down industrial development due to the heavy concentration of resources. Therefore, the government should focus on spreading industrial output by providing incentives and strongly encouraging companies to use the nation's natural resources effectively and achieve self-sustaining growth across the country. Peru also has to diversify its economy, upgrade its infrastructure and manufacturing, and allocate more funding for the growth of non-coastal industries if it wants to maintain a sustained rate of growth.

Developed countries characteristically have highly advanced industries and provide the world with high-tech products and services. The Fourth Industrial Revolution, or Industry 4.0, has arrived and is based on the use and application of artificial intelligence in almost all processes. Artificial intelligence (AI) is an old idea, but a set of important technologies is emerging from it, such as big data, cloud storage, and decision computation. Here, computers and automation will come together in a whole new way, with robotics remotely connected to computer systems equipped with intelligent algorithms that can learn and control them with very little human input. This is transforming production and changing the relationships between suppliers, producers and customers, as well as how humans and machines interacts with the world.

This industrial transformation makes it possible not only to collect information but also to analyze data in all machines, allowing faster, more flexible and more efficient processes to produce products and services of ever higher quality at a reduced cost. This manufacturing revolution is changing the economy, fueling industrial growth and reshaping the workforce, ultimately changing the competitiveness of businesses. Industry 4.0 promotes liberating technologies that will reshape all industry, and the companies that adopt it will benefit greatly from it and become more innovative, creative and adaptive.

Peru is promoting and moving toward digital transformation by allowing access and use by all its citizens; for instance, the numbers of active mobile and fixed broadband subscribers have risen over the last decade, while the number of students per computer decreased over the three-year period from 2.2 to 1.4 in 2018, which is similar to the average figures for Latin America and the Caribbean (LAC). Peru's E-Government Development Index increased to 0.65, ranking 8th in Latin America and the Caribbean (LAC) in 2018, but it still lags some way behind the top country, Uruguay, which has an index of 0.79.

Peru is still characterized by having an economy based on the export of raw materials, majoritarily minerals, followed by agricultural products and other commodities whose prices are low due to their low added value. To give added value to these raw materials, Peru needs modern and competitive industries. As we can appreciate from developed countries; the only way for a country to develop is to pursue industrialization in all areas of the economy without exception, creating industries that can add value. An ever growing number of vertical industries around the world are adopting the technologies, concepts and principles of Industry 4.0; however, if they are to be incorporated into Peruvian industry, all its companies will have to overcome challenges in the use of this technology. Entire industries are being destroyed by new technologies and business models; thus digital acceleration should be a priority for many Peruvian companies. To create a competitive industry, Peru must create production chains to increase employment and salaries and lessen the heterogeneity of the economy; and, second, it must create plans for the internationalization of companies focusing on export activities. Importantly, the pandemic has helped e-commerce to grow by 240 % and is likely to remain at this level or increase further still in the near future.

To achieve and maintain leadership in the race to build sustainable economies, industries need well trained personnel with a broad and deep working knowledge of digital technologies and related use cases in order to develop and implement digital manufacturing strategies tailored to Peru’s needs. Therefore, it is vital to promote Industry 4.0 for the the development of the country and the creation of products and services with added value.

6. Ecotourism, Restaurants, and Creative Industris

During the last few pre-pandemic years, Peru's economic growth in Latin America had attracted the attention of many experts. According to the World Bank (WB), this growth was due to solid macroeconomic fundamentals such as the country's relatively low debt relative to GDP, international reserves, and a strong Central Bank. However, the Peruvian economy contracted for the first time in over a decade in March 2020 as a result of the Covid-19 pandemic, shrinking by 16% compared to 2019 (Quigley, 2020). Faced with this situation, the Peruvian government proposed the gradual reactivation of activities divided into four phases, after considering a previous analysis that took into account the opinion of the Ministry of Health (Minsal), the evolution of the epidemiological situation, and the response capacity of the country among other factors (Supreme Decree No. 157-2020-PCM). It is relevant to mention here that some economic sectors have been hit harder than others, due to the very nature of their activities, such as the ecotourism, restaurant and catering, and creative industries. Although these sectors make an important contribution to GDP, they have experienced a very slow recovery due to the uncertain health situation, the articulation of proposed measures, and the limitations of the permitted capacity; in fact, in the fourth phase of reactivation they were only allowed to complete 50% to 60% of the total capacity allowed, resulting in a fall in income for employers. This report presents an analysis of the data collected for the planning of the technological forecast and the technological roadmap for the ecotourism, restaurant and catering, and creative industries in Peru.

6.1 The Ecotourism Industry in Peru

6.1.1 Ecotourism Industry Trends in Peru

Peru is considered one of the countries with the greatest diversity of species on the planet. It has an immense variety of fauna and flora, an endless number of butterflies and orchids, and boasts a stunning diversity of birds, mammals, fish, and reptiles. It is the third largest country in the South American continent, as well as ranking fourth in the size of its tropical forests and number of amphibian species, fifth in its species of flora and fauna, and ninth in the forestry sector. Peru’s tropical forests provide a habitat for twenty-five thousand species of plants, the greatest diversity of species, besides being a global center of endemism and genetic resources (Barrutia, 2019).

For poor local communities, ecotourism represents a fundamental way out of the limited possibilities of their customary way of life, and favors maintenance and exit from the circle of poverty (Martínez, 2018). It is said that eco tourism enables the responsible use and revaluation of the natural and cultural resources of a given territory, as these, in addition to possessing a symbolic and intangible value, constitute an important economic and socio-cultural asset of the first level for the communities that hold them.

Forests provide diverse ecosystem-oriented goods and services that are fundamental for the well-being of the population and the development of the national economy. They are considered part of the country's natural capital and are exploited through the provision of goods (e.g. timber or firewood extraction, non-timber forest products, medicinal plants, etc.) or through the provision of regulating services (e.g., carbon sequestration or water regulation), cultural heritage (e.g. scenic beauty) and support services (e.g. promoting the nitrogen cycle), according to the Millennium Ecosystem Assessment Program (2005) and the Ministry of Environment (Minam et al., 2015).

In this sense, forests contribute to the production and economic development of a country, in addition to playing a fundamental role in mitigating and adapting to climate change (INEI, 2021).

The tourism in Peru was greater and the recovery slower compared to the rest of the world due to its stricter and longer sanitary restrictions (Macroconsult, 2021)

By type of visitor, the internal flow (resident travel) presented a better recovery performance, in line with the easing of restrictions, compared to the number of arrivals of travelers (tourists and excursionists), which continued to fall in 2021, representing only 8% of the figure recorded in the pre-pandemic period (Macroconsult, 2021)

On the other hand, the Ministry of Culture (Mincul) has reported that the Machu Picchu llaqta expects to receive more than 1,111,000 visitors in 2022. It should be recalled that the admission capacity to the Historic Sanctuary of Machu Picchu was recently increased to 4,044 people per day, a figure that includes admissions via all circuits and at all times (Gob, 2022)

Another important ecotourism site close to Lima also saw a drop in the number of visits. During 2020, visitor arrivals to the Paracas National Reserve amounted to 227,760 (-53.3% variation), of whom 86% were national visitors and 14% wereforeign visitors (Ministry of Culture, 2020)

Despite the progressive recovery of the Peruvian economy, tourism operators are still experiencing a slow recovery of their activities because Peru is still facing an epidemiological situation that has led the authorities to maintain certain restrictions on economic activities in an effort to prevent and reduce the high rate of mortality and lethality caused by Covid-19 (Cdn, 2022) (See Figure 2.5).

[Figure 4-14] Activities to Be Carried Out in 2022

What activities will you offer in your tour packages? What are your recovery expectations for 2022?

***46%** Cultural tourism (31% of the packages offered in 2019).*

***29%** Nature tourism (31% of the packages offered in 2019).*

***11%** Adventure tourism (31% of the packages offered in 2019).*

***4%** Community tourism (29% of packages offered in 2019).*

***4%** Sun and beach tourism (48% of the packages offered in 2019).*

***3%** Wellness tourism (37% of packages offered in 2019).*

***3%** Fun (33% of the packages offered in 2019).*

Source: PromPeru (2022).

Peru also foresees a greater dynamism in its tourism sector this year, and to achieve this it has implemented various measures, one of which was the publication of the Emergency Plan for the Tourism Sector, which contains a series of sectorial measures designed to mitigate the serious crisis that tourism in Peru is going through as a consequence of Covid-19. This plan foresees that the country will receive more than 1,000,000 international tourists and 25,000,000 domestic trips, and directly employ 900,000 people in the tourism sector in 2022. To this end, it identifies more than sixty priority actions for coordination between the three levels of government and the private sector (El Peruano, 2022).

Peru stands out as a key destination for ecotourism because it is one of the most biodiverse countries in the world, according to Pablo Díaz (Andina, 2022).

According to the Ministry of Foreign Trade and Tourism (Mincetur), there are 158 Protected Natural Areas (ENP) in Peru, representing 16.93% of the country's territory. The ENPs are divided into the following categories: National Parks, National Reserves, Communal Reserves, National Sanctuaries, Historic Sanctuaries, Landscape Reserves, Protected Forests, Wildlife Refuges and Hunting Reserves. The site which attracted the most visits during 2018 was the Machu Picchu Historic Sanctuary, which has been declared one of the “Seven Wonders of the Modern World” (Andean, 2022).

6.1.2 Major Challenges and Suggestions

Peru faces great challenges in the sector of rural tourism in general, which was greatly affected by the Covid-19 pandemic, but there are also opportunities for consolidated tourism experiences, such as community-based tourism and ecotourism promoted by the private sector with government support, but above all for enjoyment of the outdoors.

In rural tourism, various strategic lines of action can be adoped to boost rural tourism, community-based tourism, agrotourism, and so forth. These can be developed through public-private initiatives. Tourism as an engine of rural development has been proposed by various governments around the world (Santos-Lacueva). Santos-Lacueva also indicated that one of the challenges facing rural tourism in order to continue to grow is sustainability. When reconfiguring strategies to adapt to the current situation, it is necessary to "improve the supply, the model, the products and the experiences, by incorporating sustainability criteria.

Santos-Lacueva has pointed out that now is the ideal moment to promote rural tourism among local and nearby travelers, because there is an immense potential in all the people who travel far and wide, but who neglect to get to know everything that is close to them. As great opportunities to boost rural tourism in Peru, he mentions the following:

- Consolidate the sustainable development of rural tourism: First, an analysis should be made of how sustainability criteria are incorporated into tourism; then, strategies should be developed to provide greater guarantees of sustainability and future sustainability of tourism activities in a post-Covid-19 scenario in rural areas, taking into account environmental, social and economic factors.
- Publicize the offer of community-based tourism and agritourism: Measures must be designed to enhance the value of rural tourism attractions and encourage the arrival of local visitors and other types of tourists who, under normal health and mobility conditions, might choose other types of destinations.
- Strengthen the governance of tourism destinations: The recent crisis has generated cooperation and collaboration among many tourism stakeholders. These alliances, now essential, must seek ways to strengthen these relationships beyond the health crisis.
- Training: During the health crisis and the closure of tourism businesses, there was a perfect opportunity to improve the training of all agents in the sector. Proper tourism training results in better experiences for clients and greater competitiveness for destinations.

On the other hand, according to *The Chamber* (2020), to guarantee the sustainability of rural tourism in the post-Covid-19 scenario, the strategies to be implemented should take into consideration the following challenges:

- More demanding: The global coronavirus crisis has reaffirmed and intensifiedthe trend of more demanding and committed tourists, for example, with regard to the environment, waste generation, and social responsibility. In addition, the demand for strict hygienic and sanitary standards will increase.

- Safety of the local population: The adequacy of safety protocols and communication strategies will be crucial to ensuring that the local population does not perceive the arrival of tourists as a risk to their health.
- New social values and needs: Isolation and confinement at the global level have brought with them new needs and priorities among the population. The value of all things human, of the little things and of emotions, must play a central role in the rethinking of rural tourism in the face of Covid-19.
- Tourism knowledge and research: The challenges facing tourism will increasingly require more expert and multidisciplinary knowledge. Resources must be allocated to generate knowledge and research to improve decision-making and, consequently, the competitiveness and sustainability of the destination.

6.2 The Restaurant and Catering Industry in Peru

6.2.1 Restaurant and Catering Industry Trends in Peru

As for tourism-related activities, restaurants recovered to their pre-pandemic levels by the end of 2021, mainly due to home delivery services and extended opening hours (Macroconsult, 2021)

The economic activities that generate the greatest value added within the informal sector are agriculture and fishing, transportation and communications, lodging and restaurants, and commerce (INEI, 2021)

The restaurant subsector shrank by 33.01% in November 2020, when compared to the same month of 2019, a decline which can be explained by a report on the downward trend in all food and beverage vending items, released by the National Institute of Statistics and Informatics (INEI, 2020)

Sales in these businesses were affected by temporary restrictions due to the state of health emergency decreed to reduce the spread of the pandemic (INEI, 2020).

The INEI reported that the restaurant sector shrank by 31.98% due to the lower dynamism registered in tourist restaurants, meat and grill restaurants, Creole food restaurants and sandwich shops, due to the reduced flow of customers, the continuous closure of establishments, and the limited capacity of diners, as well as changes in the business lines of some establishments. Furthermore, ice cream parlors, restaurants specializing in international cuisine, pizzerias and candy stores were also affected during the national sanitary emergency (INEI, 2020).

In November 2020, the category of other food service activities showed a variation of -27.64% due to the reduction of contracts signed by food concession companies with manufacturing companies, public and private entities, and mining and fishing companies, resulting from the reduction of working hours and shifts or the closure of branches at universities, colleges and institutes. Additionally, the supply of meals for contractors (food services to transportation companies) decreased due to the lower influx of travelers via air and land transportation companies (INEI, 2020).

In November 2020, beverage services declined by 51.13% due to the contraction of demand in bar-restaurants, cafeterias and juice shops. As for discotheques, bars and pubs, they remained closed as a control measure amid the health emergency (INEI, 2020).

In the study month, the supply of meals on demand (*catering*) decreased by 64.76% due to lower contracts for food preparation and distribution services for social, public and private events, as well as buffet and banquet services (INEI, 2020).

During the same period, restaurant and mobile food services grew by 68.73%, determined by the favorable performance of poultry restaurants, restaurants, fast food joints, chifas, coffee restaurants, cevicherías, tourist restaurants and meat and grills, as a result of the extended opening hours, greater coverage of delivery services, expansion establishments' capacity, renewal of menus (entry of new dishes), increased digital adaptation, and implementation of marketing strategies.

6.2.2 Major Challenges and Suggestions.

The gastronomic sector was one of the hardest hit by the arrival of Covid-19, resulting in the closure of almost 50% of the restaurants, as reported by the Peruvian Association of Hotels, Restaurants and Related Industries (Asociación Peruana de Hoteles, Restaurantes y Afines). However, the economic reactivation of the country and the easing of restrictions were presented as a light at the end of the tunnel, but also as a challenge for restaurant owners (Makro, 2022).

1) Challenge 1: Technology is here to stay!

Digital transformation became a priority during 2020 and 2021. As 73% of the population now have internet access according to the INEI, and given that there has been a 663% increase in the use of digital payment platforms, the challenge for restaurants is to take action to incorporate technology into their dining experience. This will not only help the business, but also leave customers with an impression of modernity (Makro, 2022).

Restaurants must continue seeking alternatives that will improve the customer experience through the use of technology. The implementation of digital menu cards, online reservations, the development of web pages with restaurant information, and the prioritization of electronic payments should not be omitted.

2) Challenge 2: Consumer habits continue to change!

The two and a half years of the pandemic have shown that customers' consumption habits evolve and adapt quickly. As people spent more time at home, they changed their relationship with food, became more aware of the importance of healthy eating habits, and started to look for better alternatives to their usual dishes.

The challenge for restaurants today is to incorporate more options into their menu to fill this gap in their gastronomic offer, without sacrificing their essence or negatively impacting their budget. A good way to start would be to replace the ingredients in their preparations or try versatile flavors and dishes that make better use of the inputs (Makro, 2022).

3) Challenge 3: Competition is increasing !

The closure of restaurants, restrictions on opening hours and seating capacity, and new consumer habits have increased competition. How can I get a customer to choose my restaurant and not the one that offers the same food next door? How can I win over the diner who has not dined out in more than a year to try my new restaurant as a dining experience?

While they remained closed, some restaurants put together proposals "for home," turning their menus into packs with dishes that can be prepared, without much difficulty, in their kitchens. This is an innovative way to share experience and build customer loyalty.

Restaurants need to think outside the box. But to do this well, it is good to train, research public preferences and new market trends, and design an all-encompassing experience. Today, customers are looking for more than just good food: they want a relationship with excellent service that makes them feel special, comfortable and safe within the restaurant space (Makro, 2022).

4) Challenge 4: The new face of delivery!

The pandemic consolidated among the public a new way of eating with convenience: home delivery. For example, from February 2020 to February of the following year, home delivery services grew by 200%. Can there now be any doubt about its relevance? However, the great challenge for restaurants at the time was the cost of implementation and logistics, as the processes were not defined; but home delivery became, for a while, their only source of income.

Given that home delivery is here to stay, it is up to restaurants to look for ways to turn it into a service that adds to their premises, evaluating costs and improving the experience to build loyalty among the public that prefers to stay at home. Exclusive promotions and team training to streamline the delivery process are some tips (Makro, 2022).

5) Challenge 5: Think more digital!

The predominance of digital can be seen not only in the way service is handled in the restaurant itself, but also in the online world. Social networks have become the new meeting point, where reviews, stories and photos converge to determine the brand image and its reach to potential customers. The generation of content, also made by the diner him/herself, is more relevant than ever.

It is desirable, then, to maintain an active profile on these platforms, establish a solid communication strategy with quality contents and engage in constant interaction with followers. In this way, not only will restaurants have greater knowledge of what people are looking for, but they will also create a comprehensive gastronomic experience even before an in-person visit (Makro, 2022).

6.3 The Creative Industries in Peru

6.3.1 Creative Industry Trends in Peru

1) Theater

The National Survey of Budgetary Programs (Enapres) defines a theater show as a live presentation of a dramatic work, performed in public or open spaces, where there is a link between artist and spectator. In addition, paid and unpaid theatrical shows are considered, as well as shows performed in educational institutions, as long as the show is not developed by students (Zúñiga, Uzuriaga & Sacramento, 2021).

In 2019, the ENAPRES was developed. Zúñiga et. al. (2021) pointed out that the results indicated that 10 out of every 100 Peruvians over 14 years of age attended at least one theater show per year. In 2016, two million people attended the theatre, and by 2019 this figure had risen to 2.4 million people.

According to this survey, for every ten people who attended a theater show, 40% attended one of these events on an annual basis, while 16.2% attended one every six months, and just 0.7% attend one once a week (Zúñiga et al., 2021).

During the years 2016-2019, attendance at theatrical shows was predominantly in urban areas. For the year 2019, 11.1% attended a theater in an urban area. The largest number of attendees was in Lima, with 965,000 people, followed by Piura with 195,000 people, Arequipa with 160,000 people, and Puno with 145,000 people. Madre de Dios was the department with the lowest number of theatergoers at just 6,000 people.

On the other hand, the survey showed that the main reasons for non-attendance at theater shows in 2019 were as follows: lack of interest (49.8%), lack of time (26.8%), lack of money (8%), no offers (10.4%), no timely information (3.2%), and others (1.9%).

2) Cinema

According to the definition of Enapres, audiovisual works are films and other video contents. Likewise, audiovisual works are cinema performances projected with a film projector, screen or other device before an audience, which due to the evolution of technology can be performed in open or closed spaces (Dirección General de Industrias Culturales y Creativas, 2021).

A study conducted by Enapres, in 2019, highlights that 30 out of every 100 Peruvians attended a movie theatre at least once per year, that is, approximately 8.3 million people. Although there was a significant increase from 2016 to 2017, the following years 2018 and 2019 saw a constant attendance (33.6% and 33.1%, respectively) (See Figure 2.23). However, as the pandemic struck in the two following years, the figures cannot be compared.

The departments with the highest attendance rates at moving screenings were Callao (53.3%), Lima (49.4%), Arequipa (42.3%) and Ucayali (39.8%), while those with the lowest attendance rates were Ayacucho (6.7%), Amazonas (7.2%) and Huancavelica (8%).

In 2019, 2.8 million people (33.8%) attended a movie every month; 1.9 million people (23%) attended once every three months, and only 4.1% of Peruvians attended every week (DGIA, 2021).

The main reasons for not attending a movie theater were as follows: lack of interest (53.9%), lack of time (16.9%), lack of money (10%), no offers (15.7%), no timely information (1.4%), and others (2.1%).

Regarding the consumption of movies through CD, Blu-ray or other means, a significant decrease can be observed in Figure 2.25, since only 25% of the Peruvian population acquired at least one movie through these media in 2019 (DGIA, 2021).

3) Music

For this study, Enapres considered the population's attendance of music shows and the consumption of music in virtual or physical format (Zúñiga, Uzuriaga, Sacramento & Sáez, 2022).

Zúñiga et al. (2022) state that attendance at music shows decreased between 2019 and 2020. In 2019, 20 out of every 100 Peruvians over 14 years of age had attended a music show during the previous twelve months. In 2020, only 15 out of every 100 Peruvians attended one of these shows.

The departments with the highest attendance rate were Huancavelica (29.4%), Puno (23.8%) and Arequipa (23%), while those with the lowest attendance rates were Ancash (8.5%), Huánuco (6%) and Lambayeque (3.5%) (Mincul, 2021).

In contrast, in 2020 only 12% of the population acquired music through CD, Blu-ray or other means (3.8% less than in 2019). It is concluded that consumption through these devices fell by 19.1% from 2016 to 2020

4) Video games

According to Enapres, in 2019, 14% of the Peruvian population had acquired video games through downloads or internet access at least once in the previous twelve months. This figure is higher than 2018, when only 12.5% obtained these video games (Zúñiga & Sacramento, 2020).

The acquisition of video games through CDs, Blu-ray or other media has been decreasing in recent years. Only 18% of the Peruvian population acquired these games through these media in 2019.

5) Books

PromPerú (2020) stated that this sector is composed of companies that offer intellectual creations, through books in different formats or related publishing products. This sector has institutions including the Peruvian Chamber of Books and the Network of University Publishers and Independent Publishers of Peru. The sector is characterized by scientific, technical and professional books, general interest books and didactic books.

According to Enapres, in Peru, the acquisition of printed books by the population aged 14 and over declined from 27.7% in 2019 to 25.5% in 2020. Although during the period 2016-2019 there were no major changes in the percentages, it is worth noting that the decrease has become noticeable since 2017, when 29.8% of the population obtained or acquired printed books (Dirección del Libro y la Lectura, 2022). In other words, there has been a decrease of 4.3 percentage points in the last four years.

6.3.2 Major Challenges and Suggestions

According to UNESCO, the main challenges facing industries in this sector, when compared to other economic activities, is that they have been affected by the sanitary emergency, physical remoteness, and the precariousness and informality that characterizes employability in the sector (DGIA, 2021).

In Peru, Mincul (2022) conducted a national survey in support of the Metropolitan Municipality of Lima to measure the impact of the pandemic on the cultural sector. The subjects of the survey were companies, associations, organizations, ventures and independent workers associated with the arts sector, museums, cultural and creative industries in Peru. The DGIA (General Directorate of Cultural Industries and Arts)(2021) states that a total of 10,452 organizations and individuals took part in the survey. One important result of this study is that it showed that the economic losses recorded between

March and June 2020 amounted to around S/ 162,967,000. More than half of the respondents registered economic losses of between S/ 1,001 and S/ 5,000, i.e. a combined loss of approximately S/ 17,296, 000 (Mincul, 2021).

Of the total number of respondents, nine out of ten organizations stated that the cultural activities they develop are their main source of economic income (DGIA, 2021). As if that were not enough, in the second quarter of 2020, six out of ten cultural workers left their occupations.

Second, another major problem in the Peruvian economy is the illegal commercialization of private property or piracy, which affects various sectors including cinema, music, toys, publishing, tobacco, textiles, shoes, food and alcoholic beverages (Consultores en Asuntos Públicos Sucursal del Perú S.A., 2020).

Peru is ranked 33rd out of fifty countries in the world ranking of intellectual property policies, developed in 2018.

In a survey conducted by the Peruvian Chamber of Books (CPL) in 2015, companies in this sector indicated that the main problems affecting their sector were piracy (53%), high input prices (52.7%), taxes (31.5%), and low demand (30%) (Consultores en Asuntos Públicos Sucursal del Perú S.A., 2020).

In addition, according to the CPL (2015), the level of sales of pirated works was estimated at 150 million soles for 2015.

Consultores en Asuntos Públicos Sucursal del Perú S.A. (2020) states that, in 2017, the International Telecommunications Union (ITU) ranked Peru 96th in the world ranking of countries in terms of access to and use of ICT.

In summary, the main challenge facing Peru is informality (commercial and labor) and the piracy of cultural products (ProChile, 2017).

Third, another technological challenge is the fact that Internet penetration is still ongoing in regions far from Lima, with bandwidths being lower than in neighboring countries (ProChile, 2017).

Consequently, in 2021, the DGIA (2022) reported that the factors that most affected the activities of cultural organizations were the cancellation of activities or programs (71.2%) and the decrease or suspension of economic income(11.2%).

Against this only 37.8% of cultural organizations developed new goods to face the pandemic, while 24.1% did not make any innovations (DGIA, 2022).

The three sectors that had to cancel their planned activities were: performing arts (76.1%), crafts and traditional arts (76.1%), and visual arts (74.9%).

According to a DGIA study (2022) conducted from 2019 to 2020, there was a drop of 54% in the monthly income of workers in the cultural sector. This result is related to the health crisis experienced in Peru. However, there was a recovery of almost 12% by 2021, yet despite this, workers argue that they did not achieve their annual targets.

Conversely, workers in the audiovisual, film and new media sector earned a higher average income in 2019, 2020 and 2021, which is attributable to the increase in job possibilities in the sector resulting from the proliferation of digital entertainment media.

Performing arts and music are the sectors that suffered the most significant decreases in 2020. This is related to the closure of several cultural spaces (DGIA, 2022).

However, regarding the positive aspects in the sector, Aguirre (2021) states that Peru has put in place appropriate legal schemes (in terms of copyright, among other regulations) and institutions (Indecopi) that provide incentives to encourage creativity. In addition, intellectual property is an appropriate tool to promote the creation, innovation and dissemination of the production processes of products and services with high knowledge components.

Inclusive, Fuertes & Badillo (2014) argue that other cultural industries such as video games and informative or interactive entertainment software are gaining strength. Traditional cultural industries are adapting to the digital era, attaching their content in digital or multimedia publications in general. These industries are being transformed by digitization in all their value chains and modifying their own media (e-book, downloads, sharing, personalized and optional menus, online reading, DTT, cross media between telephony, computer and television, etc.).

For their part, Bakhshi, Hargreaves & Mateos-Garcia (2013) noted that the fostering of creative industries requires the following:

1. A well-articulated education system, including schools, universities, colleges and training centers.
2. Tax incentives and funding programs to encourage R&D in the creative industries.
3. A financial arm to provide content to sectors that are often perceived as a high-risk segment with better access to venture capital.
4. A competition regime that protects intellectual property, provides a balanced copyright regime, prevents abuses of dominant positions, and acts quickly to remedy such abuses.
5. The question of a balanced copyright regime.

CHAPTER 5

Capacity Development Process

S&T Planning and Technological Forecasting for Prioritized Sectors in Peru

Chapter 5. Capacity Development Process

1. Kick-off Meeting(June 2, 2022)

1.1 Kick-off Meeting & In-depth Study Overview

- Date: 2nd June 2022/ 09:00-11:30 am (@Korea) / 1st June 2022 07:00-09:30 pm (@Perú)
- Working Language: English
- Venue: Virtual meeting via Zoom

1.2 Objectives and Activities

- Improve understanding regarding the project (Project outline, Project scope, Research Teams of Peruvian side and Korean side, Selection of Local consultants, etc.) among project stakeholders.
- Identify the project stakeholders and their role
- Identify policy priorities of project stakeholders by presenting the overall research scope and direction
- Identify specific needs, expectations and Diagnose the current status and challenges of the Peruvian side
- Discuss collecting and sharing data and background information
- Gain in-depth knowledge on the situation of Perú
- Review the ToR for K-Innovation Partnership Program with Perú between STEPI and CONCYTEC
- **(Deliverable)** Final Research Topic and Scope, Kick-off PPT Slides, ToR, Data and background information, etc.

1.3 Program Schedule

June 1st 07:00-09:30 pm(@Perú) / June 2nd 09:00-11:30 am(@Korea)	
Testing Equipment and Setting (06:30-07:00)	
Opening Session	*Moderator: Dr. ChiUng Song (Senior Research Fellow, STEPI)*
07:00-07:20	Introduction of Participants *- All Participants both from Korea and Perú* Group Photo
In-depth Session	*Moderator: Dr. ChiUng Song (Senior Research Fellow, STEPI)*
07:20-07:45	Peruvian STI Governance and Policy: The Current Status and Challenges (TBD) *- Presented by Alfonso Rodríguez (Senior Specialist, Research & Studies Direction, CONCYTEC)*
07:45-08:10	*Project Overview and STI policy development* *- Presented by Dr. ChiUng Song (Senior Research Fellow, STEPI)*
08:10-08:20	*Project Progress* *- Presented by Ms. So Hyun Kwon(Senior Researcher, STEPI)*
08:20-08:50	*Overview on the Methods of the Technological Forecasting* *-Presented by Dr. Johng-Ihl Lee (Professor, Department of Technology and Society, SUNY Korea)*
08:50-09:25	Q&A and discussion
Closing Session	*Moderator: Dr. ChiUng Song (Senior Research Fellow, STEPI)*
09:25-09:30	Closing Remarks *- Mr. Fernando Ortega, MBA (Director, Research & Studies, CONCYTEC)* *- Dr. ChiUng Song (Senior Research Fellow, STEPI)*

1.4 Photos


2022 K-Innovation Partnership Program with Perú
"S&T Planning and Technological Forecasting for Prioritized Sectors"
Kick-off Meeting & In-depth Study
2nd June 2022, 09:00-11:30 am @Korea / 1st June 2022, 07:00-09:30 pm @Perú
Lotte Hotel Seoul, Korea & Online
STEPI
CONCYTEC

2. Capacity Building Workshop (September 29-30, 2022)

2.1 Capacity Building Workshop Overview

1) Day 1: Lectures and Discussions on Korea's STI policy and Technological Forecasting

- Date: 28 September 2022, 07:00-09:00 pm (@Perú) / 29 September 2022, 09:00-11:00 am (@Korea)
- Working Language: English
- Venue: Virtual meeting via Zoom

2) Day 2: In-depth Lectures and Discussions on Technological Forecasting Methods

- Date: 29 September 2022, 07:00-09:00 pm (@Perú) / 30 September 2022, 09:00-11:00 am (@Korea)
- Working Language: English
- Venue: Virtual meeting via Zoom

2.2 Objectives and Activities

- To improve the capacity of CONCYTEC's practitioners and the stakeholders (government policymakers, research institutions, universities, the private sector, etc.) related to S&T planning and technological forecasting for prioritized sectors.
- To share knowledge and experience on South Korea's STI planning and technological forecasting .
- **(Deliverables)** Workshop PPT Slides, Lecture Recording

2.3 Program Schedule

September 28, 07:00-09:00 pm(@Perú)/ September 29, 09:00-11:00 am(@Korea)	
Testing Equipment and Setting (08:30-09:00)	
Session I: Lectures and Discussions on Korea's STI policy and Technological Forecasting	
09:00-09:45	*The Evolution of STI Policy* *- Presented by Dr. ChiUng Song (Senior Research Fellow, STEPI)*
09:45-10:00	Q&A and discussion
10:00-10:45	*Technology Forecasting for Policy Development and Implementation* *-Presented by Dr. Johng-Ihl Lee (Professor, Department of Technology and Society, SUNY Korea)*
10:45-11:00	Q&A and discussion
September 29, 07:00-09:00 pm(@Perú)/ September 30, 09:00-11:00 am(@Korea)	
Testing Equipment and Setting (08:30-09:00)	
Session II: In-depth Lectures and Discussions on Technological Forecasting Methods	
09:00-09:45	*Technology Foresight Activities in Korea* *- Presented by Dr. Hyun Yim (Senior Research Fellow, Center for Technology Foresight, KISTEP [Korea Institute of S&T Evaluation and Planning])*
09:45-10:00	Q&A and discussion
10:00-10:45	*Implementation of Technology Roadmap to Korean Industry Development* *-Presented by Dr. Sehee Park (CTO, Deliverdy/Former Director of KIAT [Korea Institute for Advancement of Technology])*
10:45-11:00	Q&A and discussion

2.4 Results

2.4.1 Evolution of Science and Technology Policy Presentation

The evolution of the ITS Policy shows the challenges that South Korea faced after the Korean War (1950-53) in the 1950s. The country's GDP (Gross Domestic Product) grew at an annual average rate of 4.1% from 1953 to 1961, and during this time the most urgent challenge for Korea was to free its people from chronic poverty, considering that it did not have enough tools or resources to drive industrialization.

In the 1950s, Korea had an illiteracy rate of 29% and a university enrollment rate of 29.2%. The educational level was higher compared to the income level, and there was a general consensus on the need for industrialization and the "can do" spirit among the people.

In seeking a development strategy, Korea had to look outwards for resources, technology, and markets, with human resource and industrialization policies based on innovation by imitation.

South Korea's industrialization in recent decades has been based on the production of automobiles, mobile phones, shipbuilding factories, and petrochemicals. The development strategy was an East Asian model with the following conditions:

- Lack of purchasing power in the domestic market.
- US$ 90 of GDP per capita at the beginning of the 1960s.
- Lack of non-natural resources asuch as gas and oil.
- Insufficient agricultural production.
- Lack of human resources.

The government implemented several policies to strengthen development:

- Protection of multinational companies.
- Taxes and subsidies.
- Loans and guarantees.

- Science and technological development.

To achieve development with the East Asia Economic Development Model, the following three factors were essential:

- Human Resources
- Indigenous companies
- Research and development Innovation

At the beginning, the main strategy was focused on the export of market industrialization. Technological innovation concepts originating from businessmen and the government define the role of the research sector in human capital, the existing stock of knowledge and new knowledge required to produce durable goods.

The challenges for developing countries are: how to drive industrialization and promote technological progress, how to build a research sector based on human capital, and how to build a bridge between knowledge and technology transfer.

The government's mission was to promote Science and Technology by promoting research and development activities. The Korea government's policies were focused on light industries in the 1960s, on heavy industries in the 1970s, on assembly and processing industries in the 1980s, and innovation technology industries in the 1990s. .

The Ministry of Science and Technology was created under the Office of the Prime Minister (1967), along with other institutes such as the Korea Institute of Science and Technology (1966). Furthermore, international cooperation with the USA, Japan and Germany was promoted from the 1960s onwards.

A basic strategy for achieving high levels of development over time concerns the ability to coordinate between the public and private sectors. On the one hand, the government diversified different policies from the the end of the Korean War in the mid-1950s to the 1990s, while, on the other hand, the private sector contributed to investment and innovation and pushed ahead with the 5-year plans. In the 1960s, development was based on strengthening the domestic market, both in industry and in the agricultural sector. In the 1970s, development was based on overseas markets promoting investment in free zones with tax subsidies and the allocation of labor to industrial plants situated next to shipping ports.

Another policy consisted in allocating public funds for research and development to different institutes under the supervision of the Ministry of Science and Technology. In Peru, CONCYTEC (National Council of Science and Technology; under the National Government and Subnational Governments, such as Regions and Municipalities) is mostly dependent on public funds, and fewer allocations from technical cooperation represent less than 1% of the total public budget in the last six years of National Budget approved annually by Congress.

The public funding for research and development, the more time there will be for development as a whole. The experience of South Korea in terms of development and R&D policies in recent decades shows all the steps that the government must take to increase resources in the medium and long term.

2.4.2 Presentation of Technology Forecast for Policy Development and Implementation

Research and development is associated with the resources allocated for these purposes in terms of a percentage of GDP (gross domestic product). For example, the following table shows the amount of R&D as a percentage of GDP by country in2019:

Country	GDP %
Israel	4.9
Korea	4.6
Taiwan	3.5
Sweden	3.4
Japan	3.1

This shows resources oriented towards research and development as part of the national policies of these countries.

Generally, technology and R&D show the following features:

- Technical, human, system and market risks or uncertainties
- Unpredictable results
- Linkage by unknown consumers, norms and goods.
- Disruption
- Obsolescence

Regarding the evolution of R&D Activities, four generations have been identified.

- 1^{st} generation: isolated R + D + I without systems or strategies or budget.
- 2^{nd} generation: mission oriented management by objectives(MBO), independent management, and concept as a project
- 3^{rd} generation: integrated management, technology and business strategies.
- 4^{th} generation: market involved in strategies

Technology is a strategy for modern companies, which is concerned with direction and diffusion. For example, Xerox defined the technology portfolio as having several steps.

- Step 1: Define the asset classes.
- Step 2: Propose a target allocation across portfolio segments, such as human and financial resources.
- Step 3: Categorize programs and projects into segments.
- Step 4: Evaluate programs and projects within each of the 3 R's: return factor (5 year profit, time for revenue segments and value of IP), risk factor (technical, market, and execution), and ranking.
- Step 5: Manage programs and projects to move between segments.
- Step 6: Measure, define and rebalance based on learning and the state of the business.

It is important to define technology management as the linkage of engineering, science, and management to achieve an organization's strategic and operational objectives in planning the development and implementation of technological capability. Another definition is the study of the application of ethics, sociology, and other subject areas related to engineering science and management in order to understand the interaction of social actors in the use of technologies and their development.

Technology assessment consists of a systematic analysis of the anticipated results or social, political, economic and environmental impacts of the application and/or introduction of new technologies for practical alternatives for technology policy decision-making.

For this, the valuation of a given technology is necessary; i.e. an estimate of the value of the technology in terms of monetary value, range, score and opinion regarding its technical and commercial value.

In this analysis, the market structure or segmentation should be based on the future market, investments, sensitivity test, product life, estimated risk, and costs. The cash flow method is necessary to calculate its NPV (net present value) and the internal rate of return.

Technology forecasting is a political tool for prioritizing R&D, planning the development of new technologies and products, and making strategic decisions on technology licensing, joint ventures, etc. Also, it establishes the infrastructure of technological foresight as the network of experts.

The characteristics of technological forecasting (TF) are as follows:

- It focuses on specific technologies or areas.
- The scope sometimes encompasses or is shared with others.
- The importance of TF is emphasized over time, with the number of publications more than doubling during the period 2000-2006.
- TF is culture specific.

There are various families of technological forecasting such as:

- Expert opinion: use of the Delphi method, focus groups such as panels and workshops, interviews, participatory.
- Trend analysis: use of extrapolation such as growth curve fitting, trend impact analysis, precursor analysis.
- Surveillance and intelligence methods: use of intelligence methods, environmental exploration, technological surveillance, bibliometrics as a research profile, patent analysis, text mining, statistical methods.
- Statistical methods using correlation analysis, demographics, cross-impact analysis, risk analysis, bibliometrics such as research profiles, patent analysis, and text mining.
- Modeling and simulation using agent modeling, cross-impact analysis, sustainability analysis such as life cycle analysis, causal models, diffusion modeling, complex adaptive systems modeling, system simulations such as dynamics, technological substitution.
- Scenarios with consistency checks, management, simulation such as interactive games and scenarios, field anomaly relaxation method.
- Valuation and economic decision methods such as future wheel, action analysis, cost-benefit analysis, utility analysis, and input-output analysis.
- Descriptive and matrix methods such as analogies, backcasting, impact identification checklist, innovation systems modeling, institutional analysis, road mapping, social impact assessment and multiple perspectives, analysis and organizational requirements.
- Creativity as brainstorming with nominal groups, creative workshops, vision generation, science fiction analysis.

Taking into consideration the different methods of forecasting, an analysis should be run to choose a correct method using the extent of data availability, the degree of validity of the data, the degree of similarity between the proposed technology and existing technology with degrees, i.e. small or short, medium or moderate, large or tall.

The method of technology forecasting for Peru begins by evaluating policies and priorities after taking into account the following aspects:

- Small amount of validated data.
- Good similarity between proposed and existing technology.
- Method based on information obtained from a panel of experts (Delphi method or interviews).

We can evaluate the predictions made for different technologies used in the past, such as cars, telephones, radio, television, moon landing, communications satellites, cellphones, internet, YouTube, Apple, and I-phones. None of these technologies were accepted ainitially, but after the inventors pushed them, the society accepted those who evaluated the results.

The introduction of the forecast method in Peru is based on different modules as follows:

1. Objective: To diagnose the state of science and technology in Peru. Connectivity, access to digital network, use of big data technology, digital literacy, diffusion of online purchases.
   Results: generation of evidence for technology forecasting.

2. Objective: To develop the technology forecasting methodology and survey.
   Outcome: provision of practical tools for science and technology policies.
   Result: contribution to the next steps.
3. Objective: To make technology forecasts and suggest policy directions.
   Result: application to training and academic work.

One common method of forecasting is the Delphi survey, which aims to collect and extract knowledge from a group of experts through the use of questionnaires.

Cooperation between Peru and Korea on technological forecasting requires a sustained personal network and policy binding among experts in order to build trust.

For a sustainable source of policy and values, it is necessary to have academic quality and recognition, added value for relevant institutions and participants, and a high level of government-ensured and sustained cooperation.

2.4.3 Technological Foresight Activities in Korea Presentation

Technology foresight is defined as the process of making a systematic attempt to look into the longer-term future of science, technology, the economy, and society with the goal of identifying strategic research areas and emerging generic technologies in order to generate the greatest economic and social benefits.

On the one hand, TF increases competition with new technologies, and innovations decide the competitiveness of the organization concerned. On the other hand, there are limits on public spending because not all S&T fields can be supported by governments.

The Delphi method is a well-established, expert-based technique that uses questionnaires distributed in rounds. This method has its advantages and disadvantages.

- Advantages: avoids biased discussions, collects opinions from various experts, provides feedback of expert opinions, allows free and frank communication.
- Disadvantages: low response rate, time consuming, high cost, heavy reliance on experts and statistics.

The Delphi method uses the following steps to obtain results:

1. Determining the purpose and scope of the survey
2. Making preparations
3. Running the survey
4. Obtaining the result(s)

Since 1993, Korea has acquired extensive experience in technology forecasting, which is conducted by the Korea Institute for S&T Evaluation and Planning (Kistep) every five years, and has obtained ten emerging technologies (as of 2009). Kistep provides insights into and directions for emerging S&T areas, and identifies future technologies that may have great future potential for the growth of national wealth and the improvement of the way of life and standard of living. All of this reflected in the basic CyT plan.

Korea's general science and technology forecasting framework is based on selected future technologies, taking into account future development and needs. Over time, Delphi has been the most widely used of all TF methods.

The procedures used in the 6th forecast of S&T (2020) are as follows:

1. Determine the prospects for the future society from an analysis of problems based on the determination of each trend to lead to the determination of needs.
2. Identify future technologies and innovative technologies.
3. Conduct survey and Delphi analysis.
4. Draft report.

So far, a total of 241 future technologies that address social needs have been derived, and these have been reviewed and finalized by a Steering Committee. Of the total, fifteen innovative technologies were derived, with excellent socioeconomic impact and a great domino effect.

These fifteen innovative technologies were examined for the following elements:

- The country is expected to reach the tipping point first. The tipping point is quantitatively interpreted as the moment when a product has been adopted by 16% of the total number of consumers, including innovators (2.5%) and early adopters (13.5%).
- Turning point in Korea and abroad.
- Degree of significance

- Government policy is required for the social diffusion of innovative technology.
- Describe the opportunities and negative impacts of innovative technology.
- Describe the social risks and conflicts posed by innovative technology.

The Korea-China-Japan prospective project for renewable energy, with the following three objectives:

- Increase collaboration between the three countries.
- Improve networks within and between the three countries.
- Improve the use of foresight results in strategic and political decision-making.

Four steps were adopted in this project:

- Common theme
- Delphi survey
- Future scenario
- Joint report

These three countries agreed on several important factors:

- Need to solve the high initial cost in promoting the use of renewable energy.
- Essentiality of technological advances to solve the difficulties of adopting renewable energies.
- Need to develop effective business models regardless of the energy category to promote the use of renewable energy.
- Financial incentives and environmental regulations were the two most widely favored public policies to expand the use of renewable energy.

Generally speaking, the technology forecast is an important policy tool in science and technology policy, while the Delphi survey is useful for analyzing future technologies.

2.4.4 Presentation and Implementation of the Technology Roadmap

The Techonolgy Roadmap is a blueprint for forecasting future technology for R&D programs, assessing future infrastructure and human resource demands, including future market insights, and aligning R&D and market trends with the private sector.

This model was launched in 1999 with the government-supported Samsung Advanced Institute of Technology (SAIT). Initially, the TRM was divided into three phases:

- Phase 1: Plan Design. Social consensus was necessary to explain the application of the purpose. Define the scope through the megatrend, included the strategic products and services technology tree, as well as the budget workforce.
- Phase 2: Preparation of the plan. In this phase, the mission orientation was defined using originality, market suitability, competition, urgency and economic value as the decision factors.
- Phase 3: Construction plan and review. Public use in sessions with investment plans and data updates.

Over time, several lessons have been given on the TRM (technological roadmap), which are outlined below:

- Planning: povide a sufficient time frame and ensure that the TRM is connected with other organizations.
- Implementation: Ensure that the TRD guarantees technological and economic reliability, and promote the TRM to small and medium-sized companies.

2.5 Survey Results

2.5.1 Results of evaluation of each lecture

- After the workshop, questionnaires were provided to the trainees, and they were asked to to complete evaluations of each of the lectures and a program evaluation survey.
- The questionnaire for the evaluation of the lectures consisted of 15 questions.
  - 13 multiple-choice questions: The main goal of the evaluation was to collect the trainees' opinions about the content, lecturers, and entire process (time assignment, etc.). The respondents could choose one of five possible responses to each question (strongly agree-agree-neutral-disagree-strongly disagree)
  - Two subjective questions: The trainees were asked to write freely about what they liked best about the lectures and what they felt needed to be improved.
- 13 of the 20 participants in the workshop answered the survey (response rate 65%)

[Figure 5-1] Lecture Evaluation Questionnaire

Lecture 1 — Title of Lecture		Lecturer		Name	
	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
1. The topics covered in the lecture were relevant to me.	○	○	○	○	○
2. The content was well-organized and easy to follow.	○	○	○	○	○
3. The lecture deepened my understanding of the field.	○	○	○	○	○
4. The lecture was be helpful in strengthening the institutional capacity of my organization.	○	○	○	○	○
5. The time allotted to the lecture was sufficient.	○	○	○	○	○
6. The pace of the lecture was appropriate.	○	○	○	○	○
7. The lecturer was sound in knowledge and speech.	○	○	○	○	○
8. The lecturer noticed when trainees needed help.	○	○	○	○	○
9. The objectives of the lecture were stated clearly at the beginning of the lecture.	○	○	○	○	○
10. The lecturer conveyed the topic in depth.	○	○	○	○	○
11.The Q&A and discussion helped to enhance my understanding.	○	○	○	○	○
12. I will be able to apply the knowledge learned in my work/practice.	○	○	○	○	○
13. I will recommend this lecture to others.	○	○	○	○	○
14. What did you like most about the lecture?					
15. What aspects of the lecture can be improved?					

Source: STEPI (2018), “2018 K-Innovation ODA Program”, pp. 275-276.

■ <Lecture 1. The Evolution of STI Policy (ChiUng Song)>

- The trainees were found to be generally satisfied with this lecture. The lecture allowed the trainees to understand the history and development of Korea's science and technology policy in general, and emphasized the important role played by STI in Korea’s remarkable growth and policy soundness. Notably, one trainee commented that they were particularly impressed with the development strategy from imitation to innovation, export-oriented growth strategy, OEM/ODM, and the importance of indigenous companies.

- Many were satisfied overall, but a few proposed the following suggestions:
  - Include more information about clusters, science and technology parks.
  - Provide a more detailed explanation of how Korea used foreign aid in the process of achieving rapid economic growth and innovation, and in particular, how it strengthened cooperation with the United States, Japan, and Germany.

■ <Lecture 2. Technology Forecasting for Policy Development and Implementation (Johng-Ihl Lee)>

- The results are shown to be satisfactory overall. The visions presented through a systematic analysis, and the clear explanations of the actors and processes were evaluated positively. One trainee mentioned that he/she particularly liked the explanation of technology copy (purchase) as a method of accelerating the development and heightening the importance of R&D.
- Many respondents pointed out that the length of the lecture seemed a little short, while others mentioned that they would have been able to understand the content of the lecture better if it had included practical cases of technological forecasting and innovation. Others still indicated that it should perhaps be taken into account that the influence of the WTO in the planning mentioned in the lecture worked differently in Latin America.

■ <Lecture 3. Technology Foresight Activities in Korea (Hyun Yim)>

- The trainees expressed their satisfaction with the technology foresight methodology, its importance in policy making, and the detailed explanation of the Delphi methodology. In particular, it seems that the presentation of the TF methodology in the lecture raised implications that could be sufficiently applicable to Peru. Despite their overall satisfaction, some trainees suggested that the instructor's delivery could be improved a little.

- Other questions raised pertained to the major issues that had to be overcome as a society in order to steer public, private and academic organizations towards a technology oriented country, and what had to change in the minds of national leaders, and their ways of thinking. Another trainee suggested that it would have been good to present technology forecasting with emphasis on the Delphi methodology's possible application to six priority industries in Peru.

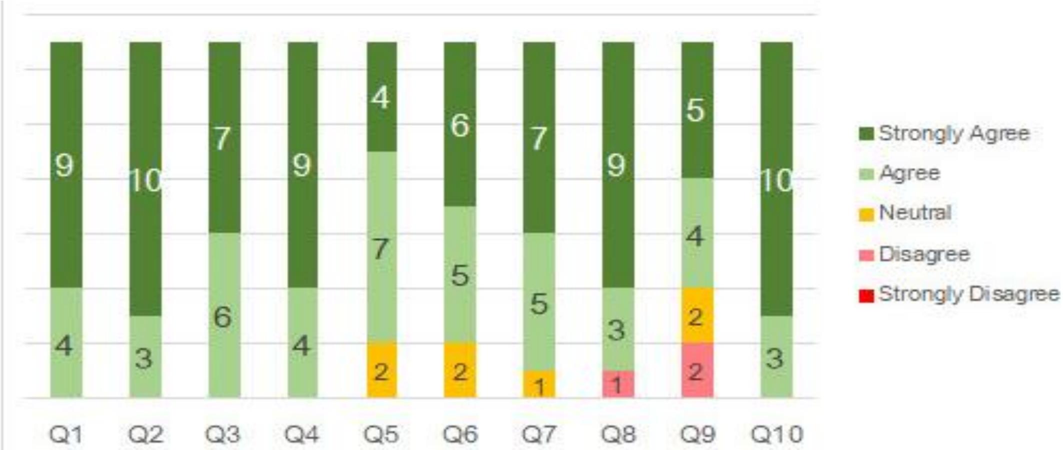
■ <Lecture 4. Application of the Technology Roadmap to Korean Industry Development (Sehee Park)>

- The results of the evaluation were found to be satisfactory overall. Many respondents answered that the explanation of planning and mapping was particularly interesting. They also requested a more detailed description of the roadmap, particularly how to strategically select and commercialize new technology.
- Many pointed out that it would be necessary to translate some examples into English. In addition, there was an opinion that the regulations on the composition and operation of the committee mentioned in the lecture could raise some implications for Peru.

2.5.2 Program Evaluation Survey

Questions	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. I expected that this training would be helpful in performing my duties.	①	②	③	④	⑤
2. The program was helpful in enhancing my capacity to perform my duties.	①	②	③	④	⑤
3. The program met the objective of institutional capacity building.	①	②	③	④	⑤
4. The program was composed according to its purpose.	①	②	③	④	⑤
5. The materials were helpful in understanding the contents of the program.	①	②	③	④	⑤
6. The topics covered were relevant to my country's S&T planning and technological forecasting needs.	①	②	③	④	⑤
7. Sufficient opportunities for participation were provided throughout the program.	①	②	③	④	⑤
8. The program has influenced me to change my mindset.	①	②	③	④	⑤
9. The duration of the program (2 days) was sufficient.	①	②	③	④	⑤
10. I would recommend this program to others.	①	②	③	④	⑤
11. What other topics or contents do you think could be addressed in the future?					
12. Are there any suggestions for improvement or requests about the overall procedure? If so, please explain them in detail (e.g. program process, contents, results, sustainability, etc.).					

[Figure 5-2] Program Evaluation Questionnaire

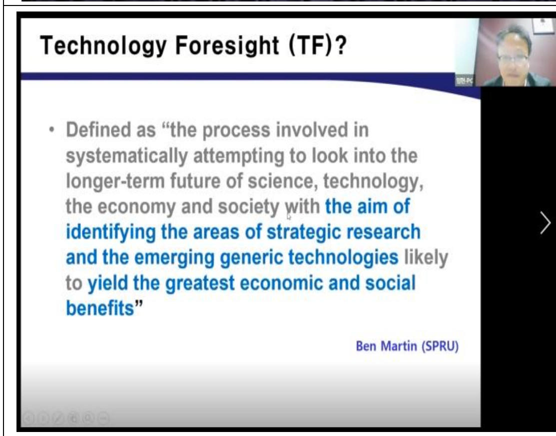


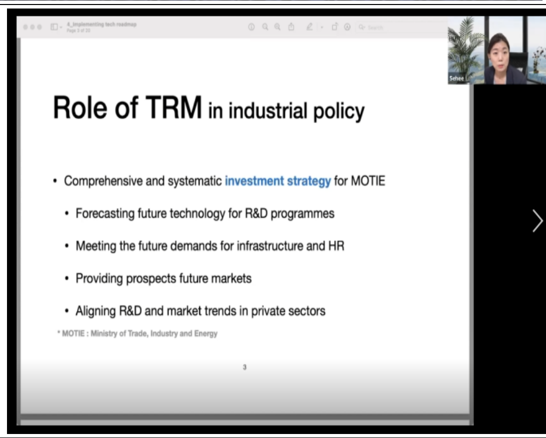
- Most of the evaluations of this workshop were positive. The trainees responded that this workshop dealt with various topics related to science and technology innovation, and showed interest in technology forecasting and the technology roadmap in particular.
- Despite their overall satisfaction with the workshop, it was suggested that a duration of two days for the workshop was insufficient.
- The following topics have been suggested for future discussion:
  - An in-depth look at how to align the major actors in public-private-academia circles from very varied perspectives. It would be good to focus on small and medium-sized enterprises (SMEs) in particular.
  - Greater focus on specific technology topics: renewable energy, mining, health, etc.
  - More in-depth content and various examples of technology foresight, methodologies, and TRM establishment.
  - Creation of an independent workshop on technology foresight design with the application of the Delphi methodology.
  - Data science, database management and identification of data collection instruments for the generation of statistics, and thus those indicators of science, technology and innovation that are useful for public policy decision-making.
- In addition, some trainees expressed their opinion on the need to provide references and materials in English, and the need for more opportunities for open discussion

between all of the lecturers and trainees before ending the sessions.

2.6 Photos and News Media Photos and News Media


2022 K-Innovation Partnership Program with Perú
"S&T Planning and Technological Forecasting for Prioritized Sectors"
Policy Practitioners' Capacity Building Workshop
28th ~29th September 2022, 07:00-09:00 pm (@Perú) / 29th ~30th September 2022, 09:00-11:00 am (@Korea)
Lotte Hotel Seoul, Korea & Online
SCIENCE AND TECHNOLOGY POLICY INSTITUTE
So Hyun Kwon
ChiUng Song
Johng-Ihl Lee


Technology Foresight (TF)?
• Defined as "the process involved in systematically attempting to look into the longer-term future of science, technology, the economy and society with the aim of identifying the areas of strategic research and the emerging generic technologies likely to yield the greatest economic and social benefits"
Ben Martin (SPRU)


Role of TRM in industrial policy
• Comprehensive and systematic investment strategy for MOTIE
• Forecasting future technology for R&D programmes
• Meeting the future demands for infrastructure and HR
• Providing prospects future markets
• Aligning R&D and market trends in private sectors
* MOTIE : Ministry of Trade, Industry and Energy

연합뉴스

과기정책연, 페루에 우리 과학기술 정책 소개

송고시간 | 2022-10-10 11:51

과기정책연, 페루에 한국 과학기술 정책 관련 지식 공유
[과학기술정책연구원 제공. 재판매 및 DB 금지]

(서울=연합뉴스) 조현영 기자 = 과학기술정책연구원(STEPI)은 지난달 말 페루에 과학기술 정책 및 기술 예측 방법에 대한 한국의 경험과 지식을 공유하는 워크숍을 열었다고 10일 밝혔다.

페루의 과학기술 기획 역량 강화를 위한 정책 컨설팅 지원 사업인 '2022 한-페루 국제기술혁신 협력사업'의 일환이다.

과기연은 이틀간 열린 행사에서 한국의 과학기술 정책 발전사와 전략을 소개하고 기술 예측 활동, 산업 기술 로드맵 수립 및 이행에 관한 우리나라의 구체적 사례를 공유했다.

김지현 과기정책연 SDGs혁신연구단장은 "이번 역량 강화 워크숍을 통해 페루의 과학기술 기획 및 기술예측 체계 수립을 위해 관련 정책 실무자 및 이해관계자의 이해 제고에 도움이 될 수 있기를 바란다"고 말했다.

hyun0@yna.co.kr

Source: Yonhap News Agency(October 10, 2022), 「과기정책연, 페루에 우리 과학기술 정책 소개」, https://www.yna.co.kr/view/AKR20221010032200017 (Search date: November 14, 2022)

11/15/22, 7:51 AM STEPI, 페루에 과학기술 기획 및 기술예측 지식·경험 공유 - 전자신문

STEPI, 페루에 과학기술 기획 및 기술예측 지식·경험 공유

발행일 : 2022-10-10 10:47

<발제하는 송치웅 아태첨단기술전략연구센터장>

과학기술정책연구원(STEPI-원장 문미옥)은 우리 시간으로 지난 9월 29일부터 30일'2022년도 한-페루 국제기술혁신협력사업(K-Innovation Partnership Program) 일환으로 페루의 과학기술 기획 및 기술예측 체계 수립 지원을 위한 역량강화 워크숍을 하이브리드 방식으로 개최했다고 10일 밝혔다.

2022 한-페루 국제기술혁신협력사업은 '페루의 국가 유망분야 과학기술 기획 및 기술예측(S&T Planning and Technological Forecasting for Prioritized Sector)' 역량 강화를 위한 정책 컨설팅 지원사업이다.

워크숍에서는 과학기술 기획 및 기술예측에 관한 한국 전문가들이 참여하여 페루 국가과학기술혁신위원회(CONCYTEC) 연구팀 및 관계자, 페루 산학연관 이해관계자 20여 명을 대상으로 과학기술정책 및 기술예측 기획과 기술예측 방법론에 대한 한국의 경험과 지식을 공유했다.

29일에는, 송치웅 아태첨단기술전략연구센터장(STEPI)이 '한국의 과학기술정책 발전사'를 주제로 우리 과학기술정책에 대한 빅푸쉬(Big Push) 전략을 시대적 배경에 맞춰 소개했다.

연구개발(R&D) 인프라 구축과 인력 양성을 위한 정부의 과감하고 대대적인 투자 및 정책지원 필요성을 강조했으며, 민간부문에 대한 전략적 지원의 필요성도 주장했다.

이어진 발제에서 이종일 교수(한국뉴욕주립대)는 '정책개발 및 실행을 위한 기술예측 방법론'이란 주제로 과학기술예측 배경과 과학기술과 R&D 관리에 대한 방법을 공유했다.

이와 함께 기술과 R&D가 갖고 있는 불확실성과 위험에 대한 인식과 기술예측을 통해 우선순위를 선정함에 있어 경제사회적 요소들을 잘 파악해야 한다는 점을 강조했다.

Source: ETNEWS(October 10, 2022), 「STEPI, 페루에 과학기술 기획 및 기술예측 지식·경험 공유」, https://www.etnews.com/20221010000069# (검색일자: Dec. 11, 2022)

3. Dissemination Workshop in Peru (October 18-20, 2022)

3.1 Dissemination Workshop Overview

- Date: October 18-20, 2022
- Working Language: English
- Venue: Lima, Peru

3.2 Objectives and Activities

- To hold technology foresight workshops for CONCYTEC's practitioners and the stakeholders (government policymakers, research institutions, universities, the private sector, etc.) related to S&T planning and technological forecasting for the prioritized sectors.
- To provide fundamental theoretical knowledge and practical sessions aimed at enhancing Peru's TF capacity.
- To share knowledge and practical experience on the evolution of South Korea's STI policy at an open forum hosted by CONCYTEC-STEPI.
- To hold a forum on Korea-Peru STI Cooperation hosted by the Embassy of South Korea in Lima and STEPI.
- To disseminate the outcomes of the 2022 K-Innovation Partnership Program with Peru.
- To discuss follow-up actions.

3.3 Program Schedule

Day 1: Monday, October 17, 2022	
Time	**Items**
11:00-14:00	(Korean delegation) Visit to the Embassy of the Republic of Korea in Peru
16:00-17:00	*Pre-meeting STEPI-CONCYTEC, site visit and meeting with CONCYTEC's Task Force*
3-Day Workshop on Technology Foresight	
Day 2: Tuesday, October 18, 2022	
10:00-12:00	Project Overview Orientation on Technology Foresight Workshop
12:00-12:30	Coffee Break
12:30-14:30	*Leader: Dr. Hyun Yim (Senior Research Fellow, KISTEP)* *Facilitators: Dr. ChiUng Song, Dr. Johng-Ihl Lee, Dr. Sehee Park, Ms. So Hyun Kwon* **Module 1 (2hrs): Hands-on Project** • Identify drivers using STEEP analysis • Evaluate drivers
17:00-20:00	Invitation Dinner by STEPI
Day 3: Wednesday, October 19, 2022	
10:00-12:00	**Module 2 (1): Hands-on Project** • Write scenarios • Draw implications for each scenario
12:00-12:30	Coffee Break

12:30-14:30	**Module 2 (2): Hands-on Project** • Write scenarios • Draw implications for each scenario
Day 4: Thursday, October 20, 2022	
10:00-12:00	**Module 3 (1): Hands-on Project** • Develop vision and goals • Identify strategic products and services using the Delphi Method • Group presentation
12:00-12:30	Coffee Break
12:30-14:30	**Module 3 (2): Hands-on Project** • Develop vision and goals • Identify strategic products and services using the Delphi Method • Group presentation

3.4 Results

This workshop was run on three business days according to the following program:

Day 1: Tuestday, October 18, 2022

- Description of project and guidance on the technological foresight workshop; identification of conductors using STEEP analysis; evaluation of drivers.

  Led by Dr. Hyun Yim, with facilitators Dr. Chiung Song, Dr. Hohng lhi Lee and Dr. Sehee Park and Ms. So Hyun Kwon.

Day 2: Wednesday, October 19, 2022

- Writing of scenarios; drawing of implications for each scenario.

Day 3 Thursday, October 20, 2022

- Development of vision and goals into a practical project; development of strategic services using the Delphi Method; group presentation.
- Identification of drivers using steep analysis, evaluation of drivers, evaluation of scenarios, implications raised when drawing each scenario, development of vision and goals, identification of strategic products services using the Delphi method.

The worksheets for the Agroindustry Sector are displayed in the excel program linked to this Report, where each label refers to each worksheet:

1. Worksheet 1 displays the drivers of agriculture in Peru, and the issues and trends identified that might be important in future; assessment of the drivers identified, and the level of impact and degree of uncertainty. The rows display the social, technological, economical, environmental and political aspects, while the columns show the of priorities from 1 to 5. Land property, security, auto supply surplus and marketing, associativity and population in rural areas have all been qualified by the work group.
2. Worksheet 2 displays the ranking of drivers identified in the group work. The columns refer to the degree of impact, while the rows refer to the level of uncertainty. This worksheet contains data from worksheet 1. Drivers of change are the key forces in the macro environment that point it up trends and issues of organizations in both the private and public sectors.
3. Worksheet 3 displays the evaluation of the drivers of changes based on two criteria: uncertainty and impact. Critical uncertainties with high uncertainty and strong impact were used to define the scenario space and to shape the scenario's story. Predetermined elements with low uncertainty and strong impact defome strategic issues needed to address along all scenarios. Axes of uncertainty in columns and posibilties, variating from secured to unsecured. Rows are deferring to food security, agroindustrial innovation and market trends.

4. Worksheet 4 displays scenarios selected by group. A scenario is a tool for displaying perceptions of different future envioremments, which can help organizations to foresee and manage the future. This instrument help to understand the developments, driving forces and key factors in planning the future in the short, medium and long term.
5. Worksheet 5 displays selected scenario A. The columns show the scenario's title, the key axes of uncertainty, and a description of the scenario.
6. Worksheet 6 displays selected scenario B. The columns show the scenario's title, the key axes of uncertainty, and a description of the scenario.
7. Worksheet 7 displays selected scenario C. The columns show the scenario's title, the key axes of uncertainty, and a description of the scenario.
8. Worksheet 8 displays the identification of products and services among the three selected scenarios A, B and C.
9. Worksheet 9 shows the priorization of strategic products and services.
10. Worksheet 10 shows the evaluation of the products and services, which were analyzed using a round Delph survey. Processed agroindustrial products show higher ratios.
11. Worksheet 11 displays the vision. The first row shows policies in the agricultural sector, such as the strong support of the Ministry of Agriculture, the large export volume of processed fruits, demand for high-quality products, added value products, the market for organic and ecological products, production of fertilizers, research and extension activities, and developed infrastructures.

As regards conclusions and recommendations, it will be useful to apply the Delphi and Steep methods to the National Agrarian University La Molina, Lima Peru, (http://www.lamolina.edu.pe), where production centers operate into input and output processes. Basically, the university runs several plants including a milk plant, agroindustrial institute and animal food plant, all of which need to increase their productivity with better outputs with high-quality inputs.

These units need to increase their income in order to cover their total costs and provide more net value. For the timebeing, these plants depend on the university budget, which comes from public funds, so increasing marketing and sales revenues will reduce their dependence on public funds, and they will eventually obtain a better university budget for undergraduate studies as a whole.

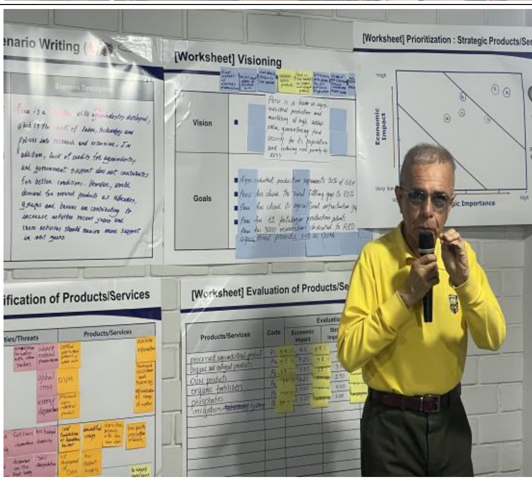
In addition, the Delphi and Steep methods could be incorporated into the curricula of various faculties (Economy, Food Industry, Science and Metereologist, Forestry, Fishing, Animal Science, Agronomy and Agriculture Engineering) as a part of their undergraduate and postgraduate courses.

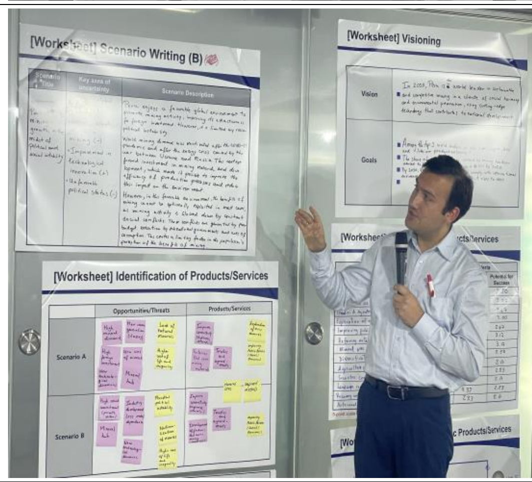
3.5 Photos

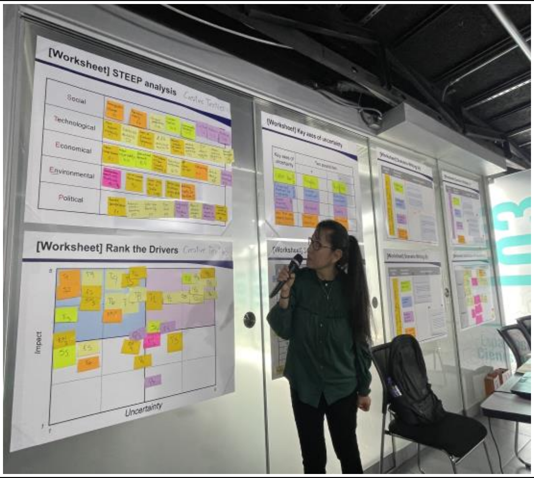

[Worksheet] Scenario selection


[Worksheet] Scenario


[Worksheet] Scenario selection


[Worksheet] Visioning
Vision
Goals
[Worksheet] Prioritization : Strategic Products/Ser
Economic Impact
[Worksheet] Evaluation of Products/Se
Products/Services


Scenario Writing (B)
[Worksheet] Visioning
Vision
Goals
[Worksheet] Identification of Products/Services
Opportunities/Threats
Products/Services
Scenario A
Scenario B


[Worksheet] STEEP analysis
Social
Technological
Economical
Environmental
Political
[Worksheet] Key axes of uncertainty
[Worksheet] Rank the Drivers
Impact
Uncertainty


[Worksheet] Scenario Writing (A)
[Worksheet] Identification of Products/Services
[Worksheet] Evaluation of Products/Services
[Worksheet] Scenario Writing (B)
[Worksheet] Visioning

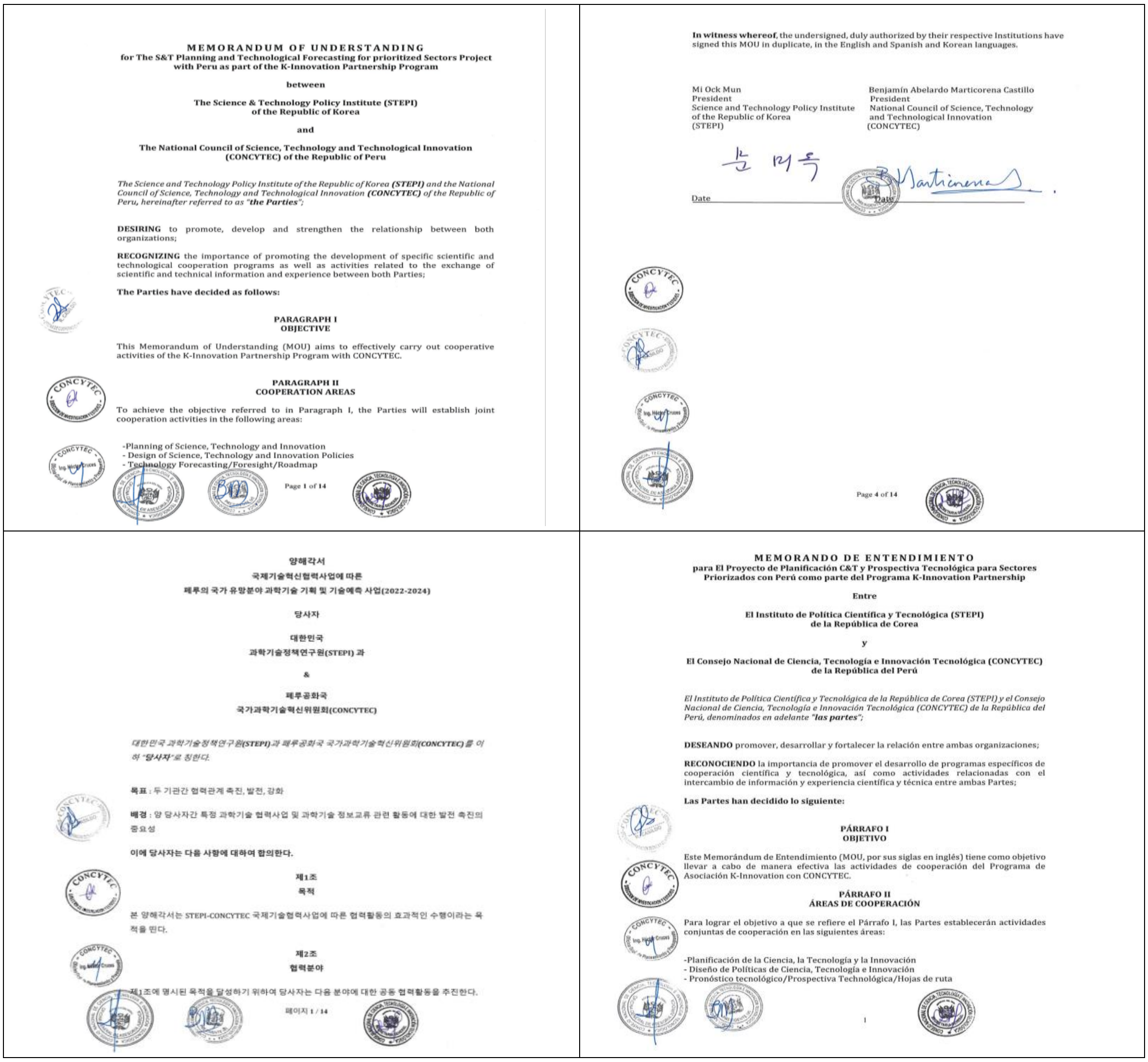

of consensus
2022 K-Innovation Partnership Program with Perú
2022 Dissemination Workshop on Technological Forecasting
CONCYTEC

4. MOU

STEPI and CONCYTEC signed the MOU for the “S&T Planning and Technological Forecasting for Prioritized Sectors” project with Peru as a part of the K-Innovation Partnership Program.

Due to the ongoing COVID-19 pandemic during the first half of 2022, both organizations discussed the MOU (Korean, English, and Spanish) via non-face-to-face meetings and e-mails, and exchanged signatures through diplomatic pouches.

[Figure 5-3] MOU between STEPI and CONCYTEC (English, Korean, Spansh)



MEMORANDUM OF UNDERSTANDING
for The S&T Planning and Technological Forecasting for prioritized Sectors Project with Peru as part of the K-Innovation Partnership Program

between

The Science & Technology Policy Institute (STEPI) of the Republic of Korea

and

The National Council of Science, Technology and Technological Innovation (CONCYTEC) of the Republic of Peru

*The Science and Technology Policy Institute of the Republic of Korea **(STEPI)** and the National Council of Science, Technology and Technological Innovation **(CONCYTEC)** of the Republic of Peru, hereinafter referred to as "**the Parties**";*

DESIRING to promote, develop and strengthen the relationship between both organizations;

RECOGNIZING the importance of promoting the development of specific scientific and technological cooperation programs as well as activities related to the exchange of scientific and technical information and experience between both Parties;

The Parties have decided as follows:

PARAGRAPH I
OBJECTIVE

This Memorandum of Understanding (MOU) aims to effectively carry out cooperative activities of the K-Innovation Partnership Program with CONCYTEC.

PARAGRAPH II
COOPERATION AREAS

To achieve the objective referred to in Paragraph I, the Parties will establish joint cooperation activities in the following areas:

-Planning of Science, Technology and Innovation
- Design of Science, Technology and Innovation Policies
- Technology Forecasting/Foresight/Roadmap

Page 1 of 14

In witness whereof, the undersigned, duly authorized by their respective Institutions have signed this MOU in duplicate, in the English and Spanish and Korean languages.

Mi Ock Mun President Science and Technology Policy Institute of the Republic of Korea (STEPI)	Benjamín Abelardo Marticorena Castillo President National Council of Science, Technology and Technological Innovation (CONCYTEC)
문미옥	B. Marticorena
Date	Date

Page 4 of 14

양해각서
국제기술혁신협력사업에 따른
페루의 국가 유망분야 과학기술 기획 및 기술예측 사업(2022-2024)

당사자

대한민국
과학기술정책연구원(STEPI) 과

&

페루공화국
국가과학기술혁신위원회(CONCYTEC)

*대한민국 과학기술정책연구원(STEPI)과 페루공화국 국가과학기술혁신위원회(CONCYTEC)를 이하 "**당사자**"로 칭한다.*

목표 : 두 기관간 협력관계 촉진, 발전, 강화

배경 : 양 당사자간 특정 과학기술 협력사업 및 과학기술 정보교류 관련 활동에 대한 발전 촉진의 중요성

이에 당사자는 다음 사항에 대하여 합의한다.

제1조
목적

본 양해각서는 STEPI-CONCYTEC 국제기술협력사업에 따른 협력활동의 효과적인 수행이라는 목적을 띤다.

제2조
협력분야

제1조에 명시된 목적을 달성하기 위하여 당사자는 다음 분야에 대한 공동 협력활동을 추진한다.

페이지 1 / 14

MEMORANDO DE ENTENDIMIENTO
para El Proyecto de Planificación C&T y Prospectiva Tecnológica para Sectores Priorizados con Perú como parte del Programa K-Innovation Partnership

Entre

El Instituto de Política Científica y Tecnológica (STEPI) de la República de Corea

y

El Consejo Nacional de Ciencia, Tecnología e Innovación Tecnológica (CONCYTEC) de la República del Perú

*El Instituto de Política Científica y Tecnológica de la República de Corea (STEPI) y el Consejo Nacional de Ciencia, Tecnología e Innovación Tecnológica (CONCYTEC) de la República del Perú, denominados en adelante "**las partes**";*

DESEANDO promover, desarrollar y fortalecer la relación entre ambas organizaciones;

RECONOCIENDO la importancia de promover el desarrollo de programas específicos de cooperación científica y tecnológica, así como actividades relacionadas con el intercambio de información y experiencia científica y técnica entre ambas Partes;

Las Partes han decidido lo siguiente:

PÁRRAFO I
OBJETIVO

Este Memorándum de Entendimiento (MOU, por sus siglas en inglés) tiene como objetivo llevar a cabo de manera efectiva las actividades de cooperación del Programa de Asociación K-Innovation con CONCYTEC.

PÁRRAFO II
ÁREAS DE COOPERACIÓN

Para lograr el objetivo a que se refiere el Párrafo I, las Partes establecerán actividades conjuntas de cooperación en las siguientes áreas:

-Planificación de la Ciencia, la Tecnología y la Innovación
- Diseño de Políticas de Ciencia, Tecnología e Innovación
- Pronóstico tecnológico/Prospectiva Technológica/Hojas de ruta

1

5. Korea-Peru S&T Cooperation Forums and Seminars

5.1 Seminar to Share South Korea's STI Policy

- Date: Friday, October 21, 2022, 09:00-12:30
- Host: STEPI, CONCYTEC
- Working Language: English, Spanish
- Venue: PUCP, Lima, Peru

5.1.1 Program

Friday, 21 October 2022	
9:00-9:30	Opening Ceremony with introductory words by President of CONCYTEC Dr. Benjamin Marticorena.
9:30-10:00	First lecture: – **Dr. ChiUng Song (Senior Research Fellow, STEPI):** The Korean STI Policy in the context of National Development
10:00-10:30	Second Lecture: – **Dr. Johng-Ihl Lee (Professor, Department of Technology and Society, SUNY Korea):** Technology Forecasting as a Policy Development Tool
10:30-11:00	Third Lecture: – **- Dr. Sehee Park (CTO, Deliverdy/ Former Director of KIAT(Korea Institute for Advancement of Technology)):** Industrial Technology Roadmapping and its application in Korea
11:00-11:30	Q&A session
11:30-12:00	Panel (3 Peruvian experts)
12:00-12:20	Q&A session
12:20-12:30	Closing Ceremony

5.1.2 Photos and News Media

Escuela de Posgrado PUCP
10월 22일 오전 2:05 ·

Estamos en el Seminario "La Experiencia de la República de Corea en el Uso de Herramientas para el Diseño de la política de Ciencia, Tecnología e Innovación", organizado por Concytec, Stepi Corea, Ceplan Perú y nuestra Escuela de Posgrado PUCP.

#Tecnologia #Información

번역 보기

Source: PUCP Facebook https://www.facebook.com/posgradopucp (Search date: November 14, 2022)

Source: STEPI Homepage(December 11, 2022)

https://www.stepi.re.kr/site/stepiko/ex/bbs/View.do?pageIndex=1&bcIdx=38884&cbIdx=1217&mode=&tgtTypeCd=SUB_CONT&searchKey=&searchSort=REG_DT (Search date: November 14, 2022)

5.2 2022 Korea-Peru S&T Cooperation Forum

- Date: Friday, October 21, 2022, 14:00-18:00
- Hosts: Embassy of the Republica of Korea in Peru, STEPI, CONCYTEC
- Working Languages: English, Spanish, Korean
- Venue: CCL, Lima, Peru

5.2.1 Program

Friday, October 21, 2022	
14:00-18:00	Forum on Korea-Peru STI Cooperation hosted by the Embassy of South Korea in Lima-STEPI-CONCYTEC (14:10-14:30) 1. Dr. ChiUng Song (Senior Research Fellow, STEPI): Innovation at the Nexus of Science and Technology along with Society and Culture (14:30-15:00) 2. Dr. Deok Soon Yim (Senior Research Fellow, STEPI): The Establishment and Management of Science and Technology Clusters in Korea (online) (15:10-15:50) *Dr. Johng-Ihl Lee (Professor, Department of Technology and Society, SUNY Korea): Technoparks in Korea : Establishment, Management and Policy Implications* (16:20-16:50) Dr. ChiUng Song (Senior Research Fellow, STEPI): Panel Discussion

5.2.2 Photos and News Media

gob.pe | Plataforma digital única del Estado Peruano

Inicio > El Estado > PCM > CONCYTEC > Noticias > Corea del Sur compartirá su experiencia en parq...

Consejo Nacional de Ciencia, Tecnología e Innovación Tecnológica

Corea del Sur compartirá su experiencia en parques científicos tecnológicos durante foro del Concytec

Nota de Prensa

El evento contará con la presencia de expertos del Centro de Capacitación en Estrategia de Alta Tecnología de Asia-Pacífico (STEPI), así como catedráticos de la Universidad de Nueva York, sede Corea.

5 de octubre de 2022 - 11:29 a. m.

El desarrollo de los parques científicos tecnológicos ha generado el interés de gobiernos regionales y universidades, sobre todo en el interior del país, que buscan contar con espacios o centros de atracción y creación de nuevas empresas innovadoras, como catalizadores de la innovación y de la transferencia tecnológica.

Source: Plataforma digital única del Estado Peruano

https://www.gob.pe/institucion/concytec/noticias/657599-corea-del-sur-compartira-su-experiencia-en-parques-cientificos-tecnologicos-durante-foro-del-concytec (Search date: November 14, 2022)

CHAPTER 6

Remarks and Implication

S&T Planning and Technological Forecasting
for Prioritized Sectors in Peru

Chapter 6. Remarks and Implication

1. Wrap up of the 1st year Activities

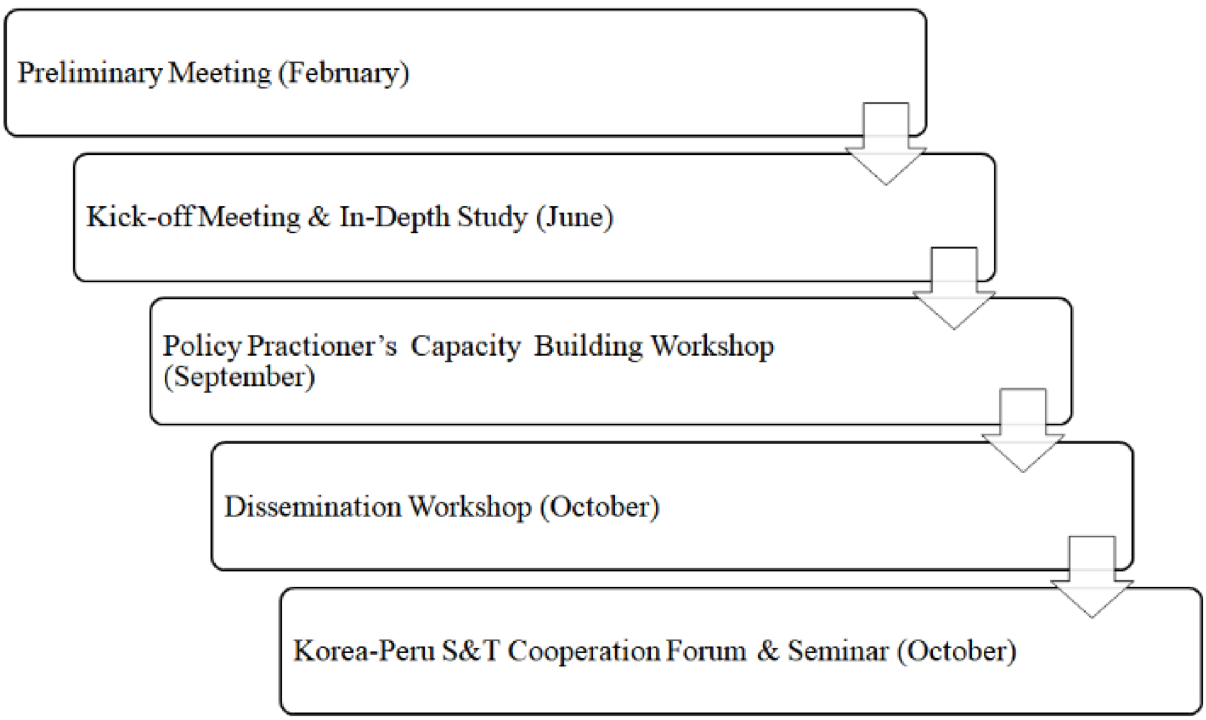
This is the first year of the K-Innovation Partnership Program with Peru, which will last until the year 2024. First, we set out to determine the scope of the project, identify the annual tasks, organize the timetable, and establish a channel of communication and dialogue with CONCYTEC. In February, the STEPI team and CONCYTEC held a preliminary meeting online and discussed the scope of the project, the annual tasks, and a provisional timetable. During this meeting, the STEPI team and CONCYTEC agreed to begin with the theme of technology foresight in the first year (2022). STEPI's team suggested that holding a workshop on technology foresight in order to provide fundamental knowledge and practical experience.

Following lengthy discussions and interaction on the MOU and TOR, the STEPI team and CONCYTEC were able to hold the Kick-off Meeting and In-Depth Study Session in June 2022. During this meeting, the two parties agreed to hold an on-site workshop on technology foresight and an S&T seminar in October of the same year in Lima. At the same time, the STEPI team proposed holding the Workshop for Building the Capacities and Competencies of Policy Practitioners in September online in order to provide more detailed knowledge and information on technology foresight.

On the basis of the Workshop for Building the Capacities and Competencies of Policy Practitioners, the STEPI team and Peruvian experts, along with the representatives of CONCYTEC, were able to share South Korea's experience of STI development as well as the concepts of the technology foresight practices of the Korean government. Korea's experience in establishing the relevant institutions, such as MOST (Ministry of Science and Technology), KIST (Korea Institute of Science and Technology), KAIST (Korean Advanced Institute of Science and Technology), and the GRIs (government research institutes) would be appropriate benchmarking case for the Peruvian government, as it has planned to establish the tentative MOSTI (Ministry of Science, Technology and Innovation) in the last few years.

At the same time, the Peruvian government also had plans to establish an organization to handle the tasks associated with technology forecasting and foresight, which is why it applied for the K-Innovation Partnershop Program earlier. Thus, the role of the STEPI team is to provide the theoretical background to technology foresight, share the practical experience of South Korea, and virtually support the capacity building of the Peruvian experts who will be involved in TF duties. The tasks and timetable for this year have been established to correspond to the very clear goal of the project. During the three-days workshop held in Lima, the Peruvian experts activley participated in and familiarized themselves with practice of technology foresight.

[Figure 6-1] Main Activities, 2022



2. Goals and Plan for the 2nd year Activities

The main task for next year (2023) will be to create the technology roadmap for the six prioritized sectors. To this end, the STEPI team has held several workshops on technology foresight as of 2022. We intend to invite 8-10 Peruvian experts to Korea to attend the Workshop for Building the Capacities and Competencies of Policy Practitioners in the first quarter of 2023. During this workshop, we will produce the first draft of the technology roadmap on the basis of this year's workshop held in Lima. At the same time, the Peruvian experts will visit a number of organizations and meet Korean experts from both the public and private sectors, which will serve as the starting point for networking on both sides.

The Peruvian experts will return home after the session in Korea with their assignments. The STEPI Team will hold online meetings from time to time to check whether the assignments are beign carried out properly. Then, the STEPI team and CONCYTEC will hold an In-depth Practice Workshop for the technology roadmap in the second quarter of 2023. At this workshop, the Peruvian partners will present the Interim Report on the Technology Roadmap for Six Sectors, and the STEPI team, with the aid of Korea experts, will examine the report and advize the Peruvian partners accordingly.

Finally, the STEPI team will travel to Peru again in order to hold the Dissemination Workshop and present the final draft of the technology roadmap. The most important and meaningful aspect of this trip is that the STEPI team and the Peruvian partners (CONCYTEC) will present the technology roadmap elaborated by the Peruvian side. We plan to visit Perui October 2023, but the timetable has not yet been fixed.

The technology roadmap will be utilized in the year 2024 when the STEPI team and Peruvian experts deal with the R&D programming and the related S&T policies. By creating this technology roadmap, the government will be able to determine which kinds of technology should be developed, by what date the prior technology should be acquired, and how much funding should be provided. As such, the technology roadmap will be the base line or starting point for introducing the R&D program and other policies.

[Figure 6-2] below shows the tometable and the main activities which the STEPI team and Peruvian experts will pursue together in order to accomplish our goals. Actually, we would like to take two business trips to Peru next year, although we may only be able to go once due to budgetary concerns.

[Figure 6-2] Main Activities, 2023

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Chapter 4

1. Agroindustry and Food Processing

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Chapter 5

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